The arithmetic trace quotient and exact heat transport

Closed images, differential correspondences, and the retained inverse-root cover

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1 The primary arithmetic object and the topology actually specified

The starting point is Connes–Consani–Moscovici, *Zeta zeros and prolate wave operators: Semilocal adelic operators*, Section 3.6, proof-PDF pages 17–19, especially Lemma 3.3, equations (3.16)–(3.22), and Proposition 3.6 [1]. The preceding arithmetic map is

$$\mathcal{E}f(x) = x^{1/2}\sum_{n\geq 1} f(nx), \qquad \mathcal{F}_\mu g(s) = \int_0^\infty g(x)x^{-is}\,\frac{dx}{x}.$$

The input consists of even Schwartz functions with $f(0) = \widehat{f}(0) = 0$. The paper's two Hermite families are

$$\psi_\ell^+ = h_{4\ell} - \frac{h_{4\ell}(0)}{h_0(0)}h_0, \qquad \psi_\ell^- = -h_{4\ell+2} + \frac{h_{4\ell+2}(0)}{h_2(0)}h_2 \quad (\ell \geq 1).$$

These signs and the minus sign in the Mellin exponent are retained. The completed function is

$$\Xi(s) = \xi(\tfrac{1}{2} + is), \qquad \xi(z) = \tfrac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z), \qquad \Xi(s) = 8H_0(2s). \tag{1}$$

The paper proves

$$\mathcal{F}_\mu\mathcal{E}\psi_\ell^\pm = R_\ell^\pm\Xi, \qquad P_\ell^\pm = -\tfrac{1}{2}(\tfrac{1}{4} + s^2)R_\ell^\pm. \tag{2}$$

It also states immediately before Proposition 3.6 that these give all polynomial multiples of Ξ. Here is the precise algebra behind the last assertion. The top summands of (3.19) and (3.20) are nonzero; their degrees are 2ℓ and $2\ell+1$. Division by $-\frac{1}{2}(\frac{1}{4}+s^2)$ gives degrees $2\ell-2$ and $2\ell-1$, respectively. The even and odd parities remain. Thus the ordered family $R_1^+, R_1^-, R_2^+, R_2^-, \ldots$ has one polynomial of each degree, with a nonzero leading coefficient. Subtracting the appropriate multiple of the polynomial of largest degree, and iterating down to degree zero, expresses every polynomial as a finite linear combination. Linear independence follows by the largest-degree coefficient of a finite relation. Consequently

$$N_{\rm ar} = \overline{\operatorname{span}\{\mathcal{F}_\mu\mathcal{E}\psi_\ell^\pm\}}^{\tau} = \overline{\mathbb{C}[s]\Xi}^{\tau}, \qquad Q_{\rm ar} = (\mathcal{H}_{\leq 1}, \tau)/N_{\rm ar}. \tag{3}$$

Here τ denotes precisely the source's topology. No equality of this closed space with a principal algebraic ideal is being asserted. Multiplication $Mf(s) = sf(s)$ is continuous in the topological ring; it preserves the dense generating subspace, and hence $N_{\rm ar}$. Its induced operator S is the operator of Proposition 3.6. We use the source's spectral assertion as an explicitly imported theorem:

$$\operatorname{Spec}(S) = \{\lambda \in \mathbb{C} : \zeta(\tfrac{1}{2} + i\lambda) = 0 \text{ at a nontrivial zero}\}. \tag{4}$$

Spectrum here means failure of a continuous two-sided inverse. No proof of the entire source spectral theorem is claimed as a new result of this manuscript.

There is a material source limitation. Page 19 calls $\mathcal{H}_{\leq 1}$ the Hadamard topological ring of entire functions of order at most one; it does not state seminorms or a convergence criterion. The cited Connes–Marcolli Proposition 2.24, printed pages 379–381, proves the polynomial-image assertion and explicitly discusses different function-space realizations [2]. It does not supply that missing specification. We therefore retain τ in every source quotient. Below, all assertions depending on another topology name and define that topology. The comparison with the source is an exact quotient diagram, rather than an identification by terminology. The bounded edition audit gives the same limitation in the primary arXiv v1 source, lines 683–704 (Proposition 3.9), and in v2, `mainc2m24fine.tex`, lines 668–689 (Proposition 3.6): each states the order class, polynomial trace image, and spectral set, without defining its topology or algebraic multiplicities. These exact edition locators supplement the author-proof locator in [1]; no claim of byte identity with a publisher's final subscription text is required. Following the book's cited Sobolev construction leads to explicit weighted Hilbert spaces in its Chapter 2, Section 9, and in Connes's original spectral realization [3]. Section 8.1 retains those spaces and proves their exact common-test comparison with Q_{ar}, including the original generator and the allowed zero multiplicities. The inspected construction does not specify the topology on all of $\mathcal{H}_{\leq 1}$ or identify it with those Hilbert topologies.

2 A globally defined heat map on the original underlying functions

For an entire function f, let $M_f(r) = \max_{|s|\leq r} |f(s)|$ and define its order by

$$\limsup_{r\to\infty} \frac{\log\log\max(e, M_f(r))}{\log r}.$$

The zero function is included. For $1 < p < 2$, $a > 0$ set

$$\|f\|_{p,a} = \sup_{s\in\mathbb{C}} |f(s)|e^{-a|s|^p}.$$

The underlying set $\mathcal{H}_{\leq 1}$ is exactly the set where all these seminorms are finite. Indeed order at most one gives, for $1 < q < p$, $M_f(r) \leq \exp(r^q)$ for sufficiently large r; the expression $r^q - ar^p$ is bounded above. Conversely finiteness gives order at most p for every $p > 1$. Write γ for this explicitly defined growth topology. It is enough to take $p = 1 + 1/j$, $j \geq 2$, and $a = 1/k$, $k \geq 1$: given $p > 1, a > 0$, select one of these $p_j < p$ and use the boundedness of $|s|^{p_j}/k - a|s|^p$. This proves equivalence of the countable family with all displayed seminorms. A sequence Cauchy in these seminorms converges uniformly on compact sets to an entire function. Taking pointwise limits in each weighted Cauchy inequality gives convergence in that seminorm and finiteness of its norm. Thus $(\mathcal{H}_{\leq 1}, \gamma)$ is a complete metrizable locally convex space.

Lemma 2.1 (Derivative bounds and the heat group). *Put $D = \partial_s$. Differentiation and multiplication by s map the original underlying set $\mathcal{H}_{\leq 1}$ into itself. For $m \geq 1$,*

$$\|D^m f\|_{p,a} \leq m! \left(\frac{epa}{m}\right)^{m/p} \|f\|_{p,a/2^{p-1}}. \tag{5}$$

For every $t \in \mathbb{C}$, the series

$$U_t f = \sum_{k=0}^{\infty} \frac{(-t/4)^k}{k!} D^{2k} f \tag{6}$$

converges in γ, locally uniformly in t; U_t is a continuous automorphism of $(\mathcal{H}_{\le 1}, \gamma)$ and $U_t U_u = U_{t+u}$, $U_t^{-1} = U_{-t}$.

Proof. Cauchy's circle formula on a circle of radius R about s gives

$$|D^m f(s)| \le m! R^{-m} \max_{|z-s|=R} |f(z)|.$$

For $b = a/2^{p-1}$ the inequality $|s+z|^p \le 2^{p-1}(|s|^p + |z|^p)$ bounds the last expression, after multiplication by $e^{-a|s|^p}$, by $m! R^{-m} e^{aR^p} \|f\|_{p,b}$. Choosing $R = (m/(ap))^{1/p}$ proves (5). For multiplication use

$$\|Mf\|_{p,a} \le \sup_{r \ge 0} r e^{-ar^p/2} \|f\|_{p,a/2}.$$

The kth heat summand is at most

$$\frac{(|t|/4)^k}{k!} (2k)! \left(\frac{epa}{2k}\right)^{2k/p} \|f\|_{p,b}.$$

Since $(2k)!/k! \le (2k)^k$, its scalar factor is at most $[C_{p,a}|t|k^{1-2/p}]^k$, whose kth root tends to zero. This proves summability, continuity, and locally uniform convergence of every time-differentiated series by the same estimate with the additional polynomial factors in k. Commutation with D follows from continuity of D. In the double composition series group terms with $n = k + \ell$. The sum of the absolute scalar coefficients at $D^{2n} f$ is $((|t| + |u|)/4)^n / n!$; the preceding estimate justifies regrouping. The binomial theorem gives U_{t+u}. The value at zero is the identity, proving the inverse. $\square$

Lemma 2.2 (Polynomial approximation)**.** *The Taylor polynomials of any $f \in \mathcal{H}_{\le 1}$ converge to f in γ. Multiplication of two elements is continuous in γ.*

Proof. If $f = \sum c_n s^n$, Cauchy's coefficient bound with parameter $b < a$ gives, for $n \ge 1$,

$$|c_n| \le \|f\|_{p,b} (epb/n)^{n/p}.$$

Also $\|s^n\|_{p,a} = (n/(epa))^{n/p}$. Therefore $\|c_n s^n\|_{p,a} \le \|f\|_{p,b} (b/a)^{n/p}$. The tail is bounded by a convergent geometric series. Finally $\|fg\|_{p,a} \le \|f\|_{p,a/2} \|g\|_{p,a/2}$. $\square$

Retain the actual real heat family, rather than substitute a polynomial:

$$\begin{aligned} \Phi(u) &= \sum_{n \ge 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ Z_t(s) &= 8H_t(2s) = 8 \int_0^\infty e^{tu^2} \Phi(u) \cos(2su)\, du, \qquad \partial_t Z_t = -\tfrac{1}{4} D^2 Z_t, Z_0 = \Xi. \end{aligned} \tag{7}$$

These are the Rodgers–Tao conventions, Section 1 [4]. For clarity the analytic exchanges in this transport can be proved directly. Since $n^2 e^{4u} \ge (n^2 + e^{4u})/2$, the absolutely convergent sums $\sum n^j e^{-\pi n^2/2}$ give constants $C, c > 0$ such that $|\Phi(u)| \le C e^{9u} e^{-ce^{4u}}$ for $u \ge 0$. For $|t| \le T$ and $|s| \le R$, every derivative of the integrand is bounded by a polynomial in u times $C \exp(Tu^2 + (2R+9)u - ce^{4u})$, which is integrable. The exponential series in t can therefore be integrated absolutely. Because $D^{2k} \cos(2su) = (-1)^k (2u)^{2k} \cos(2su)$, it gives exactly $Z_t = U_t \Xi$ with the coefficient $(-t/4)^k / k!$. In particular Z_t lies in the original order-at-most-one set for every complex t, by Lemma 2.1; it is never identically zero because U_t is invertible and $\Xi(0) \ne 0$.

3 Transport of the actual closed arithmetic image

Define τ_t on the same underlying vector space $\mathcal{H}_{\le 1}$ by declaring $U_t : (\mathcal{H}_{\le 1}, \tau) \longrightarrow (\mathcal{H}_{\le 1}, \tau_t)$ a homeomorphism. This definition specifies every open set: V is τ_t-open exactly when $U_{-t}V$ is τ-open. It uses the globally defined bijection just proved and does not require unproved continuity in τ. Set

$$N_t = U_t N_{\mathrm{ar}} = \overline{\operatorname{span}\{U_t(R_\ell^\pm \Xi)\}}^{\tau_t}, \qquad Q_t = (\mathcal{H}_{\le 1}, \tau_t)/N_t. \tag{8}$$

The closure equality follows because homeomorphisms carry closure to closure. In particular N_t is closed. The quotient map

$$\mathcal{U}_t : Q_{\mathrm{ar}} \longrightarrow Q_t, \qquad [f] \longmapsto [U_t f] \tag{9}$$

is well defined, bijective, and continuous, with continuous inverse $[g] \mapsto [U_{-t}g]$. Indeed changing f by $n \in N_{\mathrm{ar}}$ changes $U_t f$ by $U_t n \in N_t$, and the inverse gives necessity; the quotient universal property proves both continuity statements.

Theorem 3.1 (Exact arithmetic heat intertwiner)**.** *On the entire-function set,*

$$B_t = U_t M U_{-t} = M - \tfrac{t}{2} D. \tag{10}$$

This operator preserves N_t, is continuous in τ_t, and induces S_t on Q_t. It satisfies $S_t \mathcal{U}_t = \mathcal{U}_t S$ and

$$\operatorname{Spec}(S_t) = \operatorname{Spec}(S) = \{\lambda : \zeta(\tfrac{1}{2} + i\lambda) = 0 \textit{ nontrivially}\}. \tag{11}$$

The time-dependent trace functions are exactly $U_t(R_\ell^\pm \Xi) = R_\ell^\pm(B_t) Z_t$.

Proof. The product rule gives $D^{2k} M f = M D^{2k} f + 2k D^{2k-1} f$. Insert this identity in the convergent heat series to obtain $U_t M = (M - (t/2)D)U_t$, proving (10). If $n \in N_{\mathrm{ar}}$, then $B_t U_t n = U_t M n \in N_t$. Continuity is conjugation of the continuous M by the defining homeomorphism. The induced identity follows on representatives. If T is a continuous inverse of $S - \lambda$, then $\mathcal{U}_t T \mathcal{U}_t^{-1}$ is a continuous inverse of $S_t - \lambda$. Conjugating in the opposite direction proves the converse, and hence the spectrum equality using (4). Apply the polynomial identity $U_t p(M) = p(B_t) U_t$ to Ξ for the last assertion. $\square$

The heat map also transports the full multiplication, with every cross term retained. For $f, g \in \mathcal{H}_{\le 1}$ define

$$f \star_t g = U_t\big((U_{-t}f)(U_{-t}g)\big) = \sum_{n=0}^{\infty} \frac{(-t/2)^n}{n!} (D^n f)(D^n g). \tag{12}$$

We prove the series identity, including convergence. For $F_t = U_t f$, $G_t = U_t g$, the series

$$J_t = \sum_{n=0}^{\infty} \frac{(-t/2)^n}{n!} (D^n F_t)(D^n G_t)$$

converges in γ, uniformly on compact time sets: applying (5) with output norm $(p, a/2)$ to both factors bounds its nth term by

$$K \left[\frac{T}{2} \left(\frac{epa}{2} \right)^{2/p} n^{1-2/p} \right]^n \qquad (|t| \le T),$$

where K bounds the product of the two input norms $(p, a/2^p)$, and is finite by the heat estimate. The root test applies. Cauchy's formula in time permits termwise time derivatives; continuity of D permits spatial derivatives. Differentiating the coefficient contributes $-\frac{1}{2}\sum(-t/2)^n(D^{n+1}F_t)(D^{n+1}G_t)/n!$. Differentiating the factors contributes the two pure second-derivative terms with coefficient $-1/4$. The complete product rule therefore gives $\partial_t J_t = -D^2 J_t/4$ and $J_0 = fg$. Its entire time Taylor coefficients are uniquely $(-1/4)^k D^{2k}(fg)/k!$, so $J_t = U_t(fg)$. Replacing the inputs by $U_{-t}f, U_{-t}g$ proves (12). Conjugation of ordinary multiplication proves associativity, commutativity, unit 1, and continuity for τ_t, even when continuity of the series in the source topology is unknown. The identity $s \star_t f = sf - tf'/2$ recovers exactly B_t. No ideal property of N_{ar} beyond its stated polynomial invariance is required for this identity or for the quotient in Theorem 3.1.

The transported actual trace image has no common point-evaluation zero at any nonzero time. Suppose λ were such a common zero. It would annihilate each $U_t(s^n\Xi)$. For every $a \in \mathbb{C}$, the Taylor series of e^{as} converges in γ by Lemma 2.2. Multiplication, U_t, and evaluation are continuous in γ, so

$$0 = \sum_{n=0}^{\infty} \frac{a^n}{n!} U_t(s^n\Xi)(\lambda) = e^{a\lambda - ta^2/4} Z_t(\lambda - ta/2).$$

The last equality follows from the just-proved product formula and the entire Taylor formula for Z_t. At $t \neq 0$ varying a forces Z_t to vanish everywhere, a contradiction. This argument uses only the polynomial generators contained in N_t; it assumes no source-topology convergence of the exponential series. Thus the source arithmetic spectrum in (11) is carried by the explicitly transported operator and relation, even though the transported relation has no common ordinary evaluation zero.

The first two nonconstant polynomial generators make the extra terms visible without changing any polynomial coefficient:

$$\begin{aligned} U_t(s\Xi) &= sZ_t - \tfrac{t}{2}Z_t', \\ U_t(s^2\Xi) &= s^2 Z_t - tsZ_t' - \tfrac{t}{2}Z_t + \tfrac{t^2}{4}Z_t''. \end{aligned} \tag{13}$$

For $L = \partial_t + \frac{1}{4}D^2$, the complete product rule is

$$L(AP) = A(P_t + \tfrac{1}{4}P_{ss}) + (A_t + \tfrac{1}{4}A_{ss})P + \tfrac{1}{2}A_s P_s. \tag{14}$$

Thus $L(R(s)Z_t) = \frac{1}{4}R''Z_t + \frac{1}{2}R'Z_t'$. The family $R(B_t)Z_t$ solves the homogeneous heat equation, whereas the displayed expression is the full defect of replacing it by $R(s)Z_t$. For a local factorization $Z_t = AP$ with $A \neq 0$, the same formula retains the analytic unit A and its derivatives. At a simple zero $\lambda(t)$, differentiating $Z_t(\lambda(t)) = 0$ gives

$$\lambda'(t) = \frac{1}{4}\frac{Z_{ss}}{Z_s}(t, \lambda(t)) = \frac{1}{4}\frac{P_{ss}}{P_s}(t, \lambda(t)) + \frac{1}{2}\frac{A_s}{A}(t, \lambda(t)). \tag{15}$$

This statement is on the open locus $Z_s \neq 0$, on which the analytic implicit-function theorem supplies the local curve; it claims no chosen zero is simple. At a multiple zero the division is undefined, and the full equation (14) remains the equation to use.

4 Derivative domains, multivalued operators, and jet extensions

Every derivative has the full underlying domain $\mathcal{H}_{\leq 1}$ by Lemma 2.1. Continuity in the unnamed source topology is not assumed. The following gives exact source-quotient operators and their domains without imposing it. For $k \geq 1$ put

$$K_k = \{n \in N_{\mathrm{ar}} : D^k n \in N_{\mathrm{ar}}\}, \qquad W_k = (D^k N_{\mathrm{ar}} + N_{\mathrm{ar}})/N_{\mathrm{ar}} \subseteq Q_{\mathrm{ar}}.$$

The derivative correspondence is the actual linear relation

$$\mathfrak{D}_k = \{([f], [D^k f]) : f \in \mathcal{H}_{\leq 1}\} \subseteq Q_{\mathrm{ar}} \times Q_{\mathrm{ar}}. \tag{16}$$

For fixed $[f]$, its output fibre is precisely $[D^k f] + W_k$: all representatives are $f + n$ with $n \in N_{\mathrm{ar}}$, and linearity gives all and only the stated outputs. The kernel of its presenting map $f \mapsto ([f], [D^k f])$ is exactly K_k. The relation has domain all of Q_{ar}; its multivalued part at zero is exactly W_k. Thus it is the graph of an operator exactly when $D^k N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$. This is a calculated equivalence, with an explicit obstruction space, rather than an assumption of invariance.

There is no nonempty domain of quotient classes on which the same representative formula is unambiguous if $W_k \neq 0$ and every representative of a class is admitted: addition of the same offending n gives two different outputs for every class. A chosen linear lift is a different map. More precisely any vector subspace $A \subseteq \mathcal{H}_{\leq 1}$ supplies a single-valued map on its quotient image $q(A)$ exactly when $D^k(A \cap N_{\mathrm{ar}}) \subseteq N_{\mathrm{ar}}$; the identical difference-of-representatives argument proves both directions. The first heat generator is $-D^2/4$, so its ambiguity space is $-W_2/4$, with the coefficient and sign unchanged.

4.1 Exact algebraic spectral defects of the source relation

These calculations keep the actual N_{ar} and require only the already proved $MN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$. For every $\lambda \in \mathbb{C}$, division by $x = s - \lambda$ gives the algebraic identities

$$\begin{aligned} \operatorname{coker}_{\mathrm{alg}}(S - \lambda) &= \mathcal{H}_{\leq 1}/(N_{\mathrm{ar}} + x\mathcal{H}_{\leq 1}), \\ \ker(S - \lambda) &\simeq (N_{\mathrm{ar}} \cap x\mathcal{H}_{\leq 1})/(xN_{\mathrm{ar}}), \quad [n] \longmapsto [n/x]. \end{aligned} \tag{17}$$

Division belongs to the original order class: if $f(\lambda) = 0$, the filled quotient is entire, is bounded on a fixed disk, and outside $|s| \geq 2|\lambda| + 1$ its modulus is at most $2|f(s)|$. Thus $x\mathcal{H}_{\leq 1}$ is exactly the kernel of evaluation at λ. The cokernel identity follows from the image of the induced multiplication. For the second identity, changing n by xn_0 changes n/x by $n_0 \in N_{\mathrm{ar}}$; its image is in the stated kernel. Conversely any kernel representative f has $xf \in N_{\mathrm{ar}} \cap x\mathcal{H}_{\leq 1}$. Finally $[n/x] = 0$ exactly when $n \in xN_{\mathrm{ar}}$. This proves every asserted kernel and image.

The cokernel has only two possibilities. When every $n \in N_{\mathrm{ar}}$ vanishes at λ, it is one-dimensional, with its exact identification given by evaluation. Otherwise retain any $h_\lambda \in N_{\mathrm{ar}}$ with $h_\lambda(\lambda) \neq 0$. Then

$$\begin{aligned} A_\lambda f(s) &= \frac{f(s) - f(\lambda) h_\lambda(s)/h_\lambda(\lambda)}{s - \lambda}, \\ f &= \frac{f(\lambda)}{h_\lambda(\lambda)} h_\lambda + (s - \lambda) A_\lambda f \end{aligned} \tag{18}$$

proves surjectivity of $S - \lambda$, and its exact remaining kernel is

$$\ker(S - \lambda) \simeq N_{\mathrm{ar}}/((s - \lambda)N_{\mathrm{ar}} \oplus \mathbb{C}h_\lambda), \qquad n \longmapsto [A_\lambda n]. \tag{19}$$

The map is onto because a kernel class $[g]$ is the image of $n = (s - \lambda)g \in N_{\mathrm{ar}}$. Its kernel consists exactly of $n = (s - \lambda)n_0 + ch_\lambda$, $n_0 \in N_{\mathrm{ar}}$, by (18). The sum is direct by evaluation. In this second case algebraic invertibility is therefore equivalent to the exact identity $N_{\mathrm{ar}} = (s - \lambda)N_{\mathrm{ar}} \oplus \mathbb{C}h_\lambda$.

On that locus the representative formula $[f] \mapsto [A_\lambda f]$ is independent of representatives, since $A_\lambda N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$; multiplication in (18) proves the right inverse, and $A_\lambda((s-\lambda)f) = f$ proves the left inverse. These are algebraic statements; continuity of this inverse in τ has not been inserted.

The first derivative and the heat generator have exactly the same frozen invariance question for the source subspace:

$$DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \iff D^2 N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \iff -\tfrac{1}{4} D^2 N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}. \tag{20}$$

Indeed the product rule is $D^2(sn) - sD^2 n = 2Dn$. Since $sn \in N_{\mathrm{ar}}$, second-derivative invariance puts both terms on the left in N_{ar}, proving first-derivative invariance. Iteration proves the reverse implication; the retained nonzero scalar $-1/4$ proves the last equivalence. If these equivalent inclusions hold, all $D^j \Xi$ belong to N_{ar}. At each λ at least one is nonzero: otherwise the Taylor series would make Ξ identically zero. Thus every $S - \lambda$ is then algebraically surjective by (18). Its prescribed topological spectral points would be represented entirely by the kernel (19) or a noncontinuous algebraic inverse. This computes the precise alternatives without assigning either one to the source's stated spectral set.

There is also a full multiplicity consequence of algebraic derivative descent. Writing $d[f] = [Df]$ on its exact descent locus, the product rule gives $[d, S] = I$. For $0 \neq v \in \ker(S - \lambda)$,

$$(S-\lambda)d^k v = -k d^{k-1} v, \qquad (S-\lambda)^k d^k v = (-1)^k k! v \quad (k \geq 0), \tag{21}$$

with zero right side in the first identity at $k = 0$. To prove induction, commute the next d using $(S-\lambda)d = d(S-\lambda) - I$; the coefficient $-k$ becomes $-(k+1)$. Iteration proves the second identity. A finite linear dependence among $v, dv, \ldots, d^n v$, with largest nonzero coefficient at index j, becomes that coefficient times $(-1)^j j! v$ after applying $(S-\lambda)^j$, a contradiction. Thus every such eigenvector generates an infinite independent generalized eigenvector sequence. No finite source multiplicity assertion has been assumed.

Finally even a single forward nonzero heat inclusion has a calculated consequence. Fix $t \neq 0$ and $c = t/2$, and define

$$\begin{gathered} A_0 = U_t, \qquad A_{k+1} = MA_k - A_k M, \qquad A_k = P_k(D) U_t, \\ P_0(X) = 1, \qquad P_{k+1}(X) = cXP_k(X) - P_k'(X). \end{gathered} \tag{22}$$

The identity follows by induction from $[M, U_t] = cDU_t$ and $[M, D^j] = -jD^{j-1}$: commuting M through $P_k(D)U_t$ contributes $-P_k'(D)U_t + cP_k(D)DU_t$. The degree of P_k is exactly k, with leading coefficient $c^k \neq 0$. Under $U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$, induction with $MN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ gives $A_k N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$; subtracting the lower-degree terms of P_k then gives exactly

$$D^k U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \quad (k \geq 0). \tag{23}$$

In particular every derivative of the nonzero $Z_t = U_t \Xi$ lies in N_{ar}. At every point one derivative is nonzero by the entire Taylor theorem, so every $S - \lambda$ is algebraically surjective in this case too. This proof uses only the one forward inclusion. For the stronger equality $U_t N_{\mathrm{ar}} = N_{\mathrm{ar}}$, it also proves $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$, by writing each $n \in N_{\mathrm{ar}}$ as $U_t m$ in (23). None of these exact implications presumes the inclusion or equality for the actual source closure. The explicit obstruction spaces above remain the calculated source question.

Even when W_k is not closed, the smallest source-closed enlargement of the target relation subspace is explicit:

$$[f] \longmapsto [D^k f] : Q_{\mathrm{ar}} \longrightarrow \mathcal{H}_{\leq 1} / \overline{N_{\mathrm{ar}} + D^k N_{\mathrm{ar}}}^{\tau}. \tag{24}$$

It is well defined algebraically. Equip its source with the quotient of the graph topology $f \mapsto (f, D^k f)$ in $(\mathcal{H}_{\le 1}, \tau)^2$ to obtain a continuous map. This specifies a topology change explicitly, without claiming it is the original one. The lost part of the original target Q_{ar} is exactly $\overline{N_{\mathrm{ar}} + D^k N_{\mathrm{ar}}}^{\tau} / N_{\mathrm{ar}}$.

Here is an extension which retains correlated derivatives rather than quotienting them away. For $j \ge 1$ define the bijection

$$T_j(f_0, \dots, f_j) = (f_0, f_1 - Df_0, \dots, f_j - D^j f_0).$$

Its inverse is $(f_0, g_1, \dots, g_j) \mapsto (f_0, g_1 + Df_0, \dots, g_j + D^j f_0)$. Give $V_j = \mathcal{H}_{\le 1}^{j+1}$ the topology transported by T_j from $(\mathcal{H}_{\le 1}, \tau) \times (\mathcal{H}_{\le 1}, \gamma)^j$. The residual factors are explicitly given the proved Hausdorff growth topology; the source factor keeps its original topology. For $J_j f = (f, Df, \dots, D^j f)$, $T_j(J_j N_{\mathrm{ar}}) = N_{\mathrm{ar}} \oplus 0^j$, so $J_j N_{\mathrm{ar}}$ is closed. Define

$$X_j = V_j / J_j N_{\mathrm{ar}}. \tag{25}$$

Then the map

$$[f_0, \dots, f_j] \longmapsto ([f_0], f_1 - Df_0, \dots, f_j - D^j f_0)$$

is a homeomorphism $X_j \simeq Q_{\mathrm{ar}} \oplus (\mathcal{H}_{\le 1}, \gamma)^j$; independence, the inverse, and continuity follow directly from T_j and the quotient definition. In particular there is the split exact sequence

$$0 \longrightarrow \mathcal{H}_{\le 1}^j \longrightarrow X_j \xrightarrow{[f_0, \dots, f_j] \mapsto [f_0]} Q_{\mathrm{ar}} \longrightarrow 0, \qquad [f] \longmapsto [J_j f]. \tag{26}$$

Its relation on a source generator retains every product term:

$$D^k(R\Xi) = \sum_{h=0}^{k} \binom{k}{h} R^{(h)} \Xi^{(k-h)} \quad (0 \le k \le j). \tag{27}$$

The binomial formula follows by induction from the ordinary product rule, including the two edge terms at each step.

The original spectral coordinate acts on V_j by

$$(\mathcal{M}_j f)_k = s f_k + k f_{k-1} \quad (0 \le k \le j), \qquad f_{-1} = 0.$$

The identity $D^k(sf) = sD^k f + kD^{k-1} f$ proves $\mathcal{M}_j J_j f = J_j(sf)$; it therefore preserves the exact relation subspace. In the T_j coordinates it acts by S on Q_{ar} and by $g_k \mapsto s g_k + k g_{k-1}$ on the residual terms ($g_0 = 0$). This also proves continuity in the stated topology. Notice that (26) adds a free residual module: the spectrum of the entire extension is not claimed to equal the source spectrum. The arithmetic quotient and its split inclusion are explicitly preserved.

5 A fully computed analytic realization and its exact comparison

We now compute the closures and the failure of descent in specified topologies. This does not identify them with τ. Let κ be the topology of uniform convergence on compact sets on the same set $\mathcal{H}_{\le 1}$, and define

$$N_\gamma = \overline{\mathbb{C}[s]\Xi}^{\gamma}, \qquad N_\kappa = \overline{\mathbb{C}[s]\Xi}^{\kappa}.$$

Evaluations of all derivatives are continuous in both topologies by Cauchy's formula. The growth topology is finer than κ, so $N_\gamma \subseteq N_\kappa$. If λ is a zero of Ξ with its actual multiplicity m_λ, every element of either closed subspace has vanishing derivatives of orders $0, \dots, m_\lambda - 1$ at λ.

Lemma 5.1 (Entire division with its growth and topology retained)**.** *Let $g \in \mathcal{H}_{\le 1}$ with $g(0) \neq 0$. If $f \in \mathcal{H}_{\le 1}$ and $h = f/g$ extends to an entire function, then $h \in \mathcal{H}_{\le 1}$. For $1 < p < 2$, $a > 0$, put $b = a/(6 \cdot 2^p)$. There is the linear division estimate*

$$\|f/g\|_{p,a} \le \left(\frac{\max(1, \|g\|_{p,b})}{|g(0)|}\right)^3 \|f\|_{p,b}. \tag{28}$$

Multiplication by g is a topological linear isomorphism from $(\mathcal{H}_{\le 1}, \gamma)$ onto the closed subspace of functions divisible by g, with its subspace topology. Compact-topology division is also continuous.

Proof. Choose $R > 0$ with no zero of g on $|s| = R$. Factor its finitely many zeros in the disk, retaining multiplicities. The residual is holomorphic and zero-free there, so its log modulus is harmonic. For $|z| < R$ the boundary mean of $\log |Re^{i\theta} - z|$ equals $\log R$: expand $\log(1-(z/R)e^{-i\theta})$ in its uniformly convergent series and take real parts. The mean-value formula for the residual gives the exact Jensen identity

$$\frac{1}{2\pi}\int_0^{2\pi} \log |g(Re^{i\theta})|\, d\theta = \log |g(0)| + \sum_{|z|<R} m_z \log(R/|z|) \ge \log |g(0)|.$$

There is no zero at zero. With $\log^+ x = \max(0, \log x)$ and $\log^- x = \max(0, -\log x)$, the inequality $\log^+ |h| \le \log^+ |f| + \log^- |g|$ gives

$$\frac{1}{2\pi}\int_0^{2\pi} \log^+ |h(Re^{i\theta})|\, d\theta \le \log^+ M_f(R) + \log^+ M_g(R) - \log |g(0)|. \tag{29}$$

The right side is nonnegative since $M_g(R) \ge |g(0)|$. The harmonic Poisson extension of the continuous nonnegative boundary function $\log^+ |h|$ majorizes $\log |h|$ by the maximum principle and is nonnegative; hence it majorizes $\log^+ |h|$ in the disk. For $|s| \le r < R$ the Poisson kernel is at most $(R + r)/(R - r)$. Take $R \downarrow 2r$ through circles avoiding zeros. Continuity of the two maximum functions gives, for every $r > 0$,

$$\log^+ M_h(r) \le 3\big(\log^+ M_f(2r) + \log^+ M_g(2r) - \log |g(0)|\big). \tag{30}$$

No pointwise lower bound for g on a large circle has been assumed. For $f \neq 0$, put $F = \|f\|_{p,b} > 0$ and apply this to $f/F, h/F$. Then $\log^+ M_{f/F}(2r) \le b(2r)^p$, and $\log^+ M_g(2r) \le \log \max(1, \|g\|_{p,b}) + b(2r)^p$. The coefficient of r^p is $6b2^p = a$, giving (28); continuity fills in $r = 0$. The zero numerator is immediate. All p, a are allowed, so h belongs to the original order class and division is continuous. Multiplication is continuous by Lemma 2.2. The requisite vanishing jets at every zero are closed conditions; local Taylor division identifies them with entire divisibility. The estimate proves this closed set is exactly $g\mathcal{H}_{\le 1}$. For the compact topology choose $R > r$ with no boundary zero and $\delta = \min_{|s|=R} |g(s)| > 0$. The maximum principle for the filled entire quotient gives $M_h(r) \le \delta^{-1} M_f(R)$, proving compact continuity too. □

For general nonzero $g \in \mathcal{H}_{\le 1}$, choose and retain a point c with $g(c) \neq 0$ and use disks centered at c. The functions and spectral coordinate remain unchanged. The bound $|s + c|^p \le 2^{p-1}(|s|^p + |c|^p)$ proves continuity of translations by c and $-c$ in γ; compact containment proves it in κ. Thus all conclusions hold with the stated translated estimates. The original center $c = 0$ already applies to Ξ.

Proposition 5.2 (The computed closures are the same subspace)**.** *With their definitions unchanged,*

$$\begin{aligned} N_\gamma = N_\kappa &= \Xi\mathcal{H}_{\le 1} \\ &= \{f \in \mathcal{H}_{\le 1} : D^h f(\lambda) = 0 \text{ for every } \Xi(\lambda) = 0,\ 0 \le h < m_\lambda\}. \end{aligned} \tag{31}$$

This is equality of subsets of the original ring, not an identification of γ, κ, or τ.

Proof. Jet continuity puts both closures inside the displayed vanishing set and gives $N_\gamma \subseteq N_\kappa$. Local Taylor division makes $h = f/\Xi$ entire for any f in that set; the division lemma then puts h in $\mathcal{H}_{\le 1}$. Its Taylor polynomials p_n converge in γ, and multiplication gives $p_n\Xi \to f$ there. Thus the vanishing set is contained in N_γ, proving all equalities. The algebraic-ideal equality is the conclusion of this argument; it was not substituted for an unevaluated closure. □

Theorem 5.3 (Compact closure, multiplicities, and coordinate spectrum)**.** *The compact closure is exactly*

$$\begin{aligned} N_\kappa &= \{f \in \mathcal{H}_{\le 1} : f/\Xi \text{ extends to an entire function}\} \\ &= \{f \in \mathcal{H}_{\le 1} : D^h f(\lambda) = 0\ (h < m_\lambda),\ \Xi(\lambda) = 0\}. \end{aligned} \tag{32}$$

On $Q_\kappa = (\mathcal{H}_{\le 1}, \kappa)/N_\kappa$, multiplication by s has spectrum exactly the zeros of Ξ in the original s coordinate. For a zero λ of multiplicity m, there is a continuous surjection

$$Q_\kappa \longrightarrow \mathbb{C}[\epsilon]/(\epsilon^m), \qquad [f] \longmapsto \sum_{h=0}^{m-1} \frac{D^h f(\lambda)}{h!}\epsilon^h,$$

intertwining multiplication by s with $\lambda + \epsilon$.

Proof. The vanishing conditions are closed and hold on every $p\Xi$, so the closure is contained in the last displayed set. Local Taylor division at each isolated zero identifies that set with entire divisibility by Ξ. For such an f, put $g = f/\Xi \in \mathcal{O}(\mathbb{C})$. Its Taylor polynomials p_n converge uniformly on every compact; $p_n\Xi \to f$ in κ. Every $p_n\Xi$ belongs to the original order class even though no growth assertion about g is needed. This proves (32) with the closure retained. The local-jet map is independent of representatives; the polynomials $1, (s-\lambda), \dots, (s-\lambda)^{m-1}$ prove surjectivity. Leibniz's rule gives the stated action, and the finite jet evaluations are continuous.

If $\Xi(\lambda) \ne 0$, define

$$T_\lambda f(s) = \frac{f(s) - f(\lambda)\Xi(s)/\Xi(\lambda)}{s-\lambda},$$

filling in the removable value. Division by a linear factor preserves the order bound: outside a disk about λ the denominator is bounded below, and inside a disk the filled entire function is bounded. It is continuous in κ by Cauchy's formula on a larger disk applied to the numerator. For $n \in N_\kappa$ the quotient $T_\lambda n/\Xi = (n/\Xi - (n/\Xi)(\lambda))/(s-\lambda)$ is entire; thus it descends. Direct multiplication proves $(S_\kappa - \lambda)[T_\lambda f] = [f]$. Applying T_λ to $(s-\lambda)f$ gives f exactly, proving the other inverse identity. If $\Xi(\lambda) = 0$, evaluation at λ descends to a nonzero continuous functional which annihilates the image of $S_\kappa - \lambda$. Hence that operator is not surjective. This proves the spectrum claim. □

The same coordinate spectrum and the same finite-jet surjections hold for $Q_\gamma = (\mathcal{H}_{\le 1}, \gamma)/\Xi\mathcal{H}_{\le 1}$. Indeed the numerator in $T_\lambda f$ depends continuously on f in γ. Division by $s - \lambda$, on its entire-divisibility domain, is continuous by Lemma 5.1 with a nonzero center retained when needed. The already proved two inverse identities therefore give a continuous resolvent in γ as well. The evaluation obstruction at each zero is continuous in that topology too. This proves the claim, without transferring a resolvent through an unproved topology identification.

The ambient topologies κ and γ are in fact distinct. The original functions $f_n(s) = \exp(ns - n^2)$ tend to zero uniformly on each compact set. On the positive real axis the maximum of $nx - x^{3/2}$ is $4n^3/27$, attained at $x = 4n^2/9$. Since $\operatorname{Re} s \le |s|$, that axis attains the full weighted supremum, giving the exact value

$$\|f_n\|_{3/2,1} = \exp(4n^3/27 - n^2) \longrightarrow \infty.$$

Thus equality of the closed arithmetic relation subspaces and of their spectral sets does not identify the ambient topologies. The identity induces a continuous linear bijection $Q_\gamma \to Q_\kappa$; the next theorem proves that its inverse is not continuous.

5.1 The actual difference between the two analytic quotient topologies

Retain the original entire-function set $\mathcal{H}_{\le 1} = \mathcal{H}_{\le 1}$, the original coordinate s, and $\Xi(s) = \xi(\frac{1}{2} + is)$ from (1). Let Λ be the set of its distinct zeros and let $m_\lambda = \operatorname{ord}_\lambda \Xi \ge 1$ be each actual multiplicity. Put

$$I = \Xi\mathcal{H}_{\le 1}, \qquad Q_\kappa = (\mathcal{H}_{\le 1}, \kappa)/I, \qquad Q_\gamma = (\mathcal{H}_{\le 1}, \gamma)/I, \qquad \mathcal{P}_\Xi = \prod_{\lambda\in\Lambda} \mathbb{C}^{m_\lambda}.$$

The product has its ordinary product topology. Coordinates in $\mathbb{C}^{m_\lambda}$ are the actual derivatives of orders $0, \dots, m_\lambda - 1$, with no factorial rescaling. Define

$$J_\Xi f = (f^{(j)}(\lambda))_{\lambda\in\Lambda,\ 0\le j<m_\lambda}, \qquad \mathcal{V}_\Xi = J_\Xi(\mathcal{H}_{\le 1}) \subset \mathcal{P}_\Xi. \tag{33}$$

Every point of $\mathcal{V}_\Xi$ retains the entire input's arithmetic jets. The restriction to this image is essential.

Theorem 5.4 (Jet topology and its completion)**.** *The map induced by (33) is a topological linear isomorphism*

$$\overline{J}_\Xi : Q_\kappa \xrightarrow{\cong} \mathcal{V}_\Xi$$

onto the subspace topology from $\mathcal{P}_\Xi$. The image is dense and proper. Consequently Q_κ is incomplete, and its Hausdorff completion is canonically $\mathcal{P}_\Xi$ via this map. In contrast Q_γ is a Fréchet space and is complete. The identity on the same underlying quotient vector space induces a continuous linear bijection

$$\iota : Q_\gamma \longrightarrow Q_\kappa \tag{34}$$

whose inverse is not continuous.

Proof. We first construct each finite interpolation map with its constants. For a finite nonempty set $F \subset \Lambda$, write $d_F = \sum_{\lambda\in F} m_\lambda$ and set

$$B_{\lambda,F}(s) = \prod_{\substack{\nu\in F\\ \nu\ne\lambda}} (s-\nu)^{m_\nu}.$$

The empty product is 1. Since $B_{\lambda,F}(\lambda) \ne 0$, the germ $(s-\lambda)^j/(j!B_{\lambda,F}(s))$ is holomorphic at λ. Denote its Taylor polynomial through degree $m_\lambda - 1$, in powers of the unchanged difference $s - \lambda$, by $T_{\lambda,m_\lambda-1}$ of that germ. Put

$$L_{\lambda,j,F}(s) = B_{\lambda,F}(s) T_{\lambda,m_\lambda-1}\left(\frac{(s-\lambda)^j}{j!B_{\lambda,F}(s)}\right), \qquad 0 \le j < m_\lambda. \tag{35}$$

This is a polynomial of degree at most $d_F - 1$. For $\nu \in F \setminus \{\lambda\}$ it vanishes to order at least m_ν at ν. At λ its difference from $(s-\lambda)^j/j!$ vanishes to order at least m_λ. Therefore, for all $\nu \in F$ and $0 \le k < m_\nu$,

$$L^{(k)}_{\lambda,j,F}(\nu) = \begin{cases} 1, & (\nu,k) = (\lambda,j), \\ 0, & \text{otherwise.} \end{cases}$$

It follows that the linear map to polynomials

$$H_F(v)(s) = \sum_{\lambda \in F} \sum_{j=0}^{m_\lambda - 1} v_{\lambda,j} L_{\lambda,j,F}(s) \tag{36}$$

realizes every prescribed finite array of actual derivatives $v = (v_{\lambda,j})$. For each $R > 0$ define

$$M_f(R) = \max_{|s| \le R} |f(s)|, \qquad C_{R,F} = \sum_{\lambda \in F} \sum_{j=0}^{m_\lambda - 1} M_{L_{\lambda,j,F}}(R).$$

The triangle inequality gives the exact finite bound

$$M_{H_F(v)}(R) \le C_{R,F} \max_{\lambda \in F,\, j < m_\lambda} |v_{\lambda,j}|. \tag{37}$$

When F is empty set $H_F = 0$, $C_{R,F} = 0$ and the maximum over that empty set to 0.

The kernel of J_Ξ is precisely I. Indeed $J_\Xi f = 0$ means that f/Ξ has a removable singularity at each zero of Ξ by its local Taylor expansions. Away from those zeros it is already holomorphic. Thus it extends to an entire function, which belongs to the original order class $\mathcal{H}_{\le 1}$ by Lemma 5.1; consequently $f \in \Xi\mathcal{H}_{\le 1}$. The converse follows by the product rule at each zero. This proves that $\overline{J}_\Xi$ is a well-defined linear bijection onto $\mathcal{V}_\Xi$, with inverse sending the jet array of any $f \in \mathcal{H}_{\le 1}$ to its class $[f]$. The kernel calculation proves that this inverse is independent of the choice of such f.

The quotient topology on Q_κ is generated by

$$q_R([f]) = \inf_{h \in \mathcal{H}_{\le 1}} M_{f - \Xi h}(R), \qquad R > 0. \tag{38}$$

For completeness, these are seminorms because they are infima of the representatives' seminorms modulo a linear subspace. The identity $\{q_R < \epsilon\} = \pi_\kappa(\{M_f(R) < \epsilon\})$ holds for every $\epsilon > 0$, and the quotient map π_κ is open since the inverse image of the image of an open set U is the union $U + I$ of its translates. The increasing disk seminorms generate the compact-open topology; the displayed identity therefore proves the assertion about the quotient topology. Integer $R \ge 1$ suffice.

Fix $R > 0$. Choose $R' > R$ such that the circle $|s| = R'$ contains no zero of Ξ, and put $F = \{\lambda \in \Lambda : |\lambda| < R'\}$, a finite set. For $f \in \mathcal{H}_{\le 1}$ form $H = H_F((f^{(j)}(\lambda))_{\lambda \in F, j < m_\lambda})$. The function $(f - H)/\Xi$ is holomorphic throughout $|s| < R'$ by the exact finite interpolation just proved. Its Taylor polynomials $p_n(s)$ at $s = 0$ consequently converge uniformly on $|s| \le R$ to $(f - H)/\Xi$. In particular $p_n \in \mathcal{H}_{\le 1}$, and

$$M_{f - \Xi p_n}(R) \le M_H(R) + M_\Xi(R) M_{(f-H)/\Xi - p_n}(R).$$

Taking the infimum over admissible representatives and then the limit gives

$$q_R([f]) \le M_H(R) \le C_{R,F} \max_{\lambda \in F,\, j < m_\lambda} |f^{(j)}(\lambda)|. \tag{39}$$

No entire extension or growth bound for $(f-H)/\Xi$ outside this disk was used. In particular there is no additional division assumption hidden in the approximation. If F is empty, this proof gives $q_R = 0$, which is compatible with the other quotient seminorms separating points.

Conversely, fix a finite set $F \subset \Lambda$, and choose $R > \max_{\lambda \in F} |\lambda|$ and $0 < \delta < R - \max_{\lambda \in F} |\lambda|$. Cauchy's formula on the radius-δ circle about each λ gives, for every $h \in \mathcal{H}_{\leq 1}$,

$$|f^{(j)}(\lambda)| = |(f - \Xi h)^{(j)}(\lambda)| \leq j! \delta^{-j} M_{f-\Xi h}(R), \qquad \lambda \in F, \quad j < m_\lambda.$$

Here $0! = 1$, so this bound includes the zeroth derivative without altering its value. Taking infima gives

$$\max_{\lambda \in F, j < m_\lambda} |f^{(j)}(\lambda)| \leq \left(\max_{\lambda \in F, j < m_\lambda} j! \delta^{-j} \right) q_R([f]). \tag{40}$$

Equations (39) and (40) prove continuity in both directions, with the exact stated topologies.

We next verify that the zero set needed for incompleteness is indeed infinite and has an enumeration escaping every compact set. The source Hadamard product for the original even Ξ, retaining its nonzero constant $\Xi(0)$, implies that finitely many zeros would make Ξ a polynomial. This is impossible: at the original coordinate $s = -i(2n - \frac{1}{2})$, $n \geq 1$, the defining completion gives exactly

$$\Xi(-i(2n - \tfrac{1}{2})) = \xi(2n) = n(2n-1)(n-1)!\, \pi^{-n} \zeta(2n). \tag{41}$$

For integer $n \geq 2$, $\zeta(2n) > 1$ by its positive Dirichlet series, and the last $\lfloor n/2 \rfloor$ factors of $(n-1)!$ are at least $n/2$. Thus the logarithm of the right hand side is at least

$$\log(n(2n-1)) + \lfloor n/2 \rfloor \log(n/2) - n \log \pi,$$

which exceeds $d \log(2n)$ for every fixed d and all sufficiently large n. Polynomial values of degree d on these points are bounded by a constant times $(2n)^d$, a contradiction. Since Ξ is nonzero and entire, its zeros are isolated, each multiplicity is finite and each closed disk contains finitely many distinct zeros. We may therefore enumerate all distinct zeros as $\Lambda = \{\lambda_1, \lambda_2, \ldots\}$ with $|\lambda_n| \to \infty$, retaining each m_{λ_n}. This argument uses neither reality nor simplicity of any zero.

Every projection of $\mathcal{V}_\Xi$ onto finitely many full zero-jet blocks is onto, by (36). Such finite coordinate cylinders form a base of the product topology. Consequently $\mathcal{V}_\Xi$ is dense in $\mathcal{P}_\Xi$. It is proper: define the particular array $v \in \mathcal{P}_\Xi$ by

$$v_{\lambda_n,0} = \exp(|\lambda_n|^2), \qquad v_{\lambda_n,j} = 0 \quad (1 \leq j < m_{\lambda_n}). \tag{42}$$

For each $f \in \mathcal{H}_{\leq 1}$ there is a finite constant $A_f > 0$ with $M_f(r) \leq A_f \exp(r^{3/2})$ for every $r \geq 0$; this is exactly the order-at-most-one bound, extended over a bounded interval by increasing A_f. Were $J_\Xi f = v$, we would have

$$\exp(|\lambda_n|^2) \leq A_f \exp(|\lambda_n|^{3/2}) \quad \text{for all } n,$$

which contradicts $|\lambda_n| \to \infty$.

The incompleteness can be exhibited by an actual sequence of polynomial classes. Set $F_n = \{\lambda_1, \ldots, \lambda_n\}$, $H_n = H_{F_n}(v|_{F_n})$, and $x_n = [H_n] \in Q_\kappa$. For each fixed zero-jet block, $J_\Xi H_n$ agrees with v from some index onward; hence $J_\Xi H_n \to v$ in the product. Equivalently, for each fixed R, the finite set in (39) lies in every sufficiently large F_n, so that inequality gives

$$q_R(x_n - x_k) = 0 \qquad (n, k \text{ sufficiently large}).$$

Thus (x_n) is Cauchy in Q_κ. If it converged to a class $[f]$, continuity of $\overline{J}_\Xi$ and uniqueness of limits in the product would give $J_\Xi f = v$, already disproved. For example the product is metrized by the complete translation-invariant metric

$$d(v,w) = \sum_{n=1}^{\infty} 2^{-n} \min\Big(1, \max_{0 \le j < m_{\lambda_n}} |v_{\lambda_n,j} - w_{\lambda_n,j}|\Big).$$

A Cauchy sequence for this metric is Cauchy in each finite coordinate block, so has a coordinatewise limit; controlling a finite initial sum and the geometric tail proves convergence in the metric. This proves product completeness directly. The isomorphism $\overline{J}_\Xi$, being linear and continuous with a continuous linear inverse, preserves the additive uniformities. Its dense embedding in this complete product therefore gives the claimed Hausdorff completion.

Finally, we prove the completeness assertion for the stronger quotient explicitly. The growth seminorms are

$$\|f\|_{p,a} = \sup_{s \in \mathbb{C}} |f(s)| e^{-a|s|^p}, \qquad 1 < p < 2, \quad a > 0.$$

The countable family $p = 1 + 1/j$, $a = 1/k$ for integers $j \ge 2$ and $k \ge 1$ defines the same topology: for arbitrary $1 < p < 2, a > 0$ choose $p_j < p$, and the finite maximum of $r^{p_j} - ar^p$ controls $\|f\|_{p,a}$ by a constant times $\|f\|_{p_j,1}$. Each countable seminorm belongs to the original family, proving both comparisons. Enumerate this countable family as $(u_\ell)_{\ell \ge 1}$ and put $P_n = \max_{1 \le \ell \le n} u_\ell$. If (f_n) is Cauchy in all these seminorms, it is Cauchy uniformly on each compact set, and has an entire limit f. In the weighted Cauchy inequality for any fixed seminorm, passage to the pointwise limit in the second index yields the same uniform weighted bound for $f_n - f$. Thus f has all the finite growth seminorms, lies in the original $\mathcal{H}_{\le 1}$, and $f_n \to f$ in γ. This proves completeness of $(\mathcal{H}_{\le 1}, \gamma)$.

The ideal I is closed: the proved kernel description is an intersection of kernels of derivative evaluations, and Cauchy's estimate shows that each of these evaluations is continuous for γ. The quotient is therefore Hausdorff. Its topology has the countable increasing seminorm family

$$\widetilde{P}_n([f]) = \inf_{h \in I} P_n(f+h).$$

Let (y_n) be a Cauchy sequence in this quotient. Choose a strictly increasing subsequence of indices n_k so far out that

$$\widetilde{P}_k(y_{n_{k+1}} - y_{n_k}) < 2^{-k-1} \quad (k \ge 1).$$

This is possible by the Cauchy property and by taking n_k past the threshold for the kth seminorm before selecting the next index. Choose a representative $d_k \in \mathcal{H}_{\le 1}$ of each displayed difference with $P_k(d_k) < 2^{-k}$. For any fixed j, the tail $k \ge j$ satisfies $P_j(d_k) \le P_k(d_k) < 2^{-k}$. Consequently the partial sums of $\sum_{k \ge 1} d_k$ are Cauchy in γ and converge in $\mathcal{H}_{\le 1}$. Adding a representative of y_{n_1} gives a function whose quotient class is the limit of (y_{n_k}). The full Cauchy sequence (y_n) then has the same limit: for each quotient neighbourhood choose a smaller additive neighbourhood whose sum with itself is contained in it, and combine the Cauchy tail with a convergent subsequence term. The countable seminorm topology is metrizable, so this proves quotient completeness and the Fréchet assertion.

The identity $(\mathcal{H}_{\le 1}, \gamma) \to (\mathcal{H}_{\le 1}, \kappa)$ is continuous: for every R, p, a, one has $M_f(R) \le e^{aR^p} \|f\|_{p,a}$. It preserves the identical algebraic subspace I, so the quotient universal property proves continuity and bijectivity of (34). If its inverse were continuous, it would be a continuous linear map, hence uniformly continuous for the additive uniformities. The Cauchy sequence (x_n) constructed above would then be Cauchy in the complete Q_γ, would converge there, and by continuity of ι would converge in Q_κ, a contradiction. This proves that its inverse is not continuous. No invariance of metric completeness under arbitrary nonlinear homeomorphisms has been used. $\square$

The original coordinate action remains visible after completion. For $v = J_\Xi f$, the exact product rule gives

$$\big(J_\Xi(sf)\big)_{\lambda,j} = \lambda v_{\lambda,j} + j v_{\lambda,j-1}, \qquad 0 \le j < m_\lambda, \tag{43}$$

where the second term is 0 when $j = 0$. Each output coordinate depends on finitely many input coordinates, so this extends to a continuous linear operator on the entire product completion. The factors j are retained because these coordinates are actual derivatives. The completion adds precisely the nonrealizable jet arrays, including (42); it does not add an entire function of order at most one representing such an array. Neither analytic quotient topology has been identified with the source topology τ.

5.2 Exact derivative domains inside the analytic relation

Let $I = \Xi\mathcal{H}_{\le 1} = N_\gamma = N_\kappa$. For $k \ge 1$ the exact representative domain in the relation space is

$$\begin{aligned} I \cap D^{-k} I = \{\Xi h : h \in \mathcal{H}_{\le 1},\quad & [z^n]\big(A_\lambda(\lambda+z)h(\lambda+z)\big) = 0 \\ & \text{for } \max(0, k-m_\lambda) \le n < k, \text{ at every zero } \lambda\}. \end{aligned} \tag{44}$$

where the original local unit is $A_\lambda(s) = \Xi(s)/(s-\lambda)^{m_\lambda}$. To prove this, write $\Xi h = z^m \sum_{n\ge 0} b_n z^n$ at λ, with $m = m_\lambda$ and $b_n = [z^n](A_\lambda h)$. Its kth derivative is

$$\sum_{m+n\ge k} \frac{(m+n)!}{(m+n-k)!} b_n z^{m+n-k}.$$

Every factorial coefficient is nonzero. Vanishing to order at least m is equivalent to $b_n = 0$ exactly for $\max(0, k-m) \le n < k$, as stated. The derivative has the original order bound, so the vanishing condition is equivalent to membership in I by (31). This proves both directions globally, including all multiplicities and local unit coefficients.

For $k = 1$, this is simply $h(\lambda) = 0$ at every distinct zero, without changing or discarding its original multiplicity in Ξ. The derivative ambiguity has the exact presentation

$$\mathcal{H}_{\le 1}/\{h : h(\lambda) = 0 \text{ for every } \Xi(\lambda) = 0\} \xrightarrow{\cong} (DI + I)/I, \qquad [h] \longmapsto [\Xi' h]. \tag{45}$$

Indeed $D(\Xi h) = \Xi h' + \Xi' h$, and the first term lies in I. At each zero of multiplicity m, Ξ' has exactly order $m-1$, so its product with h is divisible by Ξ exactly when $h(\lambda) = 0$ for every zero. This proves the precise kernel, image, and inverse on the image in (45).

For the heat generator ($k = 2$), at any zero of multiplicity at least two the condition is $h(\lambda) = h'(\lambda) = 0$. At a simple zero it is instead

$$A_\lambda(\lambda)h'(\lambda) + A'_\lambda(\lambda)h(\lambda) = 0. \tag{46}$$

It does not require $h(\lambda) = 0$. These follow from the index range in (44): the simple zero has only the coefficient b_1 constrained. In particular the second-derivative relation domain need not be a multiplication module: at a simple zero, a local Ξh obeying (46) and with $h(\lambda) \ne 0$ has $D^2(s\Xi h)(\lambda) = 2(\Xi h)'(\lambda) \ne 0$. This is a local statement with its explicit witness condition; it does not assume the existence of a global simple zeta zero.

There is also an exact calculation of every finite projection of the ambiguity $W_k = (D^k I + I)/I$. In the factorial jet coordinates of any finite set F of distinct zeros, its image is precisely

$$\prod_{\lambda \in F} \epsilon^{\max(m_\lambda - k, 0)} \mathbb{C}[\epsilon]/(\epsilon^{m_\lambda}). \tag{47}$$

The local dimension is $\min(k, m_\lambda)$. The differentiated series above proves containment. For surjectivity, prescribe the allowed output coefficient y_j of z^j, where $\max(m-k, 0) \le j < m$. Set $b_{j+k-m} = j! y_j/(j+k)!$ and give the other coefficients $b_0, \ldots, b_{k-1}$ arbitrary values, say zero. Finite Taylor division by the nonzero unit A_λ then specifies the first k Taylor coefficients of h at λ. These finitely many specifications can be achieved by one polynomial: put $M_\lambda(s) = \prod_{\nu \in F, \nu \neq \lambda} (s-\nu)^k$; take the Taylor polynomial of degree $k-1$ of the prescribed local polynomial divided by M_λ, and sum the resulting M_λ multiples over F. At its own node each summand agrees with the prescribed polynomial modulo $(s-\lambda)^k$, and at the other nodes it vanishes to that order. This constructs the required $h \in \mathbb{C}[s] \subset \mathcal{H}_{\le 1}$ and proves (47). No claim about arbitrary infinite jet interpolation has replaced this exact finite image.

Theorem 5.5 (Direct differential and heat obstructions)**.** *Neither N_γ nor N_κ is invariant under D or D^2. For either frozen quotient, neither representative formula $[f']$ nor $[-f''/4]$ defines an operator on any nonempty class domain admitting all representatives. For every $t \neq 0$, neither $U_t N_\gamma \subseteq N_\gamma$ nor $U_t N_\kappa \subseteq N_\kappa$ holds. Ordinary multiplication by s preserves neither $U_t N_\gamma$ nor $U_t N_\kappa$.*

Proof. Choose any zero λ of the actual Ξ and let $m \ge 1$ be its multiplicity. The existence of its zeros is part of the classical arithmetic datum in the primary source; no realness or simplicity is used. Write $\Xi = (s-\lambda)^m A$ with $A(\lambda) \neq 0$. Then $\Xi' = (s-\lambda)^{m-1}(mA + (s-\lambda)A')$ does not vanish to order m. Since $\Xi \in N_\gamma \subseteq N_\kappa$, both derivative invariance assertions fail. If $m \ge 2$, Ξ'' has leading coefficient $m(m-1)A(\lambda)$ at order $m-2$, proving the second-derivative failure. If $m = 1$, use the actual polynomial multiple $n = (s-\lambda)\Xi$ instead: $n''(\lambda) = 2\Xi'(\lambda) \neq 0$. This retains the product-rule cross term and covers all multiplicities. The nonempty-domain assertion follows from the explicit fibre computation in (16).

For $a \in \mathbb{C}$, $e^{as}\Xi \in N_\gamma$ by polynomial approximation of e^{as} and continuous multiplication, and hence belongs to N_κ. The conjugation $D(e^{as} f) = e^{as}(D+a)f$ and the absolutely convergent heat and translation series give

$$U_t(e^{as}\Xi)(s) = e^{as - ta^2/4} Z_t(s - ta/2). \tag{48}$$

For completeness, the translation series for f is its Taylor series about s and sums to $f(s - ta/2)$; its local uniform convergence and the derivative estimates justify commutation of $e^{-(ta/2)D}$ with U_t. If $U_t N_\bullet \subseteq N_\bullet$ for either choice, evaluate (48) at λ. The exponential never vanishes, so $Z_t(\lambda - ta/2) = 0$ for every a. When $t \neq 0$ these arguments fill $\mathbb{C}$; this contradicts $Z_t \neq 0$. Finally if M preserved $U_t N_\bullet$, conjugation would imply $(M + (t/2)D) N_\bullet \subseteq N_\bullet$. Since M already preserves $N_\bullet$, this forces $D N_\bullet \subseteq N_\bullet$, contrary to the first part. □

The analytic moving-divisor quotient, which does carry the zeros of the actual Z_t under ordinary s, is

$$I_t = \overline{\mathbb{C}[s] Z_t}^{\kappa} = \{f \in \mathcal{H}_{\le 1} : f/Z_t \text{ is entire}\}, \qquad A_t = (\mathcal{H}_{\le 1}, \kappa)/I_t. \tag{49}$$

The division lemma and polynomial approximation further prove

$$I_t = Z_t \mathcal{H}_{\le 1} = \overline{\mathbb{C}[s] Z_t}^{\gamma} \quad (t \in \mathbb{C}).$$

For a complex t with $Z_t(0) = 0$, use the retained nonzero center c in the division lemma; Z_t is not identically zero. The analytically transported relation has the different exact principal presentation for the proved transported multiplication:

$$U_t(\Xi\mathcal{H}_{\leq 1}) = Z_t \star_t \mathcal{H}_{\leq 1}. \tag{50}$$

Indeed $U_t(\Xi h) = Z_t \star_t U_t h$, and U_t is onto. The map $h \mapsto Z_t \star_t h$ is a continuous linear isomorphism of $(\mathcal{H}_{\leq 1}, \gamma)$ onto this closed relation; its inverse on that relation is $n \mapsto U_t((U_{-t}n)/\Xi)$, continuous by the division and heat lemmas. Thus the two multiplications and their exact relation spaces are connected by a proved isomorphism with a proved inverse. The proof of Theorem 5.3 applies verbatim with the nonzero entire order-at-most-one function Z_t: it uses only local Taylor division and compact polynomial approximation. Consequently $\mathrm{Spec}(M \text{ on } A_t) = \{\lambda : Z_t(\lambda) = 0\}$, with the full local multiplicity jets, and with the explicit inverse $[f] \mapsto [(f - f(\lambda)Z_t/Z_t(\lambda))/(s - \lambda)]$ off that set. Equation (15) is the evolution of this original-coordinate spectrum on its simple-zero locus. This spectrum can move; (11) is the spectrum of the different, exactly specified transported operator B_t.

We now give the promised comparison morphisms with every kernel retained. Put $J = N_{\mathrm{ar}} \cap N_\kappa$ and $C = (\mathcal{H}_{\leq 1}, \tau \vee \kappa)/J$, where $\tau \vee \kappa$ is the topology generated by both topologies. Since each summand is closed in its own topology, J is closed in their join. The identity maps give continuous surjections

$$Q_{\mathrm{ar}} \longleftarrow C \longrightarrow Q_\kappa, \qquad \ker(C \to Q_{\mathrm{ar}}) = N_{\mathrm{ar}}/J, \quad \ker(C \to Q_\kappa) = N_\kappa/J. \tag{51}$$

Each assertion follows by taking the same representative f and testing membership in the respective relation subspace; the quotient universal property proves continuity. An algebraic common quotient is $\mathcal{H}_{\leq 1}/(N_{\mathrm{ar}} + N_\kappa)$ with the quotient topology from the join; its Hausdorff version uses $\overline{N_{\mathrm{ar}} + N_\kappa}^{\tau \vee \kappa}$. The map from C to this further quotient is continuous. Continuity of maps to it from each of the two separately topologized quotients is not asserted. No closedness of the sum is presumed. This describes the precise information lost in that further quotient.

At time t replace the two subspaces by N_t and I_t and the join by $\tau_t \vee \kappa$. The same construction gives continuous surjections from their intersection quotient to the genuine transported arithmetic quotient Q_t and to A_t, with kernels $N_t/(N_t \cap I_t)$ and $I_t/(N_t \cap I_t)$, respectively. The two generator systems are linked by the explicit formula $R(B_t)Z_t$ versus $R(s)Z_t$ and the defects (13)–(14). These are exact morphisms and closure formulas for the actual source space. Determining that either kernel vanishes for τ is not established by the inspected primary loci. Accordingly the source-frozen statements $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ and $D^2 N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ have not been decided solely from that unnamed topology. The unqualified noninvariance results above have been proved in the two explicitly specified analytic realizations.

6 Arithmetic traces and their exact Mellin obstructions

6.1 The full Schwartz trace and arithmetic principal parts

Retain the source domain

$$\mathcal{S}_0^{\mathrm{ev}} = \{\psi \in \mathcal{S}(\mathbb{R}) : \psi(-x) = \psi(x),\ \psi(0) = \widehat{\psi}(0) = 0\}, \qquad \widehat{\psi}(y) = \int_{\mathbb{R}} \psi(x) e^{-2\pi i x y}\, dx.$$

Its full trace map takes values in entire functions:

$$F_\psi(s) = \mathcal{F}_\mu \mathcal{E}\psi(s),$$
$$F_\psi(i(z - \tfrac{1}{2})) = \zeta(z)\mathcal{M}\psi(z), \quad \mathcal{M}\psi(z) = \int_0^\infty \psi(x)x^z \, \frac{dx}{x}. \tag{52}$$

The Mellin integral is holomorphic for $\operatorname{Re} z > -2$, and the factorization holds there with its removable value at $z = 1$. This statement does not impose an unproved order bound on the entire F_ψ for arbitrary Schwartz inputs.

Here is a direct proof with all analytic domains retained. For fixed $x > 0$, periodize ψ by the smooth function

$$y \longmapsto \sum_{k\in\mathbb{Z}} \psi(x(k + y)).$$

Its Fourier coefficients are $x^{-1}\widehat{\psi}(\ell/x)$: integrate on $y \in [0, 1]$, substitute $v = x(k + y)$, and sum the adjacent intervals, justified by Schwartz decay. These coefficients decrease faster than every power of ℓ, so their Fourier series converges with all derivatives and at $y = 0$ gives the Poisson identity. Evenness and the two exact zero conditions yield

$$\sum_{n\geq 1} \psi(nx) = x^{-1} \sum_{n\geq 1} \widehat{\psi}(n/x), \qquad \mathcal{E}\psi(x) = \mathcal{E}\widehat{\psi}(1/x).$$

For $x \geq 1$, a Schwartz bound of arbitrarily high order gives $|\sum \psi(nx)| \leq C_A x^{-A} \sum n^{-A}$ for $A > 1$. The same estimate after termwise differentiation follows by choosing the Schwartz exponent larger than the derivative order plus one. The Poisson identity supplies the corresponding bounds at zero. Thus $\mathcal{E}\psi(e^u)$ decreases faster than $e^{-A|u|}$ for every $A > 0$ at both ends. Its Fourier integral and each derivative in s are dominated locally uniformly in s by an integrable function. This proves that F_ψ is entire.

Since ψ is even and $\psi(0) = 0$, its Taylor formula gives $\psi(x) = O(x^2)$ at zero. Together with Schwartz decay at infinity this proves the stated holomorphy of $\mathcal{M}\psi$ by dominated differentiation, including all powers of $\log x$. For $\sigma = \operatorname{Re} z > 1$,

$$\sum_{n\geq 1} \int_0^\infty |\psi(nx)|x^\sigma \, \frac{dx}{x} = \Big(\sum_{n\geq 1} n^{-\sigma}\Big) \int_0^\infty |\psi(y)|y^\sigma \, \frac{dy}{y} < \infty.$$

Absolute interchange and the original exponent x^{-is} now give (52) on this half-plane. Moreover $\mathcal{M}\psi(1) = \int_0^\infty \psi(x)\, dx = \widehat{\psi}(0)/2 = 0$. This cancels the simple pole of ζ at 1, so the right side is holomorphic for $\operatorname{Re} z > -2$. The identity theorem extends the equality to that entire half-plane, proving the assertion.

Let $\mathcal{Z}$ denote the set of distinct nontrivial zeros ρ of ζ, with their original multiplicities m_ρ. The evenness of Ξ, immediate from the retained cosine integral, gives the exact reflected-coordinate identity

$$s = i(z - \tfrac{1}{2}), \qquad z = \tfrac{1}{2} - is, \qquad \Xi(i(z - \tfrac{1}{2})) = \xi(1 - z) = \xi(z). \tag{53}$$

At each $\rho \in \mathcal{Z}$ the completion factor $z(z - 1)\pi^{-z/2}\Gamma(z/2)/2$ is a nonvanishing holomorphic unit. Hence $\lambda_\rho = i(\rho - \frac{1}{2})$ is a zero of Ξ of exactly multiplicity m_ρ, and these are all its zeros. No reflection of the retained spectral operator s has been performed: (53) is the exact Mellin parameter dictionary for that same operator. The factorization (52) proves

$$D^j F_\psi(\lambda_\rho) = 0 \quad (\rho \in \mathcal{Z},\ 0 \leq j < m_\rho),$$
$$F_\psi \in \mathcal{H}_{\leq 1} \implies F_\psi \in \Xi\mathcal{H}_{\leq 1}. \tag{54}$$

The first assertion follows because $\mathcal{M}\psi$ is holomorphic at every ρ and $s-\lambda_\rho = i(z-\rho)$; the second then follows from the proved division theorem. It is an exact intersection assertion about the full Schwartz trace, without enlarging the order-at-most-one domain or closing it in τ.

For each ρ retain the actual local unit $a_\rho(z) = \zeta(z)/(z-\rho)^{m_\rho}$ and the space

$$\mathcal{P}_\rho^- = \left\{ \sum_{j=1}^{m_\rho} c_{-j}(z-\rho)^{-j} : c_{-j} \in \mathbb{C} \right\}.$$

The principal-part map on the original entire-function set is

$$\mathfrak{P} : \mathcal{H}_{\leq 1} \longrightarrow \prod_{\rho \in \mathcal{Z}} \mathcal{P}_\rho^-, \qquad F \longmapsto \left(\mathrm{PP}_{z=\rho} \frac{F(i(z-\frac{1}{2}))}{\zeta(z)} \right)_\rho . \tag{55}$$

The kernel of this map is exactly $\Xi\mathcal{H}_{\leq 1}$. It induces an injective linear map from the underlying vector space of each Q_γ, Q_κ onto the same specified image. Every finite projection is surjective. The map from Q_κ to its image, with the product subspace topology, is a topological linear isomorphism; the product of all principal-part spaces is its completion. From Q_γ it is a continuous bijection onto that image whose inverse is not continuous.

To prove each claim while retaining the analytic units, write $a_\rho(\rho+w)^{-1} = \sum_{\ell \geq 0} \alpha_{\rho,\ell} w^\ell$. The exact Laurent coefficient at degree $-m_\rho + q$, $0 \leq q < m_\rho$, of the ρ component is

$$c_{-m_\rho+q} = \sum_{j=0}^{q} \frac{i^j}{j!} F^{(j)}(\lambda_\rho)\alpha_{\rho,q-j}. \tag{56}$$

This follows by multiplying the two convergent local Taylor series and the retained factor w^{-m_ρ}. Its finite matrix is triangular with nonzero diagonal entries $i^j/(j!a_\rho(\rho))$. Explicitly, given a prescribed principal part $P_\rho(w)$, take the coefficients b_j through degree $m_\rho - 1$ of $a_\rho(\rho+w)w^{m_\rho}P_\rho(w)$; the required original derivatives are $j!i^{-j}b_j$. Multiplying back by a_ρ^{-1} recovers the prescribed negative powers, since the discarded error is divisible by w^{m_ρ}. This proves a finite-dimensional linear isomorphism at every zero, including its inverse and every factorial and power of i.

The vanishing of all principal parts is therefore equivalent to the vanishing of all the original multiplicity jets of F. Theorem 5.2 identifies that kernel exactly with $\Xi\mathcal{H}_{\leq 1}$. At any finite set of zeros, the required derivatives can be realized by the explicit Hermite lift (36); this proves surjectivity of every finite principal-part projection. Each block map and its inverse are continuous because their finite matrices have fixed complex coefficients. Their product is therefore a homeomorphism of the two product spaces: the inverse image of a cylinder fixing finitely many coordinates is again controlled by those finitely many blocks. Applying this product isomorphism to Theorem 5.4 proves all stated topology, completion, and comparison assertions for $\mathfrak{P}$.

This map also carries the original spectral action exactly. For $P_\rho = \mathfrak{P}_\rho(F)$ one has

$$\mathfrak{P}_\rho(sF) = \mathrm{PP}_{w=0}((\lambda_\rho + iw)P_\rho(w)). \tag{57}$$

Indeed the full Laurent quotient for sF is the quotient for F multiplied by $i(z-\frac{1}{2}) = \lambda_\rho + iw$. Its holomorphic residual contributes no negative power after this multiplication. Thus the finite block is the retained λ_ρ plus the explicitly specified nilpotent operator $P \mapsto \mathrm{PP}(iwP)$, whose nilpotency index is m_ρ: iteration on w^{-m_ρ} gives a nonzero multiple of w^{-1} at step $m_\rho - 1$ and zero at step m_ρ.

For $I = \Xi\mathcal{H}_{\le 1}$, the derivative ambiguity $W_k = (D^k I + I)/I$ has, under $\mathfrak{P}$, the exact finite image

$$\prod_{\rho\in F}\left\{\sum_{j=1}^{\min(k,m_\rho)} c_{-j}(z-\rho)^{-j} : c_{-j}\in\mathbb{C}\right\} \quad (F\subset\mathcal{Z}\text{ finite}). \tag{58}$$

The finite image in (47) consists exactly of numerator jets divisible by $w^{\max(m_\rho-k,0)}$ after the invertible substitution $s-\lambda_\rho = iw$. Multiplication by the nonvanishing unit a_ρ^{-1} preserves this divisibility and is invertible on the truncated Taylor ring. Division by w^{m_ρ} therefore yields exactly all negative powers of order at most $\min(k, m_\rho)$, proving both containment and surjectivity in the displayed finite image. In particular the complete local-unit expression, rather than only a count of vanishing orders, transports the derivative defect to the arithmetic Mellin poles.

6.2 The exact Schwartz–Mellin image

Put $\theta = x\partial_x$. For real $a\le b$, retain the finite polynomial

$$P_{a,b}(z) = \prod_{\substack{k\ge 1\\ a-1\le -2k\le b+1}} (z+2k),$$

with empty product equal to 1. Define $\mathfrak{M}_0^{\mathrm{ev}}$ to be the vector space of meromorphic functions M on $\mathbb{C}$ whose only possible poles are simple poles at $-2k$, $k\ge 1$, which satisfy $M(1)=0$, and for which all the seminorms

$$q_{a,b,N}(M) = \sup_{\substack{a\le\sigma\le b\\ v\in\mathbb{R}}} (1+|v|)^N \big|P_{a,b}(\sigma+iv)M(\sigma+iv)\big|, \qquad N=0,1,2,\ldots, \tag{59}$$

are finite. At a pole of M, the product in this formula has its removable value. The margin 1 in $P_{a,b}$ is retained in this definition. It ensures that all poles in a neighborhood of the closed strip have been removed. These seminorm conditions are equivalent to rapid decrease of M on every finite vertical strip for $|\operatorname{Im} z|\ge 1$: on that part of the strip each factor $|z+2k|$ is bounded above by a constant times $1+|v|$ and is at least $|v|$; on the remaining compact rectangle, $P_{a,b}M$ is holomorphic in a neighborhood and is bounded. In particular no bound uniform in the pole index k has been inserted into the definition. One can equivalently use the residue coordinates together with the explicit away-from-poles strip seminorms

$$r_{a,b,N,\delta}(M) = \sup_{\substack{a\le \operatorname{Re} z\le b\\ \operatorname{dist}(z,\{-2,-4,\ldots\})\ge\delta}} (1+|\operatorname{Im} z|)^N |M(z)|, \qquad 0<\delta\le \tfrac12.$$

Here is the equivalence as a statement about topologies, not only about membership. On the domain of this last supremum every factor of $P_{a,b}$ has modulus at least δ, so $r_{a,b,N,\delta}\le \delta^{-\deg P_{a,b}} q_{a,b,N}$. Residues are controlled by q as shown explicitly below. Conversely, outside the radius-δ disks about the poles, $|P_{a,b}(z)|\le C_{a,b}(1+|\operatorname{Im} z|)^{\deg P_{a,b}}$ on the strip, so the corresponding part of q is bounded by an r seminorm with this degree added to its exponent. For a pole whose radius-δ disk meets the strip, that pole is among the factors of $P_{a,b}$. The function $P_{a,b}M$ is holomorphic throughout the disk; its boundary has distance at least δ from every pole, since distinct pole locations are separated by 2. That boundary lies in the enlarged strip $a-1\le \operatorname{Re} z\le b+1$. The maximum-modulus principle bounds the product inside the disk by its boundary values, and $(1+|\operatorname{Im} z|)^N \le (1+\delta)^N$ there. An r seminorm on that enlarged strip therefore bounds the remaining part of q. There are finitely many relevant disks. These estimates prove equality of the two locally convex topologies; separate residue coordinates may be included without changing them.

The map

$$\mathcal{M} : \mathcal{S}_0^{\mathrm{ev}} \longrightarrow \mathfrak{M}_0^{\mathrm{ev}}, \qquad (\mathcal{M}\psi)(z) = \int_0^\infty \psi(x) x^z \,\frac{dx}{x} \quad (\operatorname{Re} z > -2), \tag{60}$$

is a linear bijection. Its inverse, for every fixed real $c > 1$, is

$$\psi(x) = \frac{1}{2\pi} \int_{\mathbb{R}} M(c+iv) x^{-c-iv}\, dv \quad (x > 0), \qquad \psi(0) = 0, \qquad \psi(-x) = \psi(x). \tag{61}$$

For every $k \geq 1$ its precise residue is

$$\operatorname*{Res}_{z=-2k} M(z) = \frac{\psi^{(2k)}(0)}{(2k)!}. \tag{62}$$

Both maps are continuous when the source has its ordinary Schwartz seminorms and the target has (59).

Here is a proof including the strip and reconstruction estimates. Fix a smooth function χ on $[0, \infty)$ which equals 1 on $[0, 1]$ and 0 on $[2, \infty)$. Such a function is obtained, for example, by setting $\eta(y) = 0$ for $y \leq 0$ and $\eta(y) = e^{-1/y}$ for $y > 0$, and taking $\chi(x) = \eta(2-x)/(\eta(2-x) + \eta(x-1))$. Its denominator is positive at every $x \geq 0$. Let $a_k = \psi^{(2k)}(0)/(2k)!$. For an integer $K \geq 1$ write

$$r_K(x) = \psi(x) - \chi(x) \sum_{k=1}^{K} a_k x^{2k}.$$

Evenness and $\psi(0) = 0$ show that r_K vanishes to order at least $2K + 2$ at zero. Taylor's integral remainder, applied also after differentiating, proves

$$|\theta^J r_K(x)| \leq C_{J,K}(\psi) x^{2K+2} \quad (0 < x \leq 1), \qquad J \geq 0,$$

where $C_{J,K}(\psi)$ is bounded by a constant times finitely many Schwartz seminorms of ψ. To see explicitly that all J are covered, expand $\theta^J = \sum_{\ell=0}^{J} S(J,\ell) x^\ell \partial_x^\ell$, where $S(0,0) = 1$ and $S(J+1,\ell) = \ell S(J,\ell) + S(J,\ell-1)$. For $\ell \leq 2K+2$, Taylor's remainder gives

$$|\partial_x^\ell r_K(x)| \leq \frac{x^{2K+2-\ell}}{(2K+2-\ell)!} \sup_{0 \leq y \leq 1} |\psi^{(2K+2)}(y)|.$$

For $\ell > 2K+2$, the differentiated polynomial vanishes on this interval and $x^\ell \leq x^{2K+2}$, which gives the claimed estimate using $\sup_{[0,1]} |\psi^{(\ell)}|$. At infinity $r_K = \psi$, so each $\theta^J r_K$ decreases faster than every inverse power.

The cutoff Mellin function

$$C(w) = \int_0^\infty \chi(x) x^w \,\frac{dx}{x} \quad (\operatorname{Re} w > 0)$$

has the meromorphic continuation

$$C(w) = -\frac{L(w)}{w}, \qquad L(w) = \int_1^2 \chi'(x) x^w \, dx. \tag{63}$$

The entire function L satisfies $L(0) = \chi(2) - \chi(1) = -1$, so the residue of C at 0 is 1. Repeated integration by parts with θ in the expression $L(w) = \int_1^2 (x\chi'(x))x^w \, dx/x$ gives, on every finite vertical strip, rapid decrease for $|\operatorname{Im} w| \geq 1$. Indeed $x\chi'(x)$ is smooth with support in $[1, 2]$ and all boundary derivatives vanish, and for every integer $J \geq 0$

$$w^J L(w) = (-1)^J \int_1^2 \theta^J (x\chi'(x))x^w \, \frac{dx}{x}.$$

Thus C has the same rapid decrease away from its sole pole.

For any strip $a \leq \operatorname{Re} z \leq b$, choose K with $a + 2K + 2 > 0$. On the larger half-plane $\operatorname{Re} z > -2K - 2$, the formula

$$M(z) = \int_0^\infty r_K(x)x^z \, \frac{dx}{x} + \sum_{k=1}^{K} a_k C(z + 2k) \tag{64}$$

is meromorphic and agrees with the original Mellin integral where $\operatorname{Re} z > -2$. The expressions for successive K therefore agree by the identity theorem and define one meromorphic function on $\mathbb{C}$. Formula (63) proves that its only possible poles and residues are exactly (62). For every $J \geq 0$, integration by parts gives

$$z^J \int_0^\infty r_K(x)x^z \, \frac{dx}{x} = (-1)^J \int_0^\infty \theta^J r_K(x)x^z \, \frac{dx}{x}.$$

All boundary terms vanish: at zero they are bounded by a constant times x^{a+2K+2}, and at infinity the Schwartz exponent can be chosen larger than $b + J + 1$. The absolute value of the final integral, uniformly in this strip, is at most

$$B_{K,J;a,b}(\psi) = \int_0^1 |\theta^J r_K(x)|x^{a-1} \, dx + \int_1^\infty |\theta^J r_K(x)|x^{b-1} \, dx < \infty. \tag{65}$$

For $|\operatorname{Im} z| \geq 1$, divide by $|z|^J \geq |\operatorname{Im} z|^J$. Taking J arbitrarily large proves all the required strip estimates, including the polynomial $P_{a,b}$. At bounded imaginary part, multiply (64) by $P_{a,b}$; every pole meeting the strip cancels and the resulting expression is bounded on the compact rectangle. The constants in these bounds involve only finitely many Schwartz seminorms: the first integral in (65) is at most $C_{J,K}(\psi)/(a + 2K + 2)$; on $[1, 2]$ the cutoff has fixed bounded derivatives; on $[2, \infty)$ the expansion of θ^J above uses finitely many bounds $(1 + x)^A |\psi^{(\ell)}(x)|$ with $A > b + J + 1$. This proves continuity of the forward map. Finally $M(1) = \int_0^\infty \psi(x) \, dx = \widehat{\psi}(0)/2 = 0$, retaining the factor 2 from evenness.

Conversely, let $M \in \mathfrak{M}_0^{\mathrm{ev}}$ and define (61). The integral and its ordinary x derivatives converge uniformly on each compact subinterval of $(0, \infty)$. More precisely, with the rising factorial $(z)_0 = 1$ and $(z)_j = z(z + 1) \cdots (z + j - 1)$ for $j \geq 1$, they are

$$\psi^{(j)}(x) = \frac{(-1)^j}{2\pi} \int_{\mathbb{R}} (c + iv)_j M(c + iv)x^{-c-iv-j} \, dv. \tag{66}$$

The factor $(c + iv)_j$ grows at most polynomially in v, so a strip estimate with exponent greater than $j + 1$ supplies an integrable dominating function. This proves smoothness for $x > 0$.

The inverse integral is independent of the positive choice of c. In fact integrate $M(z)x^{-z}$ around the rectangle between two positive vertical lines and heights $\pm T$. It has no poles there. On each horizontal side, its absolute value is at most a constant, depending on the fixed x and the bounded real interval, times $(1+T)^{-N}$ for arbitrary N. Thus the horizontal contributions tend to zero, and the two upward vertical integrals agree. In particular for every real $b>0$ the differentiated formula holds with c replaced by b. If

$$I_{d,j}(M)=\frac{1}{2\pi}\int_{\mathbb{R}}|(d+iv)_j M(d+iv)|\,dv \quad (d\notin\{-2,-4,-6,\dots\}), \tag{67}$$

then $I_{d,j}(M)$ is finite and

$$|\psi^{(j)}(x)|\le I_{b,j}(M)x^{-b-j} \quad (x>0,\ b>0). \tag{68}$$

This gives every required decay estimate at infinity.

For the behavior at zero fix $K\ge 1$ and take the actual left vertical line $d=-2K-1$. The rectangle between d and c has precisely the possible poles $-2,-4,\dots,-2K$. Traverse its right edge upward and its left edge downward; this is the positive orientation. The residue theorem and the same horizontal estimates give

$$\begin{aligned}\psi(x)&=\sum_{k=1}^{K}a_k x^{2k}+R_K(x), &\qquad a_k&=\operatorname*{Res}_{z=-2k}M(z),\\ R_K(x)&=\frac{1}{2\pi}\int_{\mathbb{R}}M(-2K-1+iv)x^{2K+1-iv}\,dv,\\ |R_K^{(j)}(x)|&\le I_{-2K-1,j}(M)x^{2K+1-j}.\end{aligned} \tag{69}$$

The sign of each residue is positive. The last estimate follows by differentiating the remainder integral with the exact factor $(-1)^j(-2K-1+iv)_j$; this factor is included in (67). For any given nonnegative integer j, choose K so that $2K+1>j$. Formula (69) then proves that all derivatives through order j have limits at zero given by the displayed even polynomial. These limits define a C^j extension: inductively, if $\psi^{(\ell)}$ and $\psi^{(\ell+1)}$ have these limits, integrate $\psi^{(\ell+1)}$ from a positive ε to x and let $\varepsilon\downarrow 0$; the resulting integral identity proves differentiability at zero with the asserted derivative. Because j was arbitrary, the extension is smooth, has value zero, all its odd derivatives vanish at zero, and its $2k$th derivative is $(2k)!a_k$. Reflecting it evenly across zero preserves smoothness: on the negative half-line its jth derivative is $(-1)^j\psi^{(j)}(-x)$, and the one-sided values coincide because the odd jets vanish. Together with (68), this proves $\psi\in\mathcal{S}(\mathbb{R})$ and the residue assertion.

For completeness the inversion just used also proves that the Mellin transform of this reconstructed function is the prescribed M. On a line $c>1$ put $h(u)=e^{cu}\psi(e^u)$ and $m(v)=M(c+iv)$. The formula says $h(u)=(2\pi)^{-1}\int m(v)e^{-ivu}\,dv$. Here m is continuous and integrable by the strip estimates, and h is integrable by the bounds at both ends just proved. For $\varepsilon>0$, absolute Fubini and the Gaussian integral give

$$\int_{\mathbb{R}}h(u)e^{ivu}e^{-\varepsilon u^2/2}\,du=\int_{\mathbb{R}}m(w)\frac{e^{-(v-w)^2/(2\varepsilon)}}{\sqrt{2\pi\varepsilon}}\,dw.$$

The right side tends to $m(v)$: split the integral at $|v-w|=\delta$, use continuity on the inner interval, and use the Gaussian tail and boundedness of m on its complement. The left side tends by dominated convergence to $\int h(u)e^{ivu}\,du$. Substitution $x=e^u$ proves $\mathcal{M}\psi(c+iv)=M(c+iv)$. Both sides are holomorphic on $\operatorname{Re}z>-2$, so the identity theorem proves their equality there and hence equality of their meromorphic continuations. In particular $\widehat{\psi}(0)=2\mathcal{M}\psi(1)=2M(1)=0$. Reconstruction is injective because its output has Mellin transform equal to its input. The forward map is injective as well, with the following direct uniqueness proof. If the Mellin transform of $\delta \in \mathcal{S}_0^{\mathrm{ev}}$ vanishes, the continuous integrable function $g(u) = e^{cu}\delta(e^u)$ has Fourier transform $\int g(u)e^{ivu}\,du = 0$. Put $G_\varepsilon(u) = (2\pi\varepsilon)^{-1/2}e^{-u^2/(2\varepsilon)}$. Absolute Fubini applied to the Gaussian integral gives

$$(g * G_\varepsilon)(u) = \frac{1}{2\pi}\int_{\mathbb{R}}\left(\int_{\mathbb{R}} g(w)e^{ivw}\,dw\right)e^{-\varepsilon v^2/2}e^{-ivu}\,dv = 0.$$

As $\varepsilon \downarrow 0$, the Gaussian convolution tends to $g(u)$ at every u by continuity, the unit total mass of G_ε, and its vanishing tail mass. Thus $g = 0$, $\delta = 0$ on the positive half-line, and evenness gives $\delta = 0$ everywhere. This proves that the two displayed maps are mutual inverses, not only that reconstructed inputs have the prescribed transform.

The inverse continuity can be read directly from the estimates. For nonnegative integers A, j, choose $b = A+1$ and $K \geq 1$ with $2K+1 > j$. On $|x| \geq 1$ use (68); on $|x| \leq 1$ use (69). They give

$$\sup_{x\in\mathbb{R}}(1+|x|)^A|\psi^{(j)}(x)| \leq 2^A\left(I_{b,j}(M) + I_{-2K-1,j}(M) + \sum_{\substack{1\leq k\leq K\\ 2k\geq j}}\frac{(2k)!}{(2k-j)!}|a_k|\right). \tag{70}$$

Each line integral on the right is bounded by a constant times a strip seminorm with exponent greater than $j+1$ on a strip containing its line and no pole on the line. Every one of the finitely many residues is controlled by a strip seminorm too: if $-2k$ belongs to $[a,b]$, then

$$|a_k| = \frac{|(P_{a,b}M)(-2k)|}{\left|\displaystyle\prod_{\substack{\ell\geq 1,\ \ell\neq k\\ a-1\leq -2\ell\leq b+1}}(2\ell-2k)\right|}.$$

Its denominator is a finite product of nonzero numbers. This proves continuity without a uniform assumption on the entire residue sequence and completes the characterization.

Apply this result to the exact arithmetic trace map already proved in (52). For $F \in \mathcal{H}_{\leq 1}$ define the meromorphic function

$$M_F(z) = \frac{F(i(z-\frac{1}{2}))}{\zeta(z)}.$$

Its definition uses meromorphic division, including the removable values determined by any canceled zeros or pole. Then the precise intersection of the full source trace image with its original order-at-most-one entire-function set is

$$\begin{aligned}\mathcal{A}_{\mathrm{Sch}} &:= \{F_\psi : \psi \in \mathcal{S}_0^{\mathrm{ev}}\} \cap \mathcal{H}_{\leq 1}\\ &= \{F \in \mathcal{H}_{\leq 1} : M_F \in \mathfrak{M}_0^{\mathrm{ev}}\}\\ &= \left\{\Xi h : h \in \mathcal{H}_{\leq 1},\quad A(z)h(i(z-\tfrac{1}{2})) \in \mathfrak{M}_0^{\mathrm{ev}}\right\},\\ A(z) &= \tfrac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2).\end{aligned} \tag{71}$$

For the first equality after the definition, a trace has the factorization and Mellin properties already proved. Conversely, if $M_F \in \mathfrak{M}_0^{\mathrm{ev}}$, reconstruct the unique $\psi \in \mathcal{S}_0^{\mathrm{ev}}$. In $\operatorname{Re} z > 1$ the trace factorization gives $F_\psi(i(z-\frac{1}{2})) = \zeta(z)M_F(z) = F(i(z-\frac{1}{2}))$. Both F_ψ and F are entire;

their equality on this open set proves equality everywhere. This argument retains the original s coordinate and does not assert that arbitrary Schwartz traces belong to $\mathcal{H}_{\le 1}$.

For the last equality in (71), the previous nontrivial-zero jet calculation and the proved entire division theorem give $F = \Xi h$ with $h \in \mathcal{H}_{\le 1}$ for every trace in this intersection. Conversely the retained identity $\Xi(i(z-\frac{1}{2})) = \xi(1-z) = \xi(z)$ gives exactly

$$M_{\Xi h}(z) = \tfrac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2)h(i(z-\tfrac{1}{2})),$$

including its meromorphic continuation and all its local units. The factor A has only simple poles at $-2k$, $k \ge 1$: the gamma poles at $z=0$ and the factor z cancel, the remaining gamma poles occur at the displayed negative even integers, and the factor $z-1$ gives $A(1)=0$. Thus for an arbitrary $h \in \mathcal{H}_{\le 1}$ the pole locations, orders and the equality $M_{\Xi h}(1) = 0$ are automatic; the exact additional requirement is the family of strip bounds in (59), with the actual factor A retained. No topology on the unnamed source closure is identified by this characterization.

This criterion also retains every Taylor coefficient of its arithmetic preimage. The trivial zero of ζ at $-2k$ is simple, as follows directly from the retained completion. The gamma recurrence gives a simple pole of $\Gamma(z/2)$ there with nonzero residue $2(-1)^k/k!$, while the other factors in $A(z)$ are nonzero there. On the other hand $\xi(-2k) = \xi(1+2k) > 0$: all factors in the completion at $1+2k$ are positive, including the convergent positive Dirichlet series for $\zeta(1+2k)$. Thus $\xi = A\zeta$ forces a zero of exactly order one at $-2k$. Meromorphic division now gives

$$\begin{aligned}\frac{\psi^{(2k)}(0)}{(2k)!} &= \frac{F(i(-2k-\frac{1}{2}))}{\zeta'(-2k)} \\ &= \frac{2k(2k+1)(-1)^k\pi^k}{k!}h(-i(2k+\tfrac{1}{2})) \qquad (F = \Xi h \in \mathcal{A}_{\mathrm{Sch}}).\end{aligned} \tag{72}$$

In the first line there is no extra factor i: the residue is taken in the actual coordinate z and the denominator has leading term $\zeta'(-2k)(z+2k)$. For the second line the gamma recurrence gives $\operatorname{Res}_{z=-2k}\Gamma(z/2) = 2(-1)^k/k!$. Indeed $\Gamma(w) = \Gamma(w+k+1)/[w(w+1)\cdots(w+k)]$, whose numerator is 1 at $w=-k$ and whose denominator has derivative $(-1)^k k!$ there; substitution $w = z/2$ supplies the factor 2. Multiplying this exact residue by $(-2k)(-2k-1)\pi^k/2$ and the value of the entire h gives the asserted coefficient. Thus the inclusion criterion also reconstructs the full even jet at the origin.

The inclusion $\mathcal{A}_{\mathrm{Sch}} \hookrightarrow \Xi\mathcal{H}_{\le 1}$ is the actual identity on entire functions, with exact image specified by (71). It contains $\mathbb{C}[s]\Xi$ because the retained finite Hermite trace families span precisely that polynomial image. Consequently its closures in each of the explicitly proved analytic topologies satisfy

$$\overline{\mathcal{A}_{\mathrm{Sch}}}^{\gamma} = \overline{\mathcal{A}_{\mathrm{Sch}}}^{\kappa} = \Xi\mathcal{H}_{\le 1}. \tag{73}$$

Indeed the left image lies between $\mathbb{C}[s]\Xi$ and the closed subspace $\Xi\mathcal{H}_{\le 1}$, and the closures of the smaller polynomial subspace were proved to equal that subspace. This constructs the exact inclusion, its image criterion and both analytic closures; it inserts no equality involving the unnamed topology τ.

6.3 An exact exponential orbit and its Schwartz boundary

For each unchanged complex parameter $a = u + iv$ define

$$F_a(s) = \Xi(s)e^{as}, \qquad C(z) = \tfrac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2).$$

Every F_a belongs to $\Xi\mathcal{H}_{\le 1} = N_\gamma = N_\kappa$. Its membership in the actual unclosed Schwartz trace image has the exact boundary

$$F_a \in \mathcal{F}_\mu\mathcal{E}(\mathcal{S}_0^{\mathrm{ev}}) \quad \Longleftrightarrow \quad |\operatorname{Re} a| < \pi/4. \tag{74}$$

On this open strip its unique original Schwartz input is

$$\psi_a(x) = e^{-ia/2}(4\pi^2 e^{-4ia}x^4 - 6\pi e^{-2ia}x^2)e^{-\pi e^{-2ia}x^2} \quad (x \in \mathbb{R}). \tag{75}$$

All coefficients and the factor $e^{-ia/2}$ are retained. We prove both directions, including the two boundary lines.

Write $\Theta = x\partial_x$ and $G(x) = e^{-\pi x^2}$. Direct differentiation gives

$$\psi_0 = \Theta(\Theta+1)G = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}.$$

For $\operatorname{Re} z > 0$, substitution $y = \pi x^2$ in the Euler gamma integral gives $\mathcal{M}G(z) = \frac{1}{2}\pi^{-z/2}\Gamma(z/2)$. Integration by parts gives $\mathcal{M}(\Theta f)(z) = -z\mathcal{M}f(z)$ when the boundary terms vanish. They do for G and the needed derivatives on this half-plane. Consequently $\mathcal{M}\psi_0 = C(z)$, with the exact multiplier $(-z)(1-z) = z(z-1)$.

Suppose $|u| < \pi/4$ and put $\beta = e^{-2ia}$. Then $\operatorname{Re}\beta = e^{2v}\cos(2u) > 0$, and its logarithm in the right half-plane is exactly $-2ia$. The Gaussian integral with coefficient β satisfies

$$\int_0^\infty e^{-\pi\beta x^2}x^z\,\frac{dx}{x} = \tfrac{1}{2}\pi^{-z/2}e^{iaz}\Gamma(z/2) \quad (\operatorname{Re} z > 0).$$

Indeed both sides are holomorphic in the connected right half-plane of β, locally dominated there by a positive Gaussian; they agree for positive real β by a real substitution, and the identity theorem proves equality everywhere in that half-plane. Applying $\Theta(\Theta+1)$ and retaining $e^{-ia/2}$ gives

$$\mathcal{M}\psi_a(z) = C(z)e^{ia(z-1/2)}. \tag{76}$$

The function in (75) is even, Schwartz, and zero at the origin, since its Gaussian has positive real coefficient and its polynomial starts at degree two. At $z = 1$, (76) gives $\widehat{\psi}_a(0) = 2\mathcal{M}\psi_a(1) = 0$. Thus it belongs to the exact source domain. The proved trace factorization gives

$$F_{\psi_a}(i(z-\tfrac{1}{2})) = \zeta(z)C(z)e^{ia(z-1/2)} = \Xi(i(z-\tfrac{1}{2}))e^{ai(z-1/2)} \quad (\operatorname{Re} z > 1).$$

Both sides are entire in $s = i(z-\frac{1}{2})$, so they agree everywhere. This proves sufficiency.

We next derive the exact gamma-modulus formula used for necessity. For real y, the absolutely convergent two Euler integrals, followed by $r = x_1 + x_2$, $q = x_1/(x_1+x_2)$, give

$$\begin{aligned}|\Gamma(1+iy)|^2 &= \int_0^\infty\!\!\int_0^\infty e^{-x_1-x_2}(x_1/x_2)^{iy}\,dx_1\,dx_2\\ &= \int_0^1 (q/(1-q))^{iy}\,dq = \int_{\mathbb{R}} e^{iyw}\frac{e^w}{(1+e^w)^2}\,dw.\end{aligned}$$

The middle equality uses $\int_0^\infty re^{-r}\,dr = 1$, and the last substitution is $w = \log(q/(1-q))$. For $y > 0$, integrate $f_y(w) = e^{iyw}e^w/(1+e^w)^2$ on the positively oriented rectangle with vertices $-R, R, R+2\pi i, -R+2\pi i$. The integrals on the vertical sides tend to zero, because the rational exponential factor is $O(e^{-R})$ there, uniformly in the height. The top horizontal integral is $-e^{-2\pi y}$ times the bottom integral. The only pole in the strip is at $w = \pi i$. Writing $w = \pi i + \eta$ gives

$$\frac{e^w}{(1+e^w)^2} = -\frac{1}{4\sinh^2(\eta/2)} = -\eta^{-2} + O(1),$$

so the residue of f_y is $-iye^{-\pi y}$. The residue theorem therefore gives $(1 - e^{-2\pi y}) \int_{\mathbb{R}} f_y(w)\, dw = 2\pi y e^{-\pi y}$. By conjugation for negative y and continuity at zero, we have proved for every real y

$$|\Gamma(1+iy)|^2 = \frac{\pi y}{\sinh(\pi y)}, \tag{77}$$

with value 1 at $y = 0$. This proof uses no asymptotic remainder.

If a Schwartz input had trace F_a, its Mellin transform on $\operatorname{Re} z > 1$ would have to be the quotient $C(z)e^{ia(z-1/2)}$, by the factorization and the nonvanishing of ζ on that half-plane. On the fixed line $z = 2 - 2iy$, its exact squared modulus is

$$\left| C(2-2iy)e^{ia(3/2-2iy)} \right|^2 = e^{-3v+4uy} \, \frac{(1+y^2)(1+4y^2)y}{\pi \sinh(\pi y)}. \tag{78}$$

To check every factor, $|z|^2 = 4(1+y^2)$, $|z-1|^2 = 1+4y^2$, $|\pi^{-z/2}|^2 = \pi^{-2}$, and (77) supplies the remaining gamma factor. The real part of $ia(3/2 - 2iy)$ is $2uy - 3v/2$. At $y = 0$ the right side has its removable value e^{-3v}/π^2.

For $y > 0$, $\sinh(\pi y) < e^{\pi y}/2$. If $u \geq \pi/4$, the right side of (78) is therefore greater than

$$\frac{2e^{-3v}}{\pi}(1+y^2)(1+4y^2)ye^{(4u-\pi)y},$$

which tends to infinity, including $u = \pi/4$. If $u \leq -\pi/4$, put $y = -Y$ and let $Y \to +\infty$; the same inequality with $-u$ proves divergence. But for every Schwartz input the defining integral gives the uniform bound

$$|\mathcal{M}\psi(2-2iy)| \leq \int_0^\infty |\psi(x)| x\, dx < \infty \qquad (y \in \mathbb{R}).$$

This contradiction proves necessity and the exact open boundary.

Uniqueness of the input is also exact. If two source inputs have the same trace, the Mellin transform of their difference vanishes on $\operatorname{Re} z = 2$, by the same factorization. Under $x = e^w$ this is the Fourier transform, with the retained sign change of frequency, of the integrable continuous function $e^{2w}(\psi_1 - \psi_2)(e^w)$. Fourier inversion gives zero on the positive half-line, and evenness gives zero on the negative half-line and at zero. Thus (75) is the unique original input.

On the strip $|\operatorname{Re} a| < \pi/4$ this is a holomorphic family in the Schwartz topology. Indeed on a compact subset of the strip, $\operatorname{Re}(e^{-2ia})$ has a positive lower bound. Every fixed derivative in a and in x is a polynomial in x with locally bounded coefficients times this same Gaussian. It is bounded in each Schwartz seminorm by an integrable Gaussian majorant; the Cauchy integral formula in a, with the same seminorm bounds, proves complex differentiability in that topology. Direct differentiation, with no change of the input coordinate, gives the exact source-generator equation

$$\partial_a \psi_a = -i(\Theta + \tfrac{1}{2})\psi_a, \qquad \mathcal{F}_\mu \mathcal{E}(-i(\Theta + \tfrac{1}{2})\psi_a) = sF_a(s). \tag{79}$$

The second equality also follows from $\mathcal{M}(-i(\Theta + \frac{1}{2})\psi) = i(z - \frac{1}{2})\mathcal{M}\psi$ and the same factorization. This operator is defined on every $\mathcal{S}_0^{\mathrm{ev}}$ input and preserves that space. Multiplication by x and differentiation preserve Schwartz functions, and $x\psi'$ is even when ψ is even. Its value at zero is zero. Finally integration by parts on the whole real line gives $\int x\psi'(x)\,dx = -\int \psi(x)\,dx$, with vanishing boundary terms. Hence $\int -i(\Theta+\frac{1}{2})\psi\,dx = (i/2)\int \psi\,dx = 0$. This supplies the exact morphism from the original scaling generator to multiplication by the retained spectral coordinate. In particular the inclusion of the actual Schwartz trace intersection into $\Xi\mathcal{H}_{\le 1}$ is strict: any real $a = \pi/4$ already gives an element of the latter with no Schwartz preimage. Neither this explicit separating element nor this exact generator map identifies the source closure in the unnamed topology τ.

6.4 An exact arithmetic inverse for every heated Hermite trace

Let ψ be any finite linear combination of the original $\psi_\ell^\pm$, retaining all its coefficients, and write $F_0 = \mathcal{F}_\mu\mathcal{E}\psi = p\Xi$. For $x>0$ and $t\in\mathbb{C}$ define

$$\begin{aligned} q(x) &= \sum_{n\ge 1}\psi(nx), & g_t(x) &= e^{t(\log x)^2/4}\mathcal{E}\psi(x), \\ q_t(x) &= x^{-1/2}g_t(x) = e^{t(\log x)^2/4}q(x), & \psi_t(x) &= \sum_{n\ge 1}\mu(n)q_t(nx). \end{aligned} \tag{80}$$

Here $\mu(1)=1$, $\mu(n)=0$ when a prime square divides n, and $\mu(n)=(-1)^j$ for a product of j distinct primes. For the inverse ψ_t, the symbol $\mathcal{E}\psi_t$ means the absolutely convergent trace-sum extension $x^{1/2}\sum_{m\ge 1}\psi_t(mx)$ on the positive half-line. Its domain here consists of smooth functions with rapid decrease at infinity; no Schwartz extension through zero is required. The transform and inverse assertions, with their domains, are

$$\mathcal{E}\psi_t = g_t, \qquad \mathcal{F}_\mu g_t = U_t(p\Xi) =: F_t, \qquad \int_0^\infty \psi_t(x)x^z\,\frac{dx}{x} = \frac{F_t(i(z-\frac{1}{2}))}{\zeta(z)} \quad (\operatorname{Re} z > 1). \tag{81}$$

The series for ψ_t converges locally with all derivatives on $(0,\infty)$, and ψ_t and all its derivatives decrease faster than every inverse power at infinity. No behavior at zero has been assumed for this inverse.

We prove all analytic assertions in these formulas. A finite Hermite combination and its Fourier transform are polynomials times the original Gaussian $e^{-\pi x^2}$. Thus at infinity, for every derivative order j, their derivatives are bounded by $C_j(1+x)^{d_j}e^{-\pi x^2}$. For $x\ge 1$, termwise derivatives of $\sum\psi(nx)$ are bounded by a sum of $C_j n^j(1+nx)^{d_j}e^{-\pi n^2x^2}$; splitting the Gaussian into two equal exponential factors bounds that sum by $C_j'(1+x)^{d_j}e^{-\pi x^2/2}$. This also justifies the differentiation. Both zero conditions give, by the Poisson identity,

$$\mathcal{E}\psi(x) = \mathcal{E}\widehat{\psi}(1/x).$$

Consequently, in $u=\log x$, the functions $\mathcal{E}\psi(e^u)$ and all their u derivatives have bounds of the form $C\exp(C'|u| - ce^{2|u|})$ on the two ends. Multiplication by $e^{tu^2/4}$, and any finite number of its derivatives, preserves integrability against $e^{A|u|}$ for every $A>0$, locally uniformly for t in compact subsets of $\mathbb{C}$. It follows that $\mathcal{F}_\mu g_t$ is entire in both t,s. Since $D_s^{2k}e^{-isu} = (-1)^k u^{2k}e^{-isu}$, absolute termwise integration of the time exponential gives exactly

$$\int_{\mathbb{R}} e^{tu^2/4}\mathcal{E}\psi(e^u)e^{-isu}\,du = \sum_{k\ge 0}\frac{(-t/4)^k}{k!}D_s^{2k}(p(s)\Xi(s)) = F_t(s).$$

The heat lemma identifies the same series in γ; hence $F_t\in\mathcal{H}_{\le 1}$ with the original coordinate and coefficient.

The proved estimates imply that q_t and all its x derivatives decrease faster than every power at infinity, and that q_t decreases faster than every power at zero. On a compact interval $x \in [\delta, L] \subset (0, \infty)$, choose $A > j+1$ to bound the jth derivative of the nth inverse summand by $Cn^j(nx)^{-A} \le C\delta^{-A}n^{j-A}$. The finitely many arguments below one are harmless. This proves local convergence with derivatives. On $x \ge 1$ the same bound, with arbitrarily large A, proves rapid decrease of each derivative. For a fixed $x > 0$ the double sum used below is absolutely bounded by

$$\sum_{m,n\ge 1} |q_t(mnx)| = \sum_{k\ge 1} d(k)|q_t(kx)| < \infty,$$

where $d(k) \le k$ and rapid decrease permits an exponent greater than two. Unique prime factorization gives $\sum_{n|k} \mu(n) = 1$ for $k = 1$ and $\prod_{p|k}(1-1) = 0$ for $k > 1$. Therefore

$$\sum_{m\ge 1} \psi_t(mx) = \sum_{k\ge 1} \Big(\sum_{n|k} \mu(n)\Big) q_t(kx) = q_t(x).$$

Multiplication by the retained factor $x^{1/2}$ proves the first identity of (81). It also proves uniqueness of this inverse among functions decreasing faster than every power at infinity: for any such v with $\sum_m v(mx) = q_t(x)$, the same absolutely convergent double sum applied to $\sum_n \mu(n) \sum_m v(mnx)$ equals $v(x)$.

For $\sigma = \operatorname{Re} z > 1$, absolute integration gives

$$\sum_{n\ge 1} |\mu(n)| \int_0^\infty |q_t(nx)| x^\sigma \frac{dx}{x} \le \Big(\sum_{n\ge 1} n^{-\sigma}\Big) \int_0^\infty |q_t(y)| y^\sigma \frac{dy}{y} < \infty.$$

The divisor identity also gives $(\sum \mu(n)n^{-z})\zeta(z) = 1$ by absolute multiplication of the two Dirichlet series in this half-plane. Finally

$$\int_0^\infty q_t(x) x^z \frac{dx}{x} = \int_0^\infty g_t(x) x^{z-1/2} \frac{dx}{x} = F_t(i(z - \tfrac{1}{2})),$$

because $-i\, i(z - \frac{1}{2}) = z - \frac{1}{2}$. These equalities prove the last identity of (81), including its absolute domain.

At a nontrivial zero ρ of ζ of multiplicity m, put $\lambda = i(\rho - \frac{1}{2})$ and retain $\zeta(z) = (z-\rho)^m a_\rho(z)$ with $a_\rho(\rho) \ne 0$. The retained cosine integral makes Ξ even, so its definition gives

$$\Xi(i(z - \tfrac{1}{2})) = \xi(1-z) = \xi(z).$$

At a nontrivial zero, the factor $z(z-1)\pi^{-z/2}\Gamma(z/2)/2$ is a nonvanishing holomorphic unit. Thus this reflected coordinate sends the zero at ρ to the zero at λ with exactly the same multiplicity. The meromorphic continuation of the last quotient has pole order

$$\max\{m - \operatorname{ord}_\lambda F_t, 0\}. \tag{82}$$

For the zero seed $F_t = 0$, the quotient is zero and its pole order is zero; the displayed formula uses $\operatorname{ord}_\lambda 0 = +\infty$. Indeed $s - \lambda = i(z - \rho)$, so its numerator has exactly the original order $\operatorname{ord}_\lambda F_t$, with its leading coefficient multiplied by the nonzero power of i. The unit a_ρ is retained in every principal coefficient.

For every fixed $t \ne 0$ and every zero λ of Ξ, there is a polynomial p for which $F_t(\lambda) \ne 0$: otherwise all $U_t(s^n\Xi)$ would vanish there, contrary to the no-common-zero proof in Section 3.

The exact polynomial-image calculation supplies a finite original Hermite seed for this p. For that seed the unique inverse ψ_t in (80) is unbounded near zero. To prove this last assertion, boundedness near zero together with its proved rapid decrease at infinity would make its Mellin integral holomorphic in $\operatorname{Re} z > 0$: on compact subsets, derivatives in z are dominated by powers of $|\log x| x^{\sigma-1}$ near zero with $\sigma > 0$. Its equality in $\operatorname{Re} z > 1$ would then continue as a holomorphic function through every nontrivial ρ, contradicting the pole of order m just proved at $\rho = \frac{1}{2} - i\lambda$. Thus for each nonzero heat time some actual finite Hermite trace has an explicit arithmetic inverse which fails even boundedness at the origin. This proves a concrete obstruction to Schwartz descent with a unique inverse and its poles; it does not replace the unresolved closure in τ by a different closure.

6.5 Global meromorphic arithmetic heat transport

The original coordinate dictionary permits an exact heat action on the entire arithmetic Mellin quotient, including its poles. Define

$$\begin{aligned}(\mathscr{J}F)(z) &= \frac{F(i(z-\frac{1}{2}))}{\zeta(z)}, \qquad\qquad \mathfrak{R}_\zeta = \mathscr{J}(\mathcal{H}_{\le 1}),\\ H_M(z) &= \zeta(z)M(z), \qquad (\mathscr{J}^{-1}M)(s) = \zeta(\tfrac{1}{2}-is)M(\tfrac{1}{2}-is).\end{aligned} \tag{83}$$

All divisions and products in this display are meromorphic, with their removable values retained. Thus $\mathfrak{R}_\zeta$ consists exactly of meromorphic M for which H_M extends to an entire function of order at most one. To verify the order statement in both directions, the affine coordinate maps are $z \mapsto i(z-\frac{1}{2})$ and $s \mapsto \frac{1}{2} - is$. Their moduli are bounded by $|z| + \frac{1}{2}$ and $|s| + \frac{1}{2}$, respectively. The inequality $(r+\frac{1}{2})^p \le 2^{p-1}(r^p + 2^{-p})$ for every $p > 1$ then transfers each of the defining order-class growth bounds in both directions. The displayed maps are inverse because $\frac{1}{2} - i\,i(z-\frac{1}{2}) = z$ and $i(\frac{1}{2} - is - \frac{1}{2}) = s$.

Give $\mathfrak{R}_\zeta$ the transported growth topology $\gamma_\zeta = \mathscr{J}_*\gamma$, namely the seminorms

$$\|M\|_{\zeta;p,a} = \sup_{s\in\mathbb{C}} |\zeta(\tfrac{1}{2}-is)M(\tfrac{1}{2}-is)|e^{-a|s|^p}, \qquad 1 < p < 2, \quad a > 0. \tag{84}$$

The canceled product is used at every apparent singularity. Consequently $\mathscr{J} : (\mathcal{H}_{\le 1}, \gamma) \to (\mathfrak{R}_\zeta, \gamma_\zeta)$ is a linear homeomorphism; completeness and the locally convex structure follow directly from those of $(\mathcal{H}_{\le 1}, \gamma)$. No topology has been assigned to $\mathfrak{R}_\zeta$ merely by referring to unspecified convergence of meromorphic functions.

Every $M \in \mathfrak{R}_\zeta$ is holomorphic away from zeros of ζ. At a zero of multiplicity m its pole order is at most m. At $z = 1$, where ζ has its simple pole, M is holomorphic and vanishes. These assertions follow by dividing the entire H_M by the actual local Laurent expansion of ζ. They concern all complex zeros, including the simple trivial zeros, and do not discard the local units.

Put

$$\nabla_\zeta M = M' + \frac{\zeta'}{\zeta}M = \frac{H_M'}{\zeta}, \qquad \mathcal{L}_\zeta M = \frac{1}{4}\nabla_\zeta^2 M.$$

These are globally defined meromorphic operators on $\mathfrak{R}_\zeta$; the right expression supplies every removable value. With $D = \partial_s$, the exact conjugacies are

$$\begin{aligned}\mathscr{J}D\mathscr{J}^{-1} &= -i\nabla_\zeta,\\ \mathcal{L}_\zeta M &= \frac{1}{4\zeta}(\zeta M)'' = \frac{1}{4}\left(M'' + 2\frac{\zeta'}{\zeta}M' + \frac{\zeta''}{\zeta}M\right),\\ \mathscr{J}(-D^2/4)\mathscr{J}^{-1} &= \mathcal{L}_\zeta.\end{aligned} \tag{85}$$

Indeed, $H_M(z) = F(i(z-\frac{1}{2}))$ gives $H'_M(z) = iF'(i(z-\frac{1}{2}))$ and $H''_M(z) = -F''(i(z-\frac{1}{2}))$. Division by ζ proves the first and last equalities. Expanding $(\zeta M)'' = \zeta M'' + 2\zeta' M' + \zeta'' M$ proves every coefficient in the middle one. In particular the coefficient ζ''/ζ includes both the derivative and the square of ζ'/ζ; neither term has been omitted.

For every original complex time t, define

$$\begin{aligned}\mathcal{V}_t M &= \mathscr{J} U_t \mathscr{J}^{-1} M = \frac{H_{M,t}}{\zeta},\\ H_{M,t}(z) &= \exp\left(\frac{t}{4}\partial_z^2\right) H_M(z) = \sum_{k=0}^{\infty} \frac{t^k}{4^k k!} H_M^{(2k)}(z).\end{aligned} \tag{86}$$

The last series is a definition by the proved entire-function heat calculus, not an operation on unrestricted meromorphic functions. More explicitly, its kth term equals $(-t/4)^k F^{(2k)}(i(z-\frac{1}{2}))/k!$. It therefore has exactly the growth-topology convergence already proved for $U_t F$, after the continuous affine change of argument. In particular the numerator series converges locally uniformly in (t, z); after division by ζ, it converges in every seminorm (84), locally uniformly in complex t. Each $\mathcal{V}_t$ is a continuous automorphism of $(\mathfrak{R}_\zeta, \gamma_\zeta)$, with

$$\mathcal{V}_t \mathcal{V}_v = \mathcal{V}_{t+v}, \qquad \mathcal{V}_t^{-1} = \mathcal{V}_{-t}, \qquad \partial_t \mathcal{V}_t M = \mathcal{L}_\zeta \mathcal{V}_t M.$$

These follow by conjugating the proved heat group and its differentiated series. Equivalently $\mathcal{V}_t = \exp(t\mathcal{L}_\zeta)$ in this precisely specified topology. Near a zero of ζ, local uniform convergence of the numerators also gives convergence of each Laurent coefficient and of the holomorphic remainders after the finite principal parts are removed.

The original multiplication coordinate is the actual function $i(z-\frac{1}{2})$, because $\mathscr{J}(sF) = i(z-\frac{1}{2})\mathscr{J}F$. Thus the transported operator $B_t = s - (t/2)D$ becomes

$$\mathcal{C}_t M := \mathscr{J} B_t \mathscr{J}^{-1} M = i(z-\tfrac{1}{2})M + \frac{it}{2}\nabla_\zeta M = \mathcal{V}_t\big(i(z-\tfrac{1}{2})\big)\mathcal{V}_{-t} M. \tag{87}$$

In the last expression the function in parentheses denotes multiplication. The equality follows by conjugating $B_t = U_t s U_{-t}$ and proves the retained coordinate and sign. It does not replace the original spectral value by z.

This also has an exact relation to the actual source quotient. On the same underlying set $\mathfrak{R}_\zeta$ let $\sigma_t = \mathscr{J}_* \tau_t$ and $\mathfrak{N}_t = \mathscr{J} N_t$. The already proved closedness of N_t in τ_t gives closedness of $\mathfrak{N}_t$ in σ_t. The map

$$[F]_{N_t} \longmapsto [\mathscr{J}F]_{\mathfrak{N}_t}$$

is a homeomorphism of the corresponding quotient spaces, because it and its inverse are induced by the defining homeomorphism $\mathscr{J}$ of the ambient spaces. Equation (87) intertwines the quotient spectral operators. For every scalar λ, it therefore also intertwines $B_t - \lambda$ and $\mathcal{C}_t - \lambda$, their kernels and images, and their continuous inverses whenever those exist. Their spectra are the same original arithmetic s-spectrum. This statement uses σ_t, while the entire-time heat estimates above use γ_ζ; it inserts no equality between those topologies.

6.6 Local pole cancellation with every arithmetic unit retained

Fix an actual nontrivial zero ρ of ζ, of its full multiplicity $m = m_\rho$, and use its unchanged local coordinate $w = z - \rho$. Write

$$\zeta(\rho + w) = w^m a_\rho(\rho + w), \qquad \frac{1}{a_\rho(\rho + w)} = \sum_{j \geq 0} c_j w^j, \qquad H_M(\rho + w) = \sum_{n \geq 0} b_n w^n. \tag{88}$$

Here $a_\rho(\rho) \neq 0$ and $c_0 = 1/a_\rho(\rho)$. All series converge on a sufficiently small disk. The principal coefficients of M and its heat generator are exactly

$$\begin{aligned} [w^{-m+q}]M &= \sum_{n=0}^{q} c_{q-n} b_n, \\ [w^{-m+q}]\mathcal{L}_\zeta M &= \frac{1}{4} \sum_{n=0}^{q} c_{q-n}(n+2)(n+1) b_{n+2}, \quad 0 \le q < m. \end{aligned} \tag{89}$$

In particular neither M nor its generator has a pole of order greater than m, and its potential order-m generator coefficient is $H_M''(\rho)/(4a_\rho(\rho)) = b_2/(2a_\rho(\rho))$. If that coefficient vanishes, the remaining coefficients in the display determine the full lower pole order without suppressing the unit.

There is a direct cancellation calculation for the apparently higher poles in (85). Put $v(w) = H_M(\rho + w)/a_\rho(\rho + w)$, so $M = w^{-m} v$, and temporarily write $a = a_\rho(\rho + w)$ with primes denoting the same original z derivative. Then

$$\begin{aligned} M' &= w^{-m} v' - m w^{-m-1} v, \\ M'' &= w^{-m} v'' - 2m w^{-m-1} v' + m(m+1) w^{-m-2} v, \\ \frac{\zeta'}{\zeta} &= \frac{m}{w} + \frac{a'}{a}, \\ \frac{\zeta''}{\zeta} &= \frac{m(m-1)}{w^2} + \frac{2m}{w}\frac{a'}{a} + \frac{a''}{a}. \end{aligned}$$

In $M'' + 2(\zeta'/\zeta)M' + (\zeta''/\zeta)M$, the coefficient multiplying $w^{-m-2}v$ is $m(m+1) - 2m^2 + m(m-1) = 0$. The coefficients multiplying $w^{-m-1}v'$ are $-2m + 2m = 0$, and those multiplying $w^{-m-1}(a'/a)v$ are $-2m + 2m = 0$. The complete remaining expression is

$$w^{-m}\left(v'' + 2\frac{a'}{a}v' + \frac{a''}{a}v\right) = \frac{w^{-m}}{a}(av)'' = \frac{H_M''}{\zeta}.$$

This proves the cancellation of both possible extra pole orders with every cross term and local unit present. Expanding the final numerator and $1/a$ proves (89).

The same calculation holds at every time. Set

$$b_n(t) = \frac{H_{M,t}^{(n)}(\rho)}{n!}, \qquad \lambda_\rho = i(\rho - \tfrac{1}{2}).$$

The globally convergent numerator heat series gives the exact entire functions of t

$$\begin{aligned} b_n(t) &= \sum_{k=0}^{\infty} \frac{t^k (n+2k)!}{4^k k! n!} b_{n+2k}(0) = \frac{i^n}{n!}(D^n U_t F)(\lambda_\rho), \\ \partial_t b_n(t) &= \frac{(n+2)(n+1)}{4} b_{n+2}(t), \qquad\qquad F = \mathscr{J}^{-1} M. \end{aligned} \tag{90}$$

Differentiation and evaluation are continuous on the entire growth space by the proved derivative estimates. They therefore justify termwise differentiation and evaluation of this series, including local uniform convergence in t. If $\ell_q(t) = [w^{-m+q}]\mathcal{V}_t M$, then

$$\ell_q(t) = \sum_{n=0}^{q} c_{q-n} b_n(t), \qquad \partial_t \ell_q(t) = \frac{1}{4}\sum_{n=0}^{q} c_{q-n}(n+2)(n+1) b_{n+2}(t), \quad 0 \le q < m. \tag{91}$$

Thus the evolution of the arithmetic principal parts is a calculation from the full entire numerator. It is not a closed system on only the finitely many coefficients ℓ_q; the displayed higher numerator jets are part of the exact map.

For an initial M holomorphic at ρ, write $M(\rho+w) = M_0 + M_1 w + O(w^2)$. The formula for the generator then gives the complete possible principal part

$$\mathrm{PP}_\rho(\mathcal{L}_\zeta M) = \frac{m(m-1)M_0}{4w^2} + \frac{m(m+1)M_1 + 2m(a'_\rho/a_\rho)(\rho)M_0}{4w}. \tag{92}$$

For $m = 1$ the order-two term is zero. For $m \geq 2$ this principal part vanishes exactly when $M_0 = M_1 = 0$. For $m = 1$ it vanishes exactly when $(a_\rho M)'(\rho) = 0$. These equivalences follow from the nonzero constants m, $m-1$ in the first case and $a_\rho(\rho) \neq 0$ in both cases. They describe the exact local derivative obstruction, including multiple zeros, rather than assuming their multiplicity is one.

6.7 The calculated domain for heat to retain an original Schwartz input

Let $\mathfrak{M}_0^{\mathrm{ev}}$, $P_{a,b}$, and $q_{a,b,N}$ be the explicitly proved meromorphic Mellin image and strip seminorms in (59). Its intersection

$$\mathfrak{A}_\zeta = \mathfrak{R}_\zeta \cap \mathfrak{M}_0^{\mathrm{ev}} = \mathscr{J}\mathcal{A}_{\mathrm{Sch}}$$

is the exact original Schwartz trace domain in this presentation, by (71). For each unchanged complex time t, its complete domain of retained Schwartz inputs is the following explicitly calculated subspace:

$$\begin{aligned} \mathfrak{D}_t = \Bigg\{ M \in \mathfrak{A}_\zeta : \quad & \sum_{k=0}^{\infty} \frac{t^k (n+2k)!}{4^k k! n!} b^\rho_{n+2k}(0) = 0 \quad (\rho \in \mathcal{Z},\ 0 \leq n < m_\rho), \\ & \sup_{\substack{a \leq \sigma \leq b \\ v \in \mathbb{R}}} (1+|v|)^N \left| P_{a,b}(\sigma+iv) \frac{H_{M,t}(\sigma+iv)}{\zeta(\sigma+iv)} \right| < \infty \\ & \qquad\qquad \text{for every real } a \leq b \text{ and every integer } N \geq 0 \Bigg\}. \end{aligned} \tag{93}$$

Here $\mathcal{Z}$ is the actual set of nontrivial zeros, $b^\rho_n(0) = H_M^{(n)}(\rho)/n!$, and $H_{M,t}$ is the globally convergent series in (86). The removable values in the strip expression are the values forced by the preceding zero-jet equalities and by the factors $P_{a,b}$. No unknown initial input, omitted pole cancellation, or assumed decay estimate occurs in this definition.

The exact characterization and reconstruction are

$$\begin{aligned} \mathfrak{D}_t &= \mathfrak{A}_\zeta \cap \mathcal{V}_{-t}\mathfrak{A}_\zeta, \\ M \in \mathfrak{D}_t \quad &\Longleftrightarrow \quad U_t \mathscr{J}^{-1} M \in \mathcal{A}_{\mathrm{Sch}}, \\ \psi_t(x) &= \frac{1}{2\pi} \int_{\mathbb{R}} \frac{H_{M,t}(c+iv)}{\zeta(c+iv)} x^{-c-iv}\, dv \quad (x > 0,\ c > 1), \\ \psi_t(0) &= 0, \qquad \psi_t(-x) = \psi_t(x). \end{aligned} \tag{94}$$

In the equivalence M is taken in its declared initial domain $\mathfrak{A}_\zeta$; the reconstruction formula is for $M \in \mathfrak{D}_t$. We now prove that the conditions in (93) are both necessary and sufficient, rather than leaving them as proof obligations.

The first family of equalities says exactly $H_{M,t}^{(n)}(\rho) = 0$ for $0 \le n < m_\rho$, by (90). Division by $(z-\rho)^{m_\rho} a_\rho(z)$ therefore shows that it is equivalent to holomorphy of $\mathcal{V}_t M$ at every nontrivial zero. At every trivial zero $-2k$, the zero of ζ is simple, so $H_{M,t}/\zeta$ automatically has at most a simple pole there. At $z = 1$ it automatically is holomorphic and zero, because the numerator is entire and $1/\zeta$ has a simple zero. At all remaining points it is holomorphic. Thus the first family is equivalent to all pole and value conditions in the definition of $\mathfrak{M}_0^{\mathrm{ev}}$. The second family is exactly that definition's complete finite-strip decay requirement, with its original polynomial and margin. This proves

$$M \in \mathfrak{D}_t \quad \Longleftrightarrow \quad M \in \mathfrak{A}_\zeta \text{ and } \mathcal{V}_t M \in \mathfrak{M}_0^{\mathrm{ev}}.$$

Since $\mathcal{V}_t M$ already belongs to $\mathfrak{R}_\zeta$, the last membership is precisely membership in $\mathfrak{A}_\zeta$. The group inverse proves the first identity in (94), and the proved exact trace-image characterization proves the second. The inverse Mellin theorem then proves that the displayed integral converges with the required differentiated estimates, is independent of $c > 1$, gives the unique original even Schwartz input with both zero conditions, and satisfies

$$\mathcal{F}_\mu \mathcal{E} \psi_t = \mathscr{J}^{-1} \mathcal{V}_t M = U_t \mathscr{J}^{-1} M.$$

These conclusions use the already proved inverse theorem with exactly its hypotheses verified above.

All Taylor data at the original input origin are also explicit: for $M \in \mathfrak{D}_t$ and every $k \ge 1$,

$$\frac{\psi_t^{(2k)}(0)}{(2k)!} = \operatorname*{Res}_{z=-2k} \frac{H_{M,t}(z)}{\zeta(z)} = \frac{H_{M,t}(-2k)}{\zeta'(-2k)}. \tag{95}$$

The first equality is the proved Mellin residue theorem for the reconstructed input. The second divides by the actual leading term $\zeta'(-2k)(z+2k)$; in particular it introduces no factor i into the residue coordinate z. The right side is defined as an entire function of t for every initial $M \in \mathfrak{R}_\zeta$, even at times outside $\mathfrak{D}_t$ when it cannot be interpreted as a Schwartz input's Taylor coefficient.

At $t = 0$ the definition gives all of $\mathfrak{A}_\zeta$. For every $t \ne 0$ it is a proper subspace. Indeed fix any nontrivial zero ρ and its retained spectral coordinate $\lambda_\rho = i(\rho - \frac{1}{2})$. The proved absence of a common evaluation zero among $U_t(\mathbb{C}[s]\Xi)$ gives a polynomial p for which $U_t(p\Xi)(\lambda_\rho) \ne 0$. The actual finite Hermite trace theorem places $p\Xi$ in $\mathcal{A}_{\mathrm{Sch}}$. For $M = \mathscr{J}(p\Xi)$ the computed numerator at time t is nonzero at ρ, so its quotient by ζ has a pole of the full order m_ρ there. This violates the first family in (93) and proves $M \notin \mathfrak{D}_t$. The argument is an obstruction in the actual unclosed Schwartz trace map and its calculated meromorphic heat domain; it makes no identification of the unnamed source closure.

6.8 The exact persistent heat core and all possible return times

Retain the original functions, coordinate, and analytic relation $I = \Xi\mathcal{H}_{\le 1} = N_\gamma = N_\kappa$. The obstruction can be computed for an entire orbit, not only for a chosen generator or one time. Define

$$\begin{aligned} I_{\mathrm{even}}^\infty &= \{f \in \mathcal{H}_{\le 1} : \ D^{2k} f \in I \text{ for every integer } k \ge 0\}, \\ I_{\mathrm{heat}}^\infty &= \{f \in \mathcal{H}_{\le 1} : \ U_t f \in I \text{ for every } t \in \mathbb{C}\}. \end{aligned} \tag{96}$$

Then the complete calculation is

$$I_{\mathrm{even}}^\infty = I_{\mathrm{heat}}^\infty = \{0\}. \tag{97}$$

More strongly, for every set $T \subset \mathbb{C}$ with a finite accumulation point,

$$\bigcap_{t\in T} U_{-t} I = \{0\}. \tag{98}$$

We give the complete proof, including the arithmetic obstruction to an exceptional periodic function.

First suppose $f \in I^{\infty}_{\text{even}}$. At every actual zero λ of Ξ of multiplicity m_λ, its Taylor coefficients satisfy

$$D^{2k+j} f(\lambda) = 0 \qquad (k \geq 0,\ 0 \leq j < m_\lambda).$$

If any $m_\lambda \geq 2$, the choices $j = 0, 1$ cover all nonnegative derivative orders. The entire Taylor series at that unchanged λ is zero, so $f = 0$. It remains to prove the result when all the actual zeros are simple; this is the other case of the proof, not a simplicity hypothesis. For each of these zeros, the vanishing of every even derivative and the entire Taylor expansion give the exact reflection identity

$$f(2\lambda - s) = -f(s) \qquad (s \in \mathbb{C}). \tag{99}$$

In particular f is odd about each actual zero without changing the spectral coordinate. Fix one zero λ_0. Composing the reflections at λ_0 and at λ gives

$$f(s + 2(\lambda - \lambda_0)) = f(s) \qquad (s \in \mathbb{C},\ \Xi(\lambda) = 0). \tag{100}$$

We prove the needed period assertion explicitly. Let f be a nonconstant entire function and put

$$P(f) = \{p \in \mathbb{C} :\ f(s+p) = f(s) \text{ for all } s \in \mathbb{C}\}.$$

It is an additive subgroup, closed because pointwise limits of periods remain periods by continuity of f. It has no sequence of nonzero periods tending to zero: such a sequence p_n would give $f'(s) = \lim_n (f(s+p_n) - f(s))/p_n = 0$ at every s. Thus some disk about zero contains no other period. Two periods p, q linearly independent over $\mathbb{R}$ are also impossible. Every complex number has the form $ap + bq$ with real a, b. Subtracting the integer parts of a, b puts it in the compact parallelogram $\{xp + yq : 0 \leq x, y \leq 1\}$ without changing the value of f. The resulting global bound and the Cauchy estimate $|f'(s)| \leq \sup_{\mathbb{C}} |f|/R$ for every $R > 0$ imply that f is constant, a contradiction.

If $P(f)$ contains a nonzero period p, all its periods therefore lie on the real line $\mathbb{R}p$. The set $L = \{a \in \mathbb{R} : ap \in P(f)\}$ is a closed additive subgroup of $\mathbb{R}$, contains 1, and has a positive gap about zero. Let $d = \inf(L \cap (0, \infty))$. This number is positive; a sequence approaching the infimum and closedness show $d \in L$. For any positive $a \in L$, subtract $\lfloor a/d \rfloor d$. The remainder lies in $L \cap [0, d)$ and must be zero by minimality. Negative elements are handled by changing sign. Hence $L = d\mathbb{Z}$ and

$$P(f) = \omega\mathbb{Z}, \qquad \omega = dp \neq 0.$$

This proves the full assertion about the period group used here.

Return to (100). A constant satisfying (99) is zero. If $f \neq 0$, it is therefore nonconstant. The set of distinct zeros of Ξ is infinite, as proved above from its retained product and completion, so it contains a zero different from λ_0. The period group consequently contains a nonzero period and is $\omega\mathbb{Z}$ by the preceding argument. Every actual zero must then lie in the single arithmetic progression

$$\lambda \in \lambda_0 + \tfrac{1}{2}\omega\mathbb{Z}. \tag{101}$$

The number of points in this progression in the disk $|s| \le r$ is at most $1 + 4r/|\omega|$: their integer indices lie in an interval of length at most $4r/|\omega|$, by projecting onto the line of the progression. This bound does not assume that λ_0 or ω is real.

Here is a direct contradiction with the actual completed zeta function, using no zero-counting asymptotic. Retain its exact even Hadamard product from the source,

$$\Xi(s) = \Xi(0) \prod_{\alpha \in \Lambda_+} (1 - s^2/\alpha^2),$$

where one member of each pair $\{\lambda, -\lambda\}$ is retained with its original multiplicity. In the case currently under consideration every multiplicity is one. Set $N_+(u) = \#\{\alpha \in \Lambda_+ : |\alpha| \le u\}$. There is a number $r_* > 0$ such that $N_+(u) = 0$ for $u < r_*$, since $\Xi(0) \ne 0$ and its zeros are discrete. The progression bound implies

$$N_+(u) \le Au \quad (u \ge r_*), \qquad A = \frac{1}{r_*} + \frac{4}{|\omega|}.$$

For $r > 0$, the elementary identity

$$\log(1 + r^2/v^2) = \int_v^\infty \frac{2r^2}{u(u^2 + r^2)}\, du \qquad (v > 0)$$

and integration of nonnegative summands give

$$\begin{aligned} \sum_{\alpha \in \Lambda_+} \log(1 + r^2/|\alpha|^2) &= \int_0^\infty \frac{2r^2 N_+(u)}{u(u^2 + r^2)}\, du \\ &\le 2Ar^2 \int_0^\infty \frac{du}{u^2 + r^2} = \pi A r. \end{aligned}$$

The source product, first for its finite partial products and then by its locally uniform convergence, therefore gives

$$|\Xi(s)| \le |\Xi(0)| e^{\pi A |s|} \qquad (s \in \mathbb{C}). \tag{102}$$

No lower bound on an individual product factor was used. But at the original points $s = -i(2n - \frac{1}{2})$, $n \ge 2$, the unchanged completion is

$$\Xi(-i(2n - \tfrac{1}{2})) = \xi(2n) = n(2n-1)\pi^{-n}(n-1)!\zeta(2n).$$

Every factor is positive and $\zeta(2n) > 1$ from its positive Dirichlet series. For $n \ge 4$, let $b_n = \lfloor n/2 \rfloor$. The last b_n factors of $(n-1)!$ are at least b_n, so

$$\log \left| \Xi(-i(2n - \tfrac{1}{2})) \right| \ge b_n \log b_n - n \log \pi.$$

Dividing by $2n - \frac{1}{2}$ makes the right side tend to $+\infty$, in contradiction with (102). This excludes every nonzero f in the remaining case and proves $I^\infty_{\text{even}} = \{0\}$.

For the time statements, for every actual zero and every one of its required derivative orders, define

$$h_{\lambda,j}(t) = (D^j U_t f)(\lambda), \qquad 0 \le j < m_\lambda.$$

These are entire functions of the original complex time by the proved heat series and derivative estimates. Its differentiated coefficients at zero are exactly

$$h^{(k)}_{\lambda,j}(0) = (-1/4)^k D^{2k+j} f(\lambda). \tag{103}$$

If $U_t f \in I$ for all t in a set with a finite accumulation point, every $h_{\lambda,j}$ vanishes there, and the identity theorem makes every one of them identically zero. Equation (103) then puts f in I^∞_{even} and proves (98). Taking all complex times proves $I^\infty_{\mathrm{heat}} = \{0\}$. Conversely, membership of I^∞_{even} would make all Taylor coefficients of each $h_{\lambda,j}$ zero and hence put every $U_t f$ in I, by the already proved entire division criterion. This also verifies directly the equality of the two cores, with both directions and the original coefficient retained.

In particular for each nonzero $f \in \mathcal{H}_{\le 1}$ the exact return set

$$\mathcal{T}_I(f) = \{t \in \mathbb{C} : \; U_t f \in I\} = \bigcap_{\lambda,j} \{t : h_{\lambda,j}(t) = 0\} \tag{104}$$

is closed and has no finite accumulation point. Thus it is a discrete set, possibly empty. More explicitly, at least one $h_{\lambda,j}$ is a nonzero entire function, since otherwise the core calculation would give $f = 0$; its isolated zero set already contains $\mathcal{T}_I(f)$. Every compact disk meets this return set finitely, so the set is countable by exhaustion by the integer-radius disks. Since the exact unclosed Schwartz trace intersection $\mathcal{A}_{\mathrm{Sch}}$ is a subspace of I,

$$\mathcal{T}_{\mathrm{Sch}}(f) := \{t \in \mathbb{C} : \; U_t f \in \mathcal{A}_{\mathrm{Sch}}\} \subseteq \mathcal{T}_I(f) \qquad (f \ne 0) \tag{105}$$

also has no finite accumulation point and is closed and discrete. The closedness follows because a subset of a set with no finite accumulation point is closed in $\mathbb{C}$. Its exact membership, including all strip conditions and reconstruction of the original input, was computed in Section 6.7. Consequently no nonzero original Schwartz trace follows a heat orbit through original Schwartz traces on any nontrivial time interval, even though its entire heat evolution exists for all complex times.

There is also an exact invariant-subspace conclusion. If W is a linear subspace of I with $D^2 W \subseteq W$, then $D^{2k} f \in I$ for every $f \in W$ and every $k \ge 0$. The proved core calculation forces $W = \{0\}$. The same conclusion holds for a subspace of I invariant under all U_t. These conclusions do not assume that W is closed or that its topology has any particular form. They apply to the analytic relation I and to its unclosed Schwartz trace subspace, and do not identify either with the actual source closure N_{ar} in the still unnamed τ.

6.9 The original four-term coefficient and its arithmetic return times

Retain the companion mechanism's actual coefficient operator on the unchanged entire variable s:

$$\begin{aligned}(T_D f)(s) = & -\frac{59}{275184} f(s+3) - \frac{1}{336} f'(s+3) \\ & + \frac{1}{1296} f(s + \tfrac{9}{2}) + \frac{3}{132496} f(s + \tfrac{13}{2}), \\ K_t(s) = T_D Z_t(s), & \qquad Z_t = U_t \Xi = 8 H_t(2s).\end{aligned} \tag{106}$$

Here T_D denotes this given four-term operator; it is not replaced by a differential truncation. We prove directly from its complete formula that $K_0 \ne 0$, that $K_t = U_t K_0$, and that its arithmetic return times have the precise restrictions just established.

Translations by each of the three original shifts act continuously on $(\mathcal{H}_{\le 1}, \gamma)$. For a general fixed shift h, the unchanged growth estimate is

$$\|f(\,\cdot + h)\|_{p,a} \le e^{a|h|^p} \|f\|_{p,a/2^{p-1}}, \qquad 1 < p < 2, \quad a > 0,$$

because $|s+h|^p \le 2^{p-1}(|s|^p + |h|^p)$. Differentiation is continuous by Lemma 2.1. Thus every displayed term, and T_D itself, maps $\mathcal{H}_{\le 1}$ continuously to $\mathcal{H}_{\le 1}$. Differentiation commutes with each fixed translation. Applying these continuous maps to the convergent series for U_t therefore proves term by term

$$T_D U_t = U_t T_D, \qquad K_t = U_t K_0. \tag{107}$$

In particular this calculation includes the derivative coefficient $-1/336$ at the shift 3, as well as the three function-value terms.

Nonvanishing of this particular coefficient is not assumed. Use the retained positive kernel Φ from (7), with its original formula

$$\Phi(u) = \sum_{n\geq 1}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u})e^{-\pi n^2 e^{4u}} \qquad (u \geq 0).$$

The substitution $v = 2u$ in the original cosine integral gives the exact entire Fourier representation

$$\begin{aligned} Z_t(s) &= 2\int_{\mathbb{R}} \Phi(|v|/2)e^{tv^2/4}e^{isv}\,dv,\\ K_t(s) &= 2\int_{\mathbb{R}} \Phi(|v|/2)m(v)e^{tv^2/4}e^{isv}\,dv,\\ m(v) &= \left(-\frac{59}{275184} - \frac{iv}{336}\right)e^{3iv} + \frac{e^{(9/2)iv}}{1296} + \frac{3e^{(13/2)iv}}{132496}. \end{aligned} \tag{108}$$

The coefficient 2 is required: first $8\,du = 4\,dv$ on the positive half-line, then the even extension of the cosine integral halves that coefficient. A shift by h multiplies the kernel by e^{ihv}, and the derivative multiplies it by iv for this original positive Fourier exponent. These two facts prove the second identity from the first, with the precise multiplier in the third. Every interchange and derivative is justified by the previously proved bound $|\Phi(u)| \leq Ce^{9u}e^{-ce^{4u}}$ on $u \geq 0$. On a compact set of complex t, s, this double exponential dominates the factors $e^{(\operatorname{Re} t)v^2/4}$, $e^{-\operatorname{Im}(s)v}$, and every polynomial factor from $m(v)$ and the derivatives. Thus the integrals converge absolutely and locally uniformly with all needed derivatives.

The multiplier has the exact value

$$m(0) = -\frac{59}{275184} + \frac{1}{1296} + \frac{3}{132496} = \frac{127}{219024} > 0. \tag{109}$$

Also

$$\Phi(0) = \sum_{n\geq 1} \pi n^2(2\pi n^2 - 3)e^{-\pi n^2} > 0;$$

every summand is positive and the series is absolutely convergent. The continuous integrable function $q(v) = 2\Phi(|v|/2)m(v)$ therefore has $q(0) = 2\Phi(0)\,127/219024 > 0$. If K_0 were identically zero, its Fourier transform in (108) would vanish at every real s. The Gaussian Fourier uniqueness argument already proved in Section 6.2 applies to this continuous integrable q: absolute Fubini makes every Gaussian convolution $q * G_\varepsilon$ zero, while these convolutions tend to q pointwise. This would force $q(0) = 0$, a contradiction. Hence $K_0 \neq 0$. The invertibility of U_t in (107) also proves $K_t \neq 0$ for every original complex time t.

The previous theorem now applies to this exact coefficient, not to an arbitrary replacement. In particular the sets

$$\begin{aligned} \mathcal{T}_I(K_0) &= \{t:\ T_D(8H_t(2\,\cdot)) \in \Xi\mathcal{H}_{\leq 1}\},\\ \mathcal{T}_{\mathrm{Sch}}(K_0) &= \{t:\ T_D(8H_t(2\,\cdot)) \in \mathcal{A}_{\mathrm{Sch}}\} \subseteq \mathcal{T}_I(K_0) \end{aligned} \tag{110}$$

are closed discrete subsets of $\mathbb{C}$. Their complete individual membership conditions are still the original multiplicity jets and, for the second set, the finite-strip Mellin conditions proved above. At any time outside $\mathcal{T}_I(K_0)$ there is an actual nontrivial zero ρ of multiplicity m_ρ at which

$$\frac{K_t(i(z-\frac{1}{2}))}{\zeta(z)}$$

has a pole of positive order $m_\rho - \operatorname{ord}_{i(\rho-1/2)} K_t$. Indeed nonmembership in I is exactly failure of at least one required zero jet, by the proved division theorem; local division by $(z-\rho)^{m_\rho} a_\rho(z)$ gives this pole order with the unchanged coordinate and unit. The full principal coefficients, including their time evolution, are those of Section 6.6 with the actual entire numerator $K_t(i(z-\frac{1}{2}))$ inserted.

Finally retain the original incidence factor a_{ef} in $a_{ef}K_t$. For $a_{ef} \neq 0$, scalar multiplication preserves both subspaces I and $\mathcal{A}_{\mathrm{Sch}}$ and has inverse multiplication by a_{ef}^{-1}, so it leaves both return sets unchanged and multiplies every principal coefficient by a_{ef}. For $a_{ef} = 0$, the coefficient is identically zero and belongs to both spaces at every time. This proves both incidence cases without suppressing that factor. These are exact arithmetic relations for the mechanism coefficient. They do not equate I or $\mathcal{A}_{\mathrm{Sch}}$ with N_{ar} in the unnamed source topology.

7 The inverse-root cover, the branch locus, and arithmetic fibres

The retained polynomial inverse-fibre slice is $P_{a,0,0}(r) = -2r^2 - 2a$. Its root equation is $a = -r^2$. The arithmetic coordinate is $a = 2s$, so

$$s = -r^2/2. \tag{111}$$

The finite free extension of the actual quotient is

$$\widehat{Q} = \mathbb{C}[r] \otimes_{\mathbb{C}[s]} Q_{\mathrm{ar}} = Q_{\mathrm{ar}} \oplus rQ_{\mathrm{ar}}, \qquad \mathcal{R} = \begin{pmatrix} 0 & -2S \\ 1 & 0 \end{pmatrix}, \qquad -\tfrac{1}{2}\mathcal{R}^2 = \begin{pmatrix} S & 0 \\ 0 & S \end{pmatrix}.$$

The even–odd decomposition of a polynomial in r proves freeness with basis $(1, r)$ and uniqueness coefficient by coefficient. Multiplying $q_0 + rq_1$ by r proves the displayed matrices; the finite direct-sum topology makes every induced map continuous. The involution is $(q_0, q_1) \mapsto (q_0, -q_1)$; the trace is $\operatorname{Tr}(q_0 + rq_1) = 2q_0$, with kernel rQ_{ar}. It is $\mathbb{C}[s]$-linear. It does not provide a $\mathbb{C}[r]$-module quotient with this kernel unless $S = 0$: multiplying rq by r gives $-2Sq$ in the even summand. This proves the exact limitation, rather than suppressing the odd component. The same construction at time t uses S_t and is conjugated by $\operatorname{diag}(\mathcal{U}_t, \mathcal{U}_t)$.

The original s-spectrum is unchanged by the extension: an inverse of $S - \lambda$ duplicates to a diagonal inverse, and conversely projection of an inverse of the doubled matrix to its first component, following inclusion of that component, gives an inverse of $S - \lambda$. Both directions are continuous. Moreover

$$\operatorname{Spec}(\mathcal{R}) = \{\rho \in \mathbb{C} : -\rho^2/2 \in \operatorname{Spec}(S)\}. \tag{112}$$

To prove this without an unsupported spectral-mapping citation, solve $(\mathcal{R} - \rho)(u, v) = (a, b)$. The second row gives $u = b + \rho v$ and the first becomes $-2(S + \rho^2/2)v = a + \rho b$. A continuous inverse of $S + \rho^2/2$ therefore gives a continuous inverse matrix. Conversely the right-hand side $(a, 0)$ and the second projection of an inverse matrix supply an inverse of $-2(S + \rho^2/2)$, with both identities checked by the same two rows.

Before any quotient by Ξ, let $\mathcal{O}_s$ be the sheaf of holomorphic functions in s and put $B = \mathcal{O}_s[r]/(r^2+2s)$, identified with $\mathcal{O}_s \oplus r\mathcal{O}_s$. On the punctured base $s \neq 0$, implicit differentiation gives $dr/ds = -1/r = r/(2s)$. Thus the actual lifted derivative is

$$\nabla_s \begin{pmatrix} f_0 \\ f_1 \end{pmatrix} = \begin{pmatrix} f_0' \\ f_1' + f_1/(2s) \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -2s \\ 1 & 0 \end{pmatrix}, \quad [\nabla_s, R] = -R^{-1}. \tag{113}$$

Indeed with $A = \operatorname{diag}(0, 1/(2s))$, $R' + AR - RA = \left(\begin{smallmatrix} 0 & -1 \\ 1/(2s) & 0 \end{smallmatrix}\right) = -R^{-1}$. A second application gives

$$\nabla_s^2 = \operatorname{diag}\left(D^2, D^2 + s^{-1}D - \frac{1}{4s^2}\right). \tag{114}$$

Equivalently, on a function of r the derivative is $-r^{-1}\partial_r$ and the heat generator is $-\frac{1}{4}r^{-2}\partial_r^2 + \frac{1}{4}r^{-3}\partial_r$. Squaring $-r^{-1}\partial_r$ by the product rule proves this with both signs preserved.

The branch extension is exact. The meromorphic connection sends B into $s^{-1}B$ and the connection one-form has a simple pole with residue $\operatorname{diag}(0, 1/2)$. Its maximal domain for holomorphic output at $s = 0$ is

$$\operatorname{Dom}_{\mathrm{hol}} \nabla_s = \mathcal{O}_s \oplus rs\mathcal{O}_s. \tag{115}$$

In fact the odd coefficient is $f_1' + f_1/(2s)$; its only possible pole has coefficient $f_1(0)/2$, proving necessity and sufficiency. For the square, write $f_1 = \sum_{n\geq 0} a_n s^n$. Its image has coefficient $(n^2 - 1/4)a_n s^{n-2}$. The negative powers vanish exactly when $a_0 = a_1 = 0$, so

$$\operatorname{Dom}_{\mathrm{hol}} \nabla_s^2 = \mathcal{O}_s \oplus rs^2\mathcal{O}_s.$$

For k iterates the odd coefficient is $\prod_{h=0}^{k-1}(n + 1/2 - h)a_n s^{n-k}$, whose prefactor is never zero for integral n. Thus its domain is $\mathcal{O}_s \oplus rs^k\mathcal{O}_s$. No derivation $B \to B$ extending ∂_s exists at the branch: derivation of $r^2 + 2s = 0$ would give $2r\delta(r) + 2 = 0$, forcing r to be a unit although it vanishes at the branch point. The logarithmic derivative $s\nabla_s$ does extend to all of B, with matrix $sD + \operatorname{diag}(0, 1/2)$.

These are sheaf and representative calculations; they do not assert that ∇_s descends through the frozen arithmetic relation. For a holomorphic local multiplier $R_0(s)$ the relation vector $(R_0\Xi, 0)$ acquires $(R_0'\Xi + R_0\Xi', 0)$, and the odd relation $(0, R_0\Xi)$ acquires $(0, R_0'\Xi + R_0\Xi' + R_0\Xi/(2s))$. The second iterate includes the full $2R_0'\Xi'$ term, as well as the odd $s^{-1}D - 1/(4s^2)$ terms of (114). Jet extension (25), or the moving closed quotient (8), retains these relations exactly.

Finally, the branch point is not an arithmetic spectral point at $t = 0$. The source gives $\Xi(0) \neq 0$, so its Proposition 3.6 makes S continuously invertible; then $\mathcal{R}$ is invertible by (112). In the analytic divisor realization the stronger local statement follows from Ξ being a holomorphic unit on a disk about zero: the quotient stalk is zero there. For real t, $Z_t(0) > 0$ directly from the retained kernel:

$$2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u} = \pi n^2 e^{5u}(2\pi n^2 e^{4u} - 3) > 0 \quad (u \geq 0),$$

and the positive integral converges by the bound already proved. Thus the actual real heat divisor never meets the cover branch point $s = 0$. On a simple zero curve its retained lift satisfies $r^2 = -2\lambda$, $r' = -\lambda'/r$; together with (15) these are the original-coordinate and cover equations, including the unit contribution. A nonreal r alone does not specify whether $\lambda = -r^2/2$ is nonreal.

8 Exact ingress from compact Weil tests

The accompanying retained Weil-test calculation supplies the concrete space $\mathcal{E} = C_c^\infty(\mathbb{R};\mathbb{C})$ with coordinate operator $T = -i\partial_u$ and Fourier transform

$$(\mathscr{F} f)(s) = \int_{\mathbb{R}} f(u)e^{-ius}\,du.$$

Here we give the complete heat ingress and its inverse domains; none of the following claims requires identifying the source quotient topology with a test-space topology. Integration by parts gives $\mathscr{F}(Tf) = s\mathscr{F} f$ with no endpoint term. If $\operatorname{supp} f \subset [-L, L]$, then

$$|\mathscr{F} f(s)| \le \|f\|_1 e^{L|\operatorname{Im} s|}, \qquad \|\mathscr{F} f\|_{p,a} \le \|f\|_1 \sup_{r\ge 0} e^{Lr-ar^p} < \infty.$$

Thus the Fourier transform maps the actual test space into the original entire-function set, continuously in γ on every fixed-support smooth-function space and hence in its usual inductive-limit topology.

Define, for all complex t,

$$(G_tf)(u) = e^{tu^2/4}f(u), \qquad U_t\mathscr{F} f = \mathscr{F} G_tf, \qquad T_t = G_tTG_{-t} = T + \frac{it}{2}u. \tag{116}$$

The multiplication G_t is a continuous automorphism on each fixed-support smooth-function space, with inverse G_{-t}. Indeed each derivative of the smooth multiplier is bounded on the fixed compact support, and Leibniz's finite sum bounds every output derivative seminorm by finitely many input ones. The same argument for the inverse proves the assertion. The exponential series and every u-derivative converge uniformly on a fixed compact set, locally uniformly in t; consequently the family is entire in time in that test topology. In the Fourier integral, $D^{2k}\mathscr{F} f = \mathscr{F}((-1)^k u^{2k} f)$. The compact support permits absolute interchange with the exponential series, proving the middle identity with the exact factor $1/4$. Finally differentiation of $e^{-tu^2/4}f$ gives the displayed T_t, and

$$\mathscr{F}(T_tf) = B_t\mathscr{F} f, \qquad D\mathscr{F} f = \mathscr{F}(-iuf).$$

Both equations follow directly from the integral, including their signs. The test-space version of the retained cover is therefore

$$\mathcal{R}_t^{\text{test}} = \begin{pmatrix} 0 & -2T_t \\ 1 & 0 \end{pmatrix}, \qquad (\mathcal{R}_t^{\text{test}})^2 = -2\operatorname{diag}(T_t, T_t).$$

Componentwise Fourier transformation intertwines this matrix with the representative matrix using B_t, and then with the actual quotient matrix using S_t.

To retain every possible arithmetic relation in this ingress, put

$$K = \{f \in \mathcal{E} : \mathscr{F} f \in N_{\text{ar}}\}, \qquad K_t = \{f \in \mathcal{E} : \mathscr{F} f \in N_t\}. \tag{117}$$

Then $K_t = G_tK$: by (116), $\mathscr{F} f \in U_tN_{\text{ar}}$ exactly when $\mathscr{F}(G_{-t}f) \in N_{\text{ar}}$. Thus there are injective linear maps onto their explicitly specified images

$$\mathcal{E}/K \longrightarrow Q_{\text{ar}}, \quad [f] \longmapsto [\mathscr{F} f], \qquad \mathcal{E}/K_t \longrightarrow Q_t, \quad [f] \longmapsto [\mathscr{F} f],$$

and the square formed by G_t and $\mathcal{U}_t$ commutes. The kernel statements follow exactly from (117); no claim that $K = 0$ in the unnamed source topology is inserted. To make the ingress continuous without that identification, give $\mathcal{E}$ the join η of its usual test topology with the inverse-image topology of τ under $\mathscr{F}$. Then K is closed and the first quotient map is continuous. Give the time-t test space the transported topology $(G_t)_*\eta$; the second map and the commutative square are continuous by (116). This specifies the exact topologies rather than claiming an unproved topological embedding of a quotient image equipped with its subspace topology.

There is a fully explicit inverse-domain calculation in the same test coordinates. For $\lambda \in \mathbb{C}$, $k \geq 1$, the image is

$$(T_t - \lambda)^k \mathcal{E} = \left\{ g \in \mathcal{E} : \int_{\mathbb{R}} v^j e^{-tv^2/4 - i\lambda v} g(v)\, dv = 0 \quad (0 \leq j < k) \right\}. \tag{118}$$

On that precise image the inverse is

$$[(T_t - \lambda)^{-k} g](u) = \frac{i^k e^{tu^2/4 + i\lambda u}}{(k-1)!} \int_{-\infty}^{u} (u - v)^{k-1} e^{-tv^2/4 - i\lambda v} g(v)\, dv. \tag{119}$$

To prove both directions, write $f = e^{tu^2/4 + i\lambda u} h$. Direct differentiation gives $(T_t - \lambda) f = e^{tu^2/4 + i\lambda u}(-ih')$. Iteration reduces the inverse equation to a kth primitive, giving (119) because $(-i)^k i^k = 1$. For g supported in $[a, b]$, that formula vanishes below a. Above b, expansion of $(u - v)^{k-1}$ shows it vanishes identically exactly when all k moments in (118) vanish; the binomial coefficients and the nonzero exponential do not change that equivalence. Its smoothness follows by repeated differentiation of the integral; smoothness of g with compact support gives matching derivatives at both endpoints. Necessity for any compact solution also follows by integrating $h^{(k)}$ against $1, v, \ldots, v^{k-1}$ and integrating by parts k times. Uniqueness follows since a compactly supported solution of $h^{(k)} = 0$ is a polynomial of degree at most $k - 1$ which vanishes on a ray, and hence is zero. This proves the exact image, inverse, kernel zero, and preservation of the original support interval.

In particular the inverse of $\mathcal{R}_t^{\text{test}}$ is

$$(g_0, g_1) \longmapsto (g_1, -\tfrac{1}{2} T_t^{-1} g_0), \qquad \operatorname{Dom}(\mathcal{R}_t^{\text{test}})^{-1} = \left\{ (g_0, g_1) : \int e^{-tv^2/4} g_0(v)\, dv = 0 \right\}.$$

Substitution into its two rows proves both inverse identities. Thus the branch-locus obstruction in the concrete test realization is a retained Gaussian-weighted zero-moment condition. The exact arithmetic quotient instead has invertible S_t and root operator, as proved above. Their ingress map and its kernels explain precisely how these statements coexist; one operator's inverse domain is not silently assigned to the other realization.

8.1 The source Sobolev cokernel and an exact common test-space map

The Sobolev construction cited immediately after Connes–Marcolli Proposition 2.24 is specified in Chapter 2, Section 9. It is a different, explicitly defined quotient, and supplies a useful exact comparison. We retain its weight, its generator, and its limited set of detected jets. The primary locators are Connes–Marcolli, printed pp.407–408 and 416–420 [2], and Connes, Section III, pp.10–15 and Appendix I, pp.62–67 of the author PDF [3]. Here and below $\delta > 1$.

For the rational field, choose the splitting $C_{\mathbb{Q}} = C_{\mathbb{Q},1} \times \mathbb{R}_+^*$ used in the source, give the compact factor Haar mass one, and put $x = e^u$ on the second factor. Its trivial-character component is the Hilbert space

$$B_\delta = L^2(\mathbb{R}, (1 + u^2)^{\delta/2}\, du), \qquad \|f\|_\delta^2 = \int_{\mathbb{R}} |f(u)|^2 (1 + u^2)^{\delta/2}\, du.$$

The source adelic map is

$$E_{\mathrm{ad}}\psi(g) = |g|^{1/2} \sum_{q\in\mathbb{Q}^*} \psi(qg), \quad \psi \in \mathcal{S}(\mathbb{A}_\mathbb{Q})_0 = \{\psi : \psi(0) = 0, \int \psi\, dx = 0\}.$$

Let P_1 denote averaging over $C_{\mathbb{Q},1}$, and define

$$R_\delta = \overline{P_1 E_{\mathrm{ad}}(\mathcal{S}(\mathbb{A}_\mathbb{Q})_0)}^{B_\delta}, \qquad H_{\delta,1} = B_\delta/R_\delta.$$

These are exactly the trivial-character relation space and cokernel of the source. Indeed the source completion uses the norm of its image, so the extended isometry has closed range equal to the closure of the uncompleted image. Compact averaging commutes with that equivariant closed range. If b is a limit of image vectors and satisfies $P_1 b = b$, averaging an approximating sequence gives a sequence in the averaged image converging to b. This proves that the displayed closure is the trivial-character part of the full relation space, without replacing any source relation by an orthogonal complement as a definition.

We import explicitly the following result of the source proof: under the bilinear pairing $\langle f, \eta\rangle = \int f(u)\eta(u)\, du$, the annihilator of R_δ in $B_{-\delta} = L^2(\mathbb{R}, (1+u^2)^{-\delta/2}\, du)$ is the closed linear span of

$$u^a e^{itu}, \qquad \zeta(\tfrac{1}{2} + it) = 0, \quad t \in \mathbb{R}, \quad 0 \le a < m_{-t}, \quad a < \frac{\delta - 1}{2}. \tag{120}$$

Here m_{-t} is the original multiplicity of the zero $-t$ of Ξ, equal to that of $\frac{1}{2} + it$ by the retained reflected Mellin coordinate. The source proves the finite combinations after a compact Fourier cutoff and proves that the cutoff vectors are dense by uniformly bounded approximate units. Thus the closed span, not only its finite-dimensional projections, is the imported statement. No such annihilator statement is imported for the unnamed topology τ of the entire-function ring.

The precise cutoff can also be checked directly from the defined norm:

$$\|u^a e^{itu}\|_{-\delta}^2 = \int_\mathbb{R} |u|^{2a}(1+u^2)^{-\delta/2}\, du < \infty \quad \Longleftrightarrow \quad 2a - \delta < -1.$$

Near zero the integrand is locally integrable for every integer $a \ge 0$; for $|u| \ge 1$ it is bounded above and below by positive constants times $|u|^{2a-\delta}$. Integration of this power proves the strict inequality and the logarithmic divergence at equality. Hence set

$$q_\delta = \left\lceil \frac{\delta - 1}{2} \right\rceil, \qquad n_{\delta,\lambda} = \min(m_\lambda, q_\delta). \tag{121}$$

There are exactly $n_{\delta,\lambda}$ allowed degrees at a real zero λ. The original Connes Theorem 1, Section III, p.13, states the equivalent multiplicity bound $n < (1+\delta)/2$, $n \le m_\lambda$. The inspected Connes–Marcolli printed p.408 instead displays $n < 1 + \delta/2$ in Theorem 2.47; its proof, pp.419–420, equations (2.444) and (2.448), gives precisely the stricter bound proved above. This records the primary discrepancy rather than silently identifying the two printed formulas. No assumption that an actual zeta zero has multiplicity greater than one is used.

Let $\mathcal{E} = C_c^\infty(\mathbb{R}; \mathbb{C})$ retain its usual fixed-support smooth-function inductive-limit topology, and let $\mathscr{F} f(s) = \int f(u) e^{-ius}\, du$ as above. Define

$$K_{\mathrm{full}} = \{f \in \mathcal{E} : \ D^a \mathscr{F} f(\lambda) = 0 \text{ for every zero } \lambda \text{ of } \Xi, \ 0 \le a < m_\lambda\}, \tag{122}$$

$$K_\delta = \{f \in \mathcal{E} : \ D^a \mathscr{F} f(\lambda) = 0 \text{ for every real zero } \lambda \text{ of } \Xi, \ 0 \le a < n_{\delta,\lambda}\}. \tag{123}$$

The already proved entire division theorem identifies $K_{\mathrm{full}} = \{f : \mathscr{F} f \in \Xi\mathcal{H}_{\le 1}\}$ exactly.

Theorem 8.1 (Exact dense test ingress into the source Sobolev cokernel)**.** *The map $j_\delta : \mathcal{E} \to H_{\delta,1}$ sending a test function to its Hilbert quotient class is continuous, has dense image, and has kernel exactly K_δ. It induces the continuous map*

$$J_\delta : \mathcal{E}/K_{\mathrm{full}} \longrightarrow H_{\delta,1}, \qquad [f] \longmapsto [f], \qquad \ker J_\delta = K_\delta/K_{\mathrm{full}}. \tag{124}$$

Its image is dense. These statements neither assert surjectivity nor identify the topology of its image with a quotient topology. For the source translation generator D_1, this ingress retains the original spectral coordinate through

$$j_\delta(Tf) = iD_1 j_\delta(f), \qquad T = -i\partial_u, \qquad \mathscr{F}(Tf) = s\mathscr{F}f. \tag{125}$$

Proof. The inclusion $\mathcal{E} \to B_\delta$ is continuous on every fixed-support space: on $[-L, L]$,

$$\|f\|_\delta^2 \le \left(\int_{-L}^{L} (1+u^2)^{\delta/2}\, du\right) \sup |f|^2.$$

The inductive-limit universal property and the continuous Hilbert quotient map give continuity of j_δ. Smooth compactly supported functions are dense in B_δ: truncate an arbitrary f outside $[-L, L]$, whose squared norm tail tends to zero by integrability; on the enlarged interval $[-L-1, L+1]$, the weight is bounded above and below by positive constants, and convolution with a smooth compactly supported approximate identity approximates the truncation in ordinary L^2 and therefore in the weighted norm. These convolutions have compact support and are smooth. Consequently their quotient images are dense.

The continuous dual of B_δ is exactly $B_{-\delta}$ under the displayed bilinear pairing. To check this, multiply by $(1+u^2)^{\delta/4}$ to identify B_δ isometrically with ordinary L^2; the Riesz representation there, with complex conjugation absorbed in the representing function, gives exactly the inverse-weight dual and its norm. Since R_δ is closed, a vector belongs to it exactly when every member of its annihilator vanishes on that vector. One direction is the definition. For the other, a vector outside a closed Hilbert subspace has a nonzero orthogonal component, and its inner product defines a continuous separating functional. By the explicitly imported closed-span description (120), it is enough to test the listed log-monomial characters. But

$$D^a\mathscr{F}f(-t) = (-i)^a \int u^a e^{itu} f(u)\, du, \qquad \int u^a e^{itu} f(u)\, du = i^a D^a \mathscr{F}f(-t).$$

Every factor i^a is nonzero. The coordinate $\lambda = -t$ and the proved count (121) now give $\ker j_\delta = K_\delta$. Every equation defining K_δ appears among the equations defining K_{full}, proving $K_{\mathrm{full}} \subseteq K_\delta$. Thus (124) is well defined and continuous by the quotient universal property; changing the representative changes its image by zero exactly for a member of K_δ. This proves the kernel and the dense-image assertion.

Finally the source left translation is $(V(e^\varepsilon)f)(u) = f(u-\varepsilon)$, so its infinitesimal generator on $\mathcal{E}$ is $-\partial_u$. The difference quotient converges in B_δ because the supports remain in a fixed compact set and the smooth-function difference quotient converges uniformly there. Passing to the quotient proves that $j_\delta(f)$ belongs to the domain of D_1 and $D_1 j_\delta(f) = j_\delta(-f')$. Multiplication by i gives the first equation of (125). Integration by parts in the compactly supported Fourier integral gives the second. In particular the source eigenparameter it becomes $-t$ under iD_1, exactly the same s coordinate used in the jet equations; no sign change is absorbed into the source operator. □

For completeness the unspecified source topology also has an exact comparison, requiring no unproved inclusion of its kernel in K_δ. Retain

$$K_\tau = \{f \in \mathcal{E} : \mathscr{F} f \in N_{\mathrm{ar}}\}, \qquad L_{\tau,\delta} = K_\tau \cap K_\delta.$$

Give $\mathcal{E}$ the join of its usual test topology, the inverse-image topology $\mathscr{F}^{-1}\tau$, and the seminorm $\|\cdot\|_\delta$. The last seminorm is already continuous in the test topology, but is displayed to specify every target map. Then K_τ, K_δ and $L_{\tau,\delta}$ are closed, and there are continuous maps

$$Q_{\mathrm{ar}} \longleftarrow \mathcal{E}/L_{\tau,\delta} \longrightarrow H_{\delta,1}, \qquad [f] \longmapsto [\mathscr{F} f], \quad [f] \longmapsto [f]. \tag{126}$$

Their respective kernels are exactly $K_\tau/L_{\tau,\delta}$ and $K_\delta/L_{\tau,\delta}$; the right image is dense. Indeed each composite on $\mathcal{E}$ is continuous by definition, vanishes on the intersection, and has the asserted kernel before quotienting. The quotient universal property proves the assertions, and density follows from the same set of compact tests as above. The map to the product of the two targets is injective because its kernel is precisely the intersection already divided out. No topological embedding of that image is inferred from injectivity. This is a proved relation to the actual source quotient while its named topology remains retained. The comparison also intertwines the coordinate operators: T preserves K_τ because $\mathscr{F}(Tf) = s\mathscr{F} f$ and $MN_{\mathrm{ar}} \subset N_{\mathrm{ar}}$; it preserves K_δ because

$$D^a(sF)(\lambda) = \lambda D^a F(\lambda) + aD^{a-1}F(\lambda),$$

where the second term is zero for $a = 0$ and otherwise uses an already imposed lower jet. Thus T descends to the common span, whose left map intertwines it with S and whose right map intertwines it with iD_1 on the proved domain of the ingress.

There is also a literal compatibility of the original finite Hermite traces with the adelic map. For the source's even ψ, put $\psi_{\mathrm{ad}} = \psi \otimes \mathbf{1}_{\widehat{\mathbb{Z}}}$. It lies in $\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0$ because its value and additive integral are the two original zero constraints. On the idele represented by $(x, 1, 1, \ldots)$, a rational number belongs to every $\mathbb{Z}_p$ exactly when it is an integer: write it in lowest terms and use any prime dividing its denominator for the reverse implication. Therefore

$$E_{\mathrm{ad}}\psi_{\mathrm{ad}}(x) = x^{1/2} \sum_{n \in \mathbb{Z}\setminus\{0\}} \psi(nx) = 2\mathcal{E}\psi(x).$$

Every norm-one rational idele class has a representative consisting of a real sign and finite units. Indeed, for an idele g put $q = \prod_p p^{v_p(g_p)} \in \mathbb{Q}_{>0}$, a finite product. Then $g_p/q \in \mathbb{Z}_p^*$ for every prime, while the norm-one identity gives $|g_\infty| = q$. Dividing by the diagonal rational q gives the asserted representative. The function is unchanged by the finite units; evenness removes the real sign. It therefore lies in the entire trivial-character component. The Fourier transform in $u = \log x$ is thus exactly twice the original source trace, retaining the factor 2. In particular the finite Hermite generators yield $2R_\ell^{\pm}(s)\Xi(s)$ in this explicit realization. Their logarithmic representatives lie in every B_δ by their proved rapid decay. This proves a concrete arithmetic compatibility as well as the quotient comparison; it does not identify a Sobolev topology with the topology on all of $\mathcal{H}_{\leq 1}$.

9 The retained spatial bilaplacian and the actual source quotient

The original polynomial spatial map and its inverse-fibre trajectory are

$$Q(x, y, w) = 1 + xy, \qquad \sigma(x, y, w) = \tfrac{1}{2}\{Q^3 w + y^2 Q(4 + 3xy)\},$$
$$\gamma(z) = (z^{-1}, -3z/2, 13z^2/2), \quad z > 0.$$

Here the Euclidean coordinates are exactly (x, y, w), and $\Delta = \partial_x^2 + \partial_y^2 + \partial_w^2$. The symbol $\gamma(z)$ in this section denotes this spatial curve; the topology γ on $\mathcal{H}_{\le 1}$ retains its earlier definition. Substitution gives $Q(\gamma(z)) = -1/2$ and $\sigma(\gamma(z)) = -z^2/8$. The pullback $\sigma^* : \mathcal{O}(\mathbb{C}) \longrightarrow \mathcal{O}(\mathbb{C}^3)$, $h \longmapsto h \circ \sigma$, is a continuous injective algebra homomorphism for compact convergence. Indeed, $\eta(s) = (0, 0, 2s)$ satisfies $\sigma \circ \eta(s) = s$, so $\eta^*\sigma^* = 1$. A compact set has compact image under each polynomial map, proving continuity of both pullbacks by their defining supremum seminorms. This also proves that the inverse on the image is continuous. Restriction to $\mathcal{H}_{\le 1}$ with its growth topology is continuous because that topology dominates compact convergence.

9.1 Every retained coefficient and the residue functional

Write $g = \nabla\sigma$, $H = D^2\sigma$, $N = g \cdot g$, and $L = \Delta\sigma$. For every entire h,

$$
\begin{aligned}
\Delta(h \circ \sigma) &= N h''(\sigma) + L h'(\sigma), \\
\Delta^2(h \circ \sigma) &= N^2 h^{(4)}(\sigma) + (4g^{\mathsf{T}} H g + 2NL) h^{(3)}(\sigma) \\
&\quad + (2\|H\|_{\mathrm{F}}^2 + 4g \cdot \nabla L + L^2) h''(\sigma) + (\Delta L) h'(\sigma).
\end{aligned} \tag{127}
$$

These formulas use the bilinear polynomial contractions; on the real locus $\|H\|_{\mathrm{F}}^2 = \sum_{i,j} H_{ij}^2$ is the usual squared Frobenius norm. To prove the second identity, the two product rules give

$$
\begin{aligned}
\Delta(N h'') &= (\Delta N) h'' + (2g \cdot \nabla N + NL) h^{(3)} + N^2 h^{(4)}, \\
\Delta(L h') &= (\Delta L) h' + (2g \cdot \nabla L + L^2) h'' + LN h^{(3)}.
\end{aligned}
$$

All derivatives of h are evaluated at σ. Differentiating $N = \sum_i (\partial_i \sigma)^2$ gives $\nabla N = 2Hg$ and $\Delta N = 2\sum_{i,j} H_{ij}^2 + 2g \cdot \nabla L$. Substitution proves every term in (127).

The original derivatives before restriction are

$$
\begin{aligned}
g_1 &= \tfrac{3}{2} y Q^2 w + \tfrac{1}{2} y^3 (7 + 6xy), \\
g_2 &= \tfrac{3}{2} x Q^2 w + \tfrac{1}{2} (8y + 21xy^2 + 12x^2y^3), \\
g_3 &= \tfrac{1}{2} Q^3, \\
L &= 3wQ(x^2 + y^2) + 3y^4 + 18x^2y^2 + 21xy + 4.
\end{aligned}
$$

Differentiating these polynomials and substituting the unchanged curve yields

$$
g(\gamma(z)) = \begin{pmatrix} -9z^3/32 \\ -3z/16 \\ -1/16 \end{pmatrix}, \qquad H(\gamma(z)) = \begin{pmatrix} -27z^4/4 & 3z^2/16 & -9z/16 \\ 3z^2/16 & 13/4 & 3/(8z) \\ -9z/16 & 3/(8z) & 0 \end{pmatrix}.
$$

The complete contractions are

$$
\begin{aligned}
N(\gamma(z)) &= (81z^6 + 36z^2 + 4)/1024, \\
L(\gamma(z)) &= (13 - 27z^4)/4, \\
\|H(\gamma(z))\|_{\mathrm{F}}^2 &= 729z^8/16 + 9z^4/128 + 81z^2/128 + 169/16 + 9/(32z^2), \\
(g \cdot \nabla L)(\gamma(z)) &= 9477z^8/512 - 405z^4/64 + 27z^2/128 + 81/32 + 3/(32z^2), \\
(g^{\mathsf{T}} H g)(\gamma(z)) &= -2187z^{10}/4096 + 81z^6/4096 - 81z^4/4096 + 117z^2/1024 + 9/1024, \\
(\Delta L)(\gamma(z)) &= -111z^2 + 36/z^2.
\end{aligned}
$$

Consequently the exact restricted operator is

$$(\Delta^2\sigma^* h)(\gamma(z)) = \sum_{j=1}^{4} C_j(z) h^{(j)}(-z^2/8), \tag{128}$$

where

$$\begin{aligned}
C_4(z) &= \left((81z^6 + 36z^2 + 4)/1024\right)^2,\\
C_3(z) &= -6561z^{10}/2048 + 243z^6/2048 - 135z^4/1024 + 351z^2/512 + 31/512,\\
C_2(z) &= 26973z^8/128 - 4419z^4/64 + 135z^2/64 + 669/16 + 15/(16z^2),\\
C_1(z) &= -111z^2 + 36/z^2.
\end{aligned}$$

In particular both off-diagonal Hessian occurrences and all derivatives of L have been retained.

For every entire h, continuity of its first four derivatives at zero and these four Laurent polynomials prove the limit

$$\mathfrak{r}(h) := \lim_{z\downarrow 0} z^2(\Delta^2\sigma^* h)(\gamma(z)) = 36h'(0) + \frac{15}{16}h''(0). \tag{129}$$

Indeed z^2C_4, z^2C_3 tend to zero, whereas z^2C_2 and z^2C_1 tend to $15/16$ and 36. No evenness assumption on h is needed. For any $R > 0$, Cauchy's integral formula gives

$$|\mathfrak{r}(h)| \le \left(\frac{36}{R} + \frac{15}{8R^2}\right) \sup_{|s|\le R} |h(s)|.$$

Thus $\mathfrak{r}$ is continuous for compact convergence and for the growth topology. This estimate asserts no continuity in the unnamed source topology τ.

For the actual coefficient $K_t = T_{\mathrm{D}}Z_t$ of Section 6.9, differentiation of its four original terms gives the exact value

$$\begin{aligned}
\mathfrak{r}(K_t) = &-\frac{59}{275184}\left(36Z_t'(3) + \frac{15}{16}Z_t''(3)\right)\\
&- \frac{1}{336}\left(36Z_t''(3) + \frac{15}{16}Z_t'''(3)\right)\\
&+ \frac{1}{1296}\left(36Z_t'(9/2) + \frac{15}{16}Z_t''(9/2)\right)\\
&+ \frac{3}{132496}\left(36Z_t'(13/2) + \frac{15}{16}Z_t''(13/2)\right).
\end{aligned} \tag{130}$$

For the unchanged incidence factor a_{ef} and physical diffusion $\nu\Delta$, its scaled second diffusion is $\nu^2 a_{ef}\mathfrak{r}(K_t)$ by linearity and operator composition. The limit probes spatial infinity: $|\gamma(z)| \ge z^{-1}$. Every $\sigma^* h$ is smooth at every finite real point by its entire composition. These assertions specify the domain and meaning of the observable exactly.

9.2 The full ambiguity on the actual source quotient

Set $E = \mathcal{H}_{\le 1}$, $N = N_{\mathrm{ar}}$, and $Q_{\mathrm{ar}} = (E,\tau)/N$. The original even function Ξ has $\Xi(0) > 0$, as proved from its retained positive kernel. Define

$$e(s) = \frac{8}{15\Xi(0)} s^2\Xi(s), \qquad K = \ker\mathfrak{r}, \qquad Pf = f - e\,\mathfrak{r}(f). \tag{131}$$

Then $e \in \mathbb{C}[s]\Xi \subset N$, $e'(0) = 0$, and $e''(0) = 16/15$, so $\mathfrak{r}(e) = 1$. Consequently

$$\mathfrak{r}(N) = \mathbb{C}, \qquad \{([f]_N, \mathfrak{r}(f)) : f \in E\} = Q_{\mathrm{ar}} \times \mathbb{C}. \tag{132}$$

Here is the exact representative for every pair on the right. Given any representative f of q and any $a \in \mathbb{C}$, put $f_a = f + (a - \mathfrak{r}(f))e$. Then $[f_a]_N = q$ and $\mathfrak{r}(f_a) = a$. In particular the zero class itself has representatives ae with every residue a. No nonempty set of source quotient classes admits this scalar as a value independent of all its representatives.

The strongest corresponding quotient map is explicit. Algebraically $P^2 = P$, $\operatorname{im} P = K$, $\ker P = \mathbb{C}e$, and

$$N_0 := N \cap K = P(N), \qquad P^{-1}(N_0) = N.$$

The first equality for N_0 follows because $e \in N$ and P fixes K. If $Pf \in N$, the identity $f = Pf + e\mathfrak{r}(f)$ proves $f \in N$, establishing the inverse-image equality. Give K the quotient topology $\tau_P = P_*\tau$: a subset $V \subset K$ is open exactly when $P^{-1}(V)$ is open in (E, τ). This is the topology of $(E, \tau)/\mathbb{C}e$ under its displayed algebraic isomorphism with K. In particular it is a vector topology. Since N is closed, the same inverse-image equality proves that N_0 is closed in (K, τ_P). There is a homeomorphism

$$(K, \tau_P)/N_0 \xrightarrow{[k]\mapsto[k]_N} (E, \tau)/N. \tag{133}$$

For completeness, let $q_N : E \to E/N$ and $q_0 : K \to K/N_0$ denote the quotient maps. The proposed map A satisfies $Aq_0P = q_N$. Since q_0P is a composition of quotient maps and hence is quotient, this identity proves continuity of A. The inverse is induced by q_0P, because its kernel is N; the quotient property of q_N proves its continuity. These maps are inverse since $Pf - f \in N$ and $Pk = k$. This proves the assertion without using τ-continuity of $\mathfrak{r}$ or of P as an endomorphism of (E, τ).

There is also a proved comparison using the inherited topology: the identity $(K, \tau|_K) \to (K, \tau_P)$ is continuous. Indeed for V open in τ_P, $V = K \cap P^{-1}(V)$, which is inherited-open. It induces a continuous bijection $(K, \tau|_K)/N_0 \to Q_{\mathrm{ar}}$. No equality of these two topologies on K is asserted. For the growth topology the stronger result is available: $\mathfrak{r}$ and P are continuous by the Cauchy bound, so $f \mapsto (Pf, \mathfrak{r}(f))$ and $(k, a) \mapsto k + ae$ are inverse continuous linear maps

$$(E, \gamma) \simeq (K, \gamma|_K) \oplus \mathbb{C}. \tag{134}$$

The continuity of $a \mapsto ae$ follows from each defining seminorm. This proves the full split topology at precisely the topology where its hypotheses have been established.

9.3 Every finite jet at the original coordinate origin

The residue calculation is a particular finite-jet phenomenon. For $d \geq 0$ retain the derivative coordinates $J_d f = (f(0), f'(0), \dots, f^{(d)}(0))$. Write the convergent germ $1/\Xi(s) = \sum_{n\geq 0} b_n s^n$; its coefficients are given without any change of Ξ by

$$b_0 = \Xi(0)^{-1}, \qquad b_n = -\Xi(0)^{-1} \sum_{k=1}^{n} \frac{\Xi^{(k)}(0)}{k!} b_{n-k}.$$

For $a = (a_0, \dots, a_d)$ define the entire polynomial trace

$$E_d a(s) = \Xi(s) \sum_{n=0}^{d} \left(\sum_{j=0}^{n} \frac{a_j}{j!} b_{n-j} \right) s^n. \tag{135}$$

Multiplying its truncated Taylor series by that of Ξ gives $E_d a(s) = \sum_{j=0}^{d} a_j s^j/j! + O(s^{d+1})$, because the defining reciprocal convolution has coefficient one at zero and zero at each positive degree. Therefore $J_d E_d = 1$ and $\operatorname{im} E_d \subset \mathbb{C}[s]\Xi \subset N$. This proves $J_d(N) = \mathbb{C}^{d+1}$, with its complete finite-dimensional section, rather than only the scalar image. For any $q = [f]_N$ and any jet a, the representative $f + E_d(a - J_d f)$ realizes (q, a).

Put $P_d = 1 - E_d J_d$, $K_d = \ker J_d$, and $N_d = N \cap K_d$. The same identities, now with $\ker P_d = \operatorname{im} E_d$, prove $P_d(N) = N_d$ and $P_d^{-1}(N_d) = N$. Thus $(K_d, (P_d)_*\tau)/N_d \simeq Q_{\mathrm{ar}}$ by the two quotient-map arguments just given. Cauchy's bounds $|f^{(j)}(0)| \le j! R^{-j} \sup_{|s|\le R} |f(s)|$ prove the continuous growth-topology decomposition $(E, \gamma) \simeq (K_d, \gamma|_{K_d}) \oplus \mathbb{C}^{d+1}$. All factorials are retained in the derivative coordinates and in the section.

9.4 Heat transport of the retained residue and its quotient

The scalar observable also has an exact heat transport. Define

$$\mathfrak{r}_t = \mathfrak{r} U_{-t}, \qquad e_t = U_t e, \qquad P_t = 1 - e_t \mathfrak{r}_t = U_t P U_{-t}, \qquad K_t^{\mathrm{res}} = \ker \mathfrak{r}_t.$$

These definitions retain $U_t = \exp(-tD^2/4)$ and hence give the convergent formula

$$\mathfrak{r}_t(f) = 36 \sum_{k=0}^{\infty} \frac{(t/4)^k}{k!} f^{(2k+1)}(0) + \frac{15}{16} \sum_{k=0}^{\infty} \frac{(t/4)^k}{k!} f^{(2k+2)}(0). \tag{136}$$

The earlier heat-series proof gives convergence in the growth topology, locally uniformly in t; applying the two continuous derivative evaluations gives the displayed convergence and its value. In particular $\mathfrak{r}_t U_t = \mathfrak{r}$, $\mathfrak{r}_t(e_t) = 1$, and $e_t \in N_t$. Also $U_t K = K_t^{\mathrm{res}}$ and $U_t N_0 = N_t \cap K_t^{\mathrm{res}}$ by these identities and invertibility of U_t.

Give K_t^{res} the topology $(P_t)_*\tau_t$. It equals $(U_t)_*\tau_P$: for $V \subset K_t^{\mathrm{res}}$, the identity $P_t U_t = U_t P$ gives $U_{-t}(P_t^{-1}(V)) = P^{-1}(U_{-t}V)$, which is exactly the equivalence of the two defining open-set conditions. Thus U_t gives a homeomorphism between the residue-zero quotient in (133) and

$$(K_t^{\mathrm{res}}, (P_t)_*\tau_t)/(N_t \cap K_t^{\mathrm{res}}).$$

The latter is homeomorphic to $(E, \tau_t)/N_t$ by $[k] \mapsto [k]_{N_t}$, with inverse $[f] \mapsto [P_t f]$; the two quotient universal properties apply exactly as above. The two homeomorphisms commute with the already proved heat homeomorphism of the actual source quotient because both send $[f]$ to $[U_t f]$. Finally, for each $q = [f]_{N_t}$ and $a \in \mathbb{C}$, the unchanged class has the representative $f + e_t(a - \mathfrak{r}_t(f))$ with transported residue a. This computes the full transported scalar relation without identifying $\mathfrak{r}_t$ with the fixed physical functional $\mathfrak{r}$.

9.5 Every one-point finite jet of the moving actual relation

The complete finite-jet statement persists on the actual transported source relation, at every point once the time is nonzero. Throughout this subsection the derivative coordinate is the original s, the time may be complex, and $N_t = U_t N_{\mathrm{ar}}$ retains its proved closure topology τ_t.

Proposition 9.1 (Full finite jets on the moving source relation)**.** *For every $t \in \mathbb{C} \setminus \{0\}$, $\lambda \in \mathbb{C}$, and integer $d \ge 0$, put*

$$J_{\lambda,d} f = (D^j f(\lambda))_{j=0}^{d}.$$

Then

$$J_{\lambda,d}(U_t(\mathbb{C}[s]\Xi)) = J_{\lambda,d}(N_t) = \mathbb{C}^{d+1}. \tag{137}$$

There is a continuous linear right inverse of this jet map from the usual finite-dimensional space into $(\mathcal{H}_{\le 1}, \tau_t)$, whose image consists of finite linear combinations of the actual transported polynomial traces.

Proof. Suppose a complex linear functional

$$Lf = \sum_{j=0}^{d} c_j D^j f(\lambda)$$

annihilates $U_t(\mathbb{C}[s]\Xi)$. We first justify the full exponential generating calculation in the established growth topology, without invoking convergence in τ or τ_t. For every $1 < p < 2$, $a > 0$, and $n \ge 1$, direct maximization gives

$$\|s^n\Xi\|_{p,a} \le \|\Xi\|_{p,a/2}\left(\frac{2n}{ape}\right)^{n/p}. \tag{138}$$

Indeed the maximum of $r^n \exp(-ar^p/2)$ occurs at $r^p = 2n/(ap)$. The $n = 0$ term is controlled directly by $\|\Xi\|_{p,a}$. For every $A > 0$, the sum of the right side times $A^n/n!$ converges: the elementary bound $n! \ge (n/\mathrm{e})^n$ makes its n-th root tend to zero. Each fixed derivative of the parameter power series adds only a polynomial factor in n, together with a fixed shift of its power. The same root estimate proves locally uniform convergence of all those differentiated series in every growth seminorm. Thus

$$\alpha \longmapsto e^{\alpha s}\Xi(s) = \sum_{n=0}^{\infty} \frac{\alpha^n}{n!} s^n \Xi(s)$$

is entire with values in $(\mathcal{H}_{\le 1}, \gamma)$, with the displayed parameter derivatives. The proved continuity of U_t in γ, and Cauchy's estimate for every derivative evaluation in L, justify applying LU_t to every series. All its Taylor coefficients vanish by the assumed annihilation. Hence $LU_t(e^{\alpha s}\Xi) = 0$ for every $\alpha \in \mathbb{C}$.

The exact heat and exponential identity, retaining its original sign and coefficient, is

$$U_t(e^{\alpha s}\Xi)(s) = e^{\alpha s - t\alpha^2/4} Z_t(s - t\alpha/2).$$

It also follows directly by applying the convergent heat series to $D(e^{\alpha s} f) = e^{\alpha s}(D + \alpha) f$ and expanding the commuting operators D and the scalar α. Leibniz's rule therefore gives

$$0 = e^{\alpha\lambda - t\alpha^2/4} \sum_{j=0}^{d} c_j \sum_{k=0}^{j} \binom{j}{k} \alpha^{j-k} Z_t^{(k)}(\lambda - t\alpha/2). \tag{139}$$

The exponential is nonzero. Set $w = \lambda - t\alpha/2$, $b = 2/t$, and $x = w - \lambda$. Since $t \ne 0$, w ranges over all of $\mathbb{C}$, and $\alpha = -bx$. Substitution in (139) gives

$$\sum_{j=0}^{d} c_j \mathcal{N}_j Z_t = 0, \qquad \mathcal{N}_j = \sum_{k=0}^{j} \binom{j}{k} (-bx)^{j-k} D_w^k. \tag{140}$$

The polynomial factors in this formula multiply after the indicated derivatives are taken. They do not commute with D_w; in particular this formula is not an uncorrected power of $D_w - bx$.

Here is the exact correction at every order. Define

$$A = D_w - bx, \qquad H_j(X; b) = j! \sum_{\ell=0}^{\lfloor j/2 \rfloor} \frac{(b/2)^\ell X^{j-2\ell}}{\ell!(j-2\ell)!}. \tag{141}$$

Then $\mathcal{N}_j = H_j(A; b)$. To prove this coefficient identity on any entire f, use

$$G(u, w) = e^{-bu(w-\lambda)} f(w + u) = \sum_{j \geq 0} \frac{u^j}{j!} \mathcal{N}_j f(w).$$

Differentiating the left side in both variables gives $\partial_u G = (A + bu)G$. Comparing its Taylor coefficients yields

$$\mathcal{N}_0 = 1, \quad \mathcal{N}_1 = A, \quad \mathcal{N}_{j+1} = A\mathcal{N}_j + jb\mathcal{N}_{j-1} \quad (j \geq 1).$$

The explicitly displayed polynomials have exponential generating series $\exp(uX + bu^2/2)$ and hence satisfy the same recurrence and initial values. Induction proves the claimed identity. For example, $\mathcal{N}_2 = A^2 + b$ and $\mathcal{N}_3 = A^3 + 3bA$, including the terms created by noncommutation. This argument compares finite-order differential operators coefficient by coefficient and makes no claim that multiplication by an exponential quadratic preserves $\mathcal{H}_{\leq 1}$.

In the coordinate w, retain the proved conjugated source operator $B_t = w - tD_w/2 = U_t M U_{-t}$. Then

$$A = -\frac{2}{t}(B_t - \lambda), \qquad U_{-t} A U_t = -\frac{2}{t}(M - \lambda).$$

All the finite-order operators in (140) act continuously in γ. Applying U_{-t} to that identity and using $Z_t = U_t \Xi$ therefore gives the exact entire identity

$$\mathcal{P}(w)\Xi(w) = 0, \qquad \mathcal{P}(w) = \sum_{j=0}^{d} c_j H_j\left(-\frac{2}{t}(w - \lambda); \frac{2}{t}\right). \tag{142}$$

If c_J is the last nonzero coefficient, $H_J(X; b)$ is monic of degree J, while every preceding polynomial has smaller degree. Thus $\mathcal{P}$ has leading coefficient $c_J(-2/t)^J \neq 0$. But $\Xi(0) \neq 0$, so (142) forces $\mathcal{P}$ to vanish on a disk about zero and hence to be the zero polynomial, a contradiction. The annihilating functional is therefore zero.

A proper linear subspace of $\mathbb{C}^{d+1}$ has a nonzero annihilating functional: extend a basis of the subspace to a basis of the whole space and take a coordinate functional on one of the added basis vectors. The result just proved therefore gives the first equality in (137). Since every transported polynomial trace belongs to N_t, it gives the second equality too.

Here is a finite construction of the promised section. Enumerate $h_n = U_t(s^n \Xi)$, $n = 0, 1, \ldots$. Choose n_0 to be the first index whose jet column is nonzero, and successively choose n_ℓ to be the first index whose jet column is outside the span of the previously selected columns. Full rank, already proved, makes this selection terminate at $d + 1$ columns. Define

$$G_{j\ell} = D^j h_{n_\ell}(\lambda) \quad (0 \leq j, \ell \leq d), \qquad \mathcal{S}_{t,\lambda,d}(v) = \sum_{\ell=0}^{d} h_{n_\ell} (G^{-1} v)_\ell. \tag{143}$$

The determinant of G is nonzero by its construction; equivalently $G^{-1} = \operatorname{adj}(G)/\det G$. Direct multiplication gives $J_{\lambda,d} \mathcal{S}_{t,\lambda,d} = 1$. Its image is a finite-dimensional subspace of $U_t(\mathbb{C}[s]\Xi)$. Each scalar coefficient in the finite sum is a continuous linear function of v, and scalar multiplication and addition are continuous in the source vector topology τ_t. This proves continuity of the section into precisely $(\mathcal{H}_{\leq 1}, \tau_t)$, as claimed. □

There is an exact topological quotient for these representatives, without requiring the jet map itself to be τ_t-continuous. For the fixed $t \neq 0$, λ, and d above, put

$$K_{\lambda,d} = \ker J_{\lambda,d}, \qquad P_{t,\lambda,d} = 1 - \mathcal{S}_{t,\lambda,d} J_{\lambda,d}, \qquad N_{t,\lambda,d} = N_t \cap K_{\lambda,d},$$
$$\theta_{t,\lambda,d} = (P_{t,\lambda,d})_* \tau_t.$$

Here the last topology is the quotient topology on $K_{\lambda,d}$; a set is open exactly when its inverse image under $P_{t,\lambda,d}$ is τ_t-open. Since the section has image in N_t and is a right inverse, direct substitution gives

$$P_{t,\lambda,d}^2 = P_{t,\lambda,d}, \qquad P_{t,\lambda,d}(N_t) = N_{t,\lambda,d}, \qquad P_{t,\lambda,d}^{-1}(N_{t,\lambda,d}) = N_t.$$

The last equality proves that $N_{t,\lambda,d}$ is closed in the stated quotient topology. There is a homeomorphism

$$(K_{\lambda,d}, \theta_{t,\lambda,d})/N_{t,\lambda,d} \xrightarrow{[k]\mapsto[k]_{N_t}} Q_t, \qquad [f]_{N_t} \longmapsto [P_{t,\lambda,d} f] \tag{144}$$

as its inverse. Well-definedness and both inverse identities follow from the three displayed relations and $f - P_{t,\lambda,d} f \in N_t$. The forward map is continuous because its composition with the quotient map $P_{t,\lambda,d}$ and the further quotient by $N_{t,\lambda,d}$ is the original continuous quotient $\mathcal{H}_{\le 1} \to Q_t$. The reverse map is continuous because it is induced by that continuous composite quotient map out of $(\mathcal{H}_{\le 1}, \tau_t)$ with kernel N_t. As before, restriction of $P_{t,\lambda,d}$ to its range is the identity, proving the continuous identity map from the inherited-topology range to the quotient-topology range. This induces a continuous bijection of their quotients; inverse continuity for the inherited topology has not been inserted.

Finally apply the proposition with $\lambda = 0$ and $d = 2$. The explicit choice

$$\varepsilon_t = \mathcal{S}_{t,0,2}(0, 0, 16/15) \quad (t \neq 0), \qquad \varepsilon_0 = \frac{8s^2 \Xi(s)}{15\Xi(0)} \tag{145}$$

belongs to N_t and satisfies $\mathfrak{r}(\varepsilon_t) = 36\varepsilon_t'(0) + (15/16)\varepsilon_t''(0) = 1$. At $t = 0$ this is the previously computed polynomial trace; at every nonzero time the section fixes the two required derivatives to 0 and 16/15. Therefore, at every complex time including zero,

$$\mathfrak{r}(N_t) = \mathbb{C}, \qquad \{([f]_{N_t}, \mathfrak{r}(f)) : f \in \mathcal{H}_{\le 1}\} = Q_t \times \mathbb{C}. \tag{146}$$

For each pair $([f]_{N_t}, a)$ the exact representative $f + (a - \mathfrak{r}(f))\varepsilon_t$ proves surjectivity onto that pair. Put $P_t^{\mathfrak{r}} = 1 - \varepsilon_t \mathfrak{r}$ and retain $K = \ker \mathfrak{r}$. The identities

$$P_t^{\mathfrak{r}}(N_t) = N_t \cap K, \qquad (P_t^{\mathfrak{r}})^{-1}(N_t \cap K) = N_t$$

give, by the same two quotient-map arguments, the homeomorphism

$$(K, (P_t^{\mathfrak{r}})_* \tau_t)/(N_t \cap K) \cong Q_t, \qquad [k] \mapsto [k]_{N_t}, \quad [f]_{N_t} \mapsto [P_t^{\mathfrak{r}} f]. \tag{147}$$

This retains the actual transported source topology and the exact physical residue at each fixed time. No regular dependence of the selected finite pivot columns on t is needed or asserted. In particular, the full scalar ambiguity holds without a statement that the original frozen relation N_{ar} is invariant under heat, and without identifying that relation with $\Xi\mathcal{H}_{\le 1}$.

10 The Cartan construction and an exact antiunitary reversal

The supplied modular–Cartan manuscript [5] contains both a sector conjugation preserving a logarithmic generator and a standard-form involution reversing the left-minus-right generator. The following construction retains these two equations explicitly. It constructs the negative radial direction, its domains, and its maps to the arithmetic heat space and moving source quotients.

10.1 The generator equation determines the reflected parameter

Let $\mathscr{H}$ be a complex Hilbert space, K a self-adjoint operator, and E_K its projection-valued spectral measure. Define

$$\begin{aligned} T(\zeta) &= e^{-i\zeta K}, \qquad \zeta = a + ib \in \mathbb{C}, \\ \operatorname{Dom} T(\zeta) &= \left\{ f \in \mathscr{H} : \int_{\mathbb{R}} e^{2b\lambda}\, d\langle E_K(\lambda) f, f\rangle < \infty \right\}. \end{aligned} \tag{148}$$

This is a closed densely defined normal operator for every b, including every negative b. Density follows because the spectral cutoffs $E_K([-n, n])f$ belong to this domain and converge to f. Closedness follows from the spectral multiplication representation.

Proposition 10.1. *Let J be an antiunitary involution and suppose the equality of self-adjoint operators $JKJ = \epsilon K$ holds, where $\epsilon \in \{1, -1\}$; thus it includes equality of domains. Then*

$$JT(\zeta)J = T(-\epsilon\overline{\zeta}), \qquad Je^{zK}J = e^{\epsilon\overline{z}K}. \tag{149}$$

Both equalities include the transported operator domains. In particular, with $\Delta = e^{-K}$, the Tomita relation $J\Delta J = \Delta^{-1}$ is equivalent to $JKJ = -K$ and gives

$$J\Delta^{i(a+ib)}J = \Delta^{i(a-ib)}, \qquad Je^{bK}J = e^{-bK} \quad (b \in \mathbb{R}). \tag{150}$$

Proof. The uniqueness of the spectral resolution gives $JE_K(B)J = E_{\epsilon K}(B)$ for every Borel set B. For a simple scalar function ϕ, antilinearity gives $J\phi(K)J = \overline{\phi}(\epsilon K)$. Uniform approximation proves this for bounded Borel functions. For an arbitrary finite-valued Borel function, apply the bounded identity to $\phi\,\mathbf{1}_{\{|\phi|\le n\}}$ and pass to the spectral integral. The squared-integrability condition for $|\phi|$ is transported by J, proving equality of the full domains as well as of the values. Taking $\phi(\lambda) = e^{-i\zeta\lambda}$ and $e^{z\lambda}$ proves (149). Applying the same real-valued functional calculus to $-\log$ proves the equivalence involving Δ, whose kernel is zero. Substitution of $\epsilon = -1$ proves (150). $\square$

The Cartan logarithm in the supplied construction is

$$\mathbf{L}_\ell(T) = \frac{1}{2\ell}\log(T^*T), \qquad \ell > 0.$$

For the operator in (148), multiplication of the spectral functions, including the product domain, gives $T(a+ib)^*T(a+ib) = e^{2bK}$. Indeed the product domain is precisely the set on which the squared multiplier $e^{2b\lambda}$ is square-integrable; its integrability also implies that of $e^{b\lambda}$ by Cauchy–Schwarz. Consequently,

$$\mathbf{L}_\ell(T(a+ib)) = \begin{cases} (b/\ell)K & b \neq 0, \quad \text{with domain } \operatorname{Dom} K, \\ 0 & b = 0, \quad \text{with domain } \mathscr{H}. \end{cases} \tag{151}$$

This follows by composing the real spectral functions $\log(e^{2b\lambda}) = 2b\lambda$. The zero slice has been stated separately: the everywhere-defined $\log I$ must not be replaced by a zero operator restricted to $\operatorname{Dom} K$.

10.2 The original heat multiplier and a genuine modular double

Retain the original Fourier variable v and set

$$\begin{aligned}\mathscr{H}_0 &= L^2(\mathbb{R}, dv), \qquad (C_0 f)(v) = \overline{f(-v)},\\ (Af)(v) &= v^2 f(v), \qquad \operatorname{Dom} A = \{f : v^2 f \in L^2\},\\ G_t &= e^{tA/4},\\ \operatorname{Dom} G_t &= \left\{ f : \int_{\mathbb{R}} e^{(\operatorname{Re} t)v^2/2} |f(v)|^2 \, dv < \infty \right\}.\end{aligned} \tag{152}$$

The multiplication operator A is nonnegative self-adjoint, C_0 is an antiunitary involution, and $C_0 A C_0 = A$ with the displayed domain. Thus $C_0 G_t C_0 = G_{\bar{t}}$, which fixes each real heat parameter. For $\operatorname{Re} t \le 0$, G_t is bounded with norm one. For $\operatorname{Re} t > 0$, it is densely defined and unbounded: unit vectors supported in $[n, n+1]$ have image norm at least $e^{(\operatorname{Re} t)n^2/4}$. Its domain is proper, since $f(v) = e^{-(\operatorname{Re} t)v^2/8}$ belongs to L^2 and not to that domain. Multiplication gives the exact inverse statement

$$\operatorname{Ran} G_t = \operatorname{Dom} G_{-t}, \qquad G_{-t} G_t f = f \quad (f \in \operatorname{Dom} G_t). \tag{153}$$

For the converse inclusion in the range statement, take $f = G_{-t} g$ for $g \in \operatorname{Dom} G_{-t}$; then $G_t f = g \in L^2$, proving $f \in \operatorname{Dom} G_t$. The common invariant space $\mathscr{D} = C_c^\infty(\mathbb{R})$ is a graph core for each G_t: truncate a vector in its weighted L^2 graph norm, and approximate the truncations by smooth compact functions, using equivalence of the weighted and unweighted norms on each fixed compact interval.

There is no antiunitary J_0 on this one sector satisfying $J_0 A J_0 = -A$. Such an equality would make both A and $-A$ nonnegative by preservation of real quadratic forms, forcing $\langle Af, f\rangle = 0$ for every $f \in \operatorname{Dom} A$. A nonzero smooth function supported in $(1, 2)$ contradicts this. The exact construction retaining this original A is as follows.

Theorem 10.2. *On $\widehat{\mathscr{H}} = \mathscr{H}_0 \oplus \mathscr{H}_0$, define*

$$\begin{aligned}\widehat{K} &= \begin{pmatrix} A & 0 \\ 0 & -A \end{pmatrix}, \quad \widehat{\Delta} = \begin{pmatrix} e^{-A} & 0 \\ 0 & e^{A} \end{pmatrix},\\ J(f, g) &= (C_0 g, C_0 f), \quad \widehat{G}_t = e^{t\widehat{K}/4} = \begin{pmatrix} G_t & 0 \\ 0 & G_{-t} \end{pmatrix}.\end{aligned} \tag{154}$$

Here $\operatorname{Dom} \widehat{K} = \operatorname{Dom} A \oplus \operatorname{Dom} A$, $\operatorname{Dom} \widehat{\Delta} = \mathscr{H}_0 \oplus \operatorname{Dom} e^{A}$, *and* $\operatorname{Dom} \widehat{G}_t = \operatorname{Dom} G_t \oplus \operatorname{Dom} G_{-t}$. *Then*

$$J^2 = 1, \quad J\widehat{K}J = -\widehat{K}, \quad J\widehat{\Delta}J = \widehat{\Delta}^{-1}, \quad J\widehat{G}_t J = \widehat{G}_{-\bar{t}}. \tag{155}$$

These are the modular data of an explicitly constructed standard closed real subspace of $\widehat{\mathscr{H}}$.

Proof. The block formulas and $C_0 A C_0 = A$ prove all four identities, including their domains. For the last assertion put $Q = e^{-A/2}$. This is a bounded positive injective operator, with dense range, commuting with C_0. Define the closed antilinear operator

$$S = J\widehat{\Delta}^{1/2}, \qquad \operatorname{Dom} S = \mathscr{H}_0 \oplus \operatorname{Dom} Q^{-1}, \qquad S(f, g) = (C_0 Q^{-1} g, C_0 Q f).$$

Its domain is invariant under S, and direct substitution gives $S^2(f, g) = (f, g)$. Its fixed real subspace is exactly

$$V = \{(f, C_0 Q f) : f \in \mathscr{H}_0\}. \tag{156}$$

Both implications follow by substituting the second component in $S(f,g)=(f,g)$. The graph defining V is closed because C_0Q is bounded. If a vector in V equals i times a vector in V, its first components give $f=ig$, and its second components give $-iC_0Qg=iC_0Qg$; injectivity of Q gives $f=g=0$. Hence $V\cap iV=\{0\}$.

Moreover $V+iV=\mathscr{H}_0\oplus\operatorname{Ran}Q=\operatorname{Dom}S$. To prove the nontrivial inclusion, for $x\in\mathscr{H}_0$ and $y\in\operatorname{Ran}Q$ put $b=C_0Q^{-1}y$, $f=(x+b)/2$ and $g=(x-b)/(2i)$. Then

$$(f,C_0Qf)+i(g,C_0Qg)=(f+ig,C_0Q(f-ig))=(x,y).$$

This domain is dense since $\operatorname{Ran}Q$ is dense. Thus V is standard: it is closed real, has zero intersection with iV, and has dense complex span. On that span $S(v+iw)=v-iw$ for $v,w\in V$, as follows from antilinearity and $S|_V=1$. Its polar decomposition is exactly $S=J\widehat{\Delta}^{1/2}$. This proves the claimed modular real-subspace construction, including its canonical involution. $\square$

For $\widehat{T}(\zeta)=e^{-i\zeta\widehat{K}}$, the Cartan logarithm is (151) with $K=\widehat{K}$, and $\widehat{G}_t=\widehat{T}(it/4)$. In particular the involution sends every real t to $-t$. For nonzero real t, the double is unbounded in one of its two sectors; its domain remains dense. The inclusion $I_+f=(f,0)$ retains the original multiplier exactly:

$$\widehat{G}_tI_+f=I_+G_tf\quad(f\in\operatorname{Dom}G_t).$$

No Hilbert completion of the unnamed arithmetic topology is required for this construction.

10.3 Exact ingress and involution on the actual moving quotients

On the original entire-function space $\mathcal{H}_{\le 1}$ put $(C_Eh)(s)=\overline{h(\overline{s})}$. This is an antilinear involution and preserves each previously defined growth seminorm, since $s\mapsto\overline{s}$ preserves $|s|$. For the Fourier transform already used in the test ingress,

$$(F\phi)(s)=\int_{\mathbb{R}}\phi(v)e^{-isv}\,dv,$$

the change of variable $v\mapsto -v$ proves $C_EF=FC_0$ on $\mathscr{D}$. Differentiating the integral and summing the absolutely uniformly convergent exponential series on the compact support proves

$$U_tF=FG_t,\qquad C_EU_t=U_{\overline{t}}C_E. \tag{157}$$

The second equality also follows directly from the convergent entire-space series for U_t and the real coefficient $-1/4$. Thus on $\mathcal{H}_{\le 1}\oplus\mathcal{H}_{\le 1}$ the everywhere-defined maps

$$\widehat{U}_t=(U_t,U_{-t}),\qquad \mathcal{J}(h,k)=(C_Ek,C_Eh)$$

satisfy $\mathcal{J}\widehat{U}_t\mathcal{J}=\widehat{U}_{-\overline{t}}$ for every complex t. The exact diagram is expressed by

$$(F\oplus F)\widehat{G}_t=\widehat{U}_t(F\oplus F),\qquad (F\oplus F)J=\mathcal{J}(F\oplus F)\quad\text{on }\mathscr{D}\oplus\mathscr{D}.$$

Here $\widehat{G}_t$ on the test space is the restriction of the closed Hilbert operator constructed above.

Keep the actual source topology τ and actual closed trace image N_{ar}. Define its conjugate presentation by $\tau^c=(C_E)_*\tau$ and $N^c=C_EN_{\mathrm{ar}}$. Since Ξ is real entire, $C_E(\mathbb{C}[s]\Xi)=\mathbb{C}[s]\Xi$, and therefore $N^c=\overline{\mathbb{C}[s]\Xi}^{\tau^c}$. This defines the conjugate topology explicitly without assuming C_E is continuous for the unnamed original τ.

Proposition 10.3. *For each complex t form the topological quotient*

$$\begin{aligned}&\widehat{N}_t = U_t N_{\mathrm{ar}} \oplus U_{-t} N^c,\\ &\widehat{Q}_t = \big((\mathcal{H}_{\le 1}, \tau_t) \oplus (\mathcal{H}_{\le 1}, (\tau^c)_{-t})\big)/\widehat{N}_t, \qquad \tau_t = (U_t)_*\tau, \quad (\tau^c)_{-t} = (U_{-t})_*\tau^c.\end{aligned} \tag{158}$$

The map $\widehat{U}_t$ induces a complex-linear homeomorphism $\overline{\widehat{U}}_t : \widehat{Q}_0 \to \widehat{Q}_t$. The involution $\mathcal{J}$ induces an antilinear homeomorphism

$$\overline{\mathcal{J}}_t : \widehat{Q}_t \longrightarrow \widehat{Q}_{-\bar{t}}, \qquad \overline{\mathcal{J}}_{-\bar{t}}\overline{\mathcal{J}}_t = 1, \qquad \overline{\mathcal{J}}_t\overline{\widehat{U}}_t = \overline{\widehat{U}}_{-\bar{t}}\overline{\mathcal{J}}_0. \tag{159}$$

For $\iota_t(\phi, \psi) = [F\phi, F\psi] \in \widehat{Q}_t$,

$$\begin{aligned}\ker \iota_t &= F^{-1}(U_t N_{\mathrm{ar}}) \oplus F^{-1}(U_{-t} N^c),\\ \iota_t \widehat{G}_t &= \overline{\widehat{U}}_t \iota_0, \qquad \overline{\mathcal{J}}_t \iota_t = \iota_{-\bar{t}} J.\end{aligned} \tag{160}$$

The inverse images here are taken inside $\mathscr{D}$.

Proof. The direct-sum heat map transports both the specified topology and the specified closed subspace exactly, proving its quotient homeomorphism. Next

$$C_E U_{-t} N^c = U_{-\bar{t}} N_{\mathrm{ar}}, \qquad C_E U_t N_{\mathrm{ar}} = U_{\bar{t}} N^c.$$

The same identities applied to the definitions of the transported topologies prove that $\mathcal{J}$ is an antilinear homeomorphism of the two ambient spaces in (158). It transports their closed relation spaces exactly, so it induces the asserted quotient map. Applying it twice is the identity on representatives. The conjugacy follows from $\mathcal{J}\widehat{U}_t = \widehat{U}_{-\bar{t}}\mathcal{J}$. The kernel formula is exactly the condition that both Fourier components lie in the two relation spaces. Finally (157) proves both equations in (160) on representatives. □

Thus the constructed Hilbert involution is antiunitary and its actual source-quotient image is a proved antilinear homeomorphism. No inner product is asserted on $\widehat{Q}_t$. For real t, this construction provides the requested negative parameter and its exact quotient target. It does not assert the distinct frozen inclusion $U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$.

10.4 The exact viscosity sign and tensor Fourier map

Let $\nu > 0$ be the original viscosity and let $\theta \ge 0$ be physical elapsed time. On $L^2(\mathbb{R}^3, dq)$ define $A_\nu = -\nu\Delta_q$ by the unitary Fourier transform

$$(\mathcal{F}_3 f)(k) = (2\pi)^{-3/2} \int_{\mathbb{R}^3} f(q) e^{-ik\cdot q}\, dq.$$

Its domain is exactly $\{f : |k|^2\mathcal{F}_3 f \in L^2\}$, and its multiplier is $\nu|k|^2$. Integration by parts verifies this differential expression on Schwartz functions, and the spectral multiplication domain supplies the closed realization. On the frequency tensor product $L^2(\mathbb{R})^{\otimes 3} = L^2(\mathbb{R}^3)$ one has

$$\mathcal{F}_3 e^{-\theta A_\nu} \mathcal{F}_3^{-1} = e^{-\nu\theta|k|^2} = G_{-4\nu\theta} \otimes G_{-4\nu\theta} \otimes G_{-4\nu\theta}. \tag{161}$$

The equality holds first on simple tensors, where multiplication of the three factors gives the middle multiplier, and then on all of L^2 by their common boundedness and tensor density. On $\mathscr{D}^{\otimes 3}$ the original unnormalized Fourier transform $F^{\otimes 3}$ is $(2\pi)^{3/2}\mathcal{F}_3$ with the variables renamed; this scalar is retained. Each transformed test is entire in the three coordinates, and (157) gives $U^{\otimes 3}_{-4\nu\theta}$ on that entire tensor test space. This proves the sign and coefficient in the viscous bridge. The multiplier also acts componentwise on vector fields and preserves the divergence-free subspace: in Fourier coordinates $k \cdot \widehat{u}(k) = 0$ is unchanged by scalar multiplication. Its inverse at $\theta > 0$ has precisely the domain $\{f : e^{\nu\theta|k|^2}\mathcal{F}_3 f \in L^2\}$. These identities identify the linear viscous operator, including its inverse domain; they do not replace the nonlinear velocity, pressure, or forcing equations.

10.5 The public viscous source and its exact proof status

The inspected public manuscript [6] has two preserved revisions. Its earlier 165-page PDF and its later 166-page PDF have different recorded byte hashes. The later revision is the one served after 19:09:35 UTC on 8 September 2026; its first three pages were inspected directly. No equivalence of the two complete proofs is asserted. The pinned source repository commit is recorded separately in the bibliography and provenance.

Theorem 1.1 in that manuscript is an imported source claim: for every $\nu > 0$ it asserts existence of $f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty); \mathbb{R}^3)$, a compact spatial set K, and smooth u, p on $\mathbb{R}^3 \times [0, 1)$ satisfying

$$\partial_\theta u + (u \cdot \nabla)u - \nu\Delta u + \nabla p = f, \qquad \nabla \cdot u = 0, \qquad u(\cdot, 0) = 0,$$

with the support and norm conditions

$$\operatorname{supp} u(\cdot,\theta) \cup \operatorname{supp} p(\cdot,\theta) \subseteq K, \qquad \sup_{0\le\theta<1} \|u(\cdot,\theta)\|_2 < \infty, \qquad \limsup_{\theta\uparrow 1} \|u(\cdot,\theta)\|_\infty = \infty.$$

It also asserts nonexistence of a global smooth competitor with the same force and initial datum and uniformly bounded kinetic energy. These assertions have not been independently proved in this fragment or checked here by a Lean kernel. The identities in (161) are independently proved above and match the exact ordinary Laplacian and positive viscosity in this source equation.

The updated manuscript credits the preceding constructions of Diego Córdoba and Luis Martínez-Zoroa, and the hypodissipative work with Zheng. Separately, the primary statement [7] attributes announced smooth-forcing results for incompressible porous media, Boussinesq, and three-dimensional incompressible Euler to Levent Alpöge and Tristan Buckmaster, and credits the Córdoba–Martínez-Zoroa programme. Those are distinct source attributions. The operator calculations here do not decide any private-data or appropriation allegation.

10.6 The retained spatial pullback, exact diffusion leakage, and feedback

Keep the original Cartesian coordinates $q = (x, y, w)$ and polynomials

$$Q = 1 + xy, \qquad \sigma(q) = \tfrac{1}{2}\big[Q^3 w + y^2 Q(4 + 3xy)\big], \qquad \eta(s) = (0, 0, 2s).$$

Put $\mathscr{H} = \mathcal{O}(\mathbb{C}_s)$ and $\mathscr{G} = \mathcal{O}(\mathbb{C}^3_q)$, with uniform convergence on compact sets, and define the actual pullback and section maps

$$\mathcal{I} = \sigma^* : \mathscr{H} \longrightarrow \mathscr{G}, \qquad E_\eta = \eta^* : \mathscr{G} \longrightarrow \mathscr{H}.$$

Substitution gives $\sigma\eta(s) = s$, so $E_\eta \mathcal{I} = I$. If $K \subset \mathbb{C}^3$ is compact, then $\sigma(K)$ is compact and $\sup_K |\mathcal{I}h| = \sup_{\sigma(K)} |h|$. The corresponding identity for η proves continuity of both maps. Thus $P = \mathcal{I}E_\eta$ is a continuous projection and

$$\mathscr{G} = \mathcal{I}\mathscr{H} \oplus \ker E_\eta. \tag{162}$$

Indeed $F = \mathcal{I}E_\eta F + (F - \mathcal{I}E_\eta F)$; the second summand is in $\ker E_\eta$, and $\mathcal{I}h \in \ker E_\eta$ implies $h = 0$. These decomposition maps and their inverse, addition, are continuous. In particular $\mathcal{I}\mathscr{H} = \ker(I - P)$ is closed. This analytic splitting does not identify the source topology τ with a compact-open topology.

Retain

$$D = \partial_s, \quad \Delta = \partial_x^2 + \partial_y^2 + \partial_w^2, \quad g = \nabla\sigma, \quad N = \sum_{i=1}^{3} g_i^2, \quad L = \Delta\sigma.$$

The sum defining N is polynomial over $\mathbb{C}$ and is the Euclidean squared norm on real q. Coordinate differentiation gives

$$\partial_i(\mathcal{I}h) = g_i\mathcal{I}h', \qquad \partial_i^2(\mathcal{I}h) = g_i^2\mathcal{I}h'' + (\partial_i^2\sigma)\mathcal{I}h', \qquad \Delta\mathcal{I}h = N\mathcal{I}h'' + L\mathcal{I}h'.$$

The full coefficients before any evaluation are

$$\begin{aligned}
g_1 &= \tfrac{3}{2}yQ^2w + \tfrac{1}{2}y^3(7 + 6xy),\\
g_2 &= \tfrac{3}{2}xQ^2w + \tfrac{1}{2}(8y + 21xy^2 + 12x^2y^3),\\
g_3 &= \tfrac{1}{2}Q^3,\\
L &= 3wQ(x^2 + y^2) + 3y^4 + 18x^2y^2 + 21xy + 4.
\end{aligned}$$

At $\eta(s)$ these give $g = (0, 0, 1/2)$, $N = 1/4$, $L = 4$. Consequently the compressed operator and the full complementary component are

$$A = E_\eta\Delta\mathcal{I} = \tfrac{1}{4}D^2 + 4D : \mathscr{H} \longrightarrow \mathscr{H}, \tag{163}$$

$$W = (I - P)\Delta\mathcal{I}, \qquad Wh = (N - \tfrac{1}{4})\mathcal{I}h'' + (L - 4)\mathcal{I}h' : \mathscr{H} \longrightarrow \ker E_\eta, \tag{164}$$

$$\Delta\mathcal{I} = \mathcal{I}A + W.$$

Every map is continuous because composition, differentiation, and multiplication by a fixed polynomial are continuous on the specified spaces.

Proposition 10.4. *The exact kernel of W is the constant functions. Equivalently,*

$$\{h \in \mathscr{H} : \Delta\mathcal{I}h \in \mathcal{I}\mathscr{H}\} = \mathbb{C}.$$

Proof. On the entire two-parameter family $(x, 0, 2s)$, direct substitution gives

$$\sigma(x, 0, 2s) = s, \quad N(x, 0, 2s) = \tfrac{1}{4} + 9x^2s^2, \quad L(x, 0, 2s) = 4 + 6x^2s.$$

Thus $Wh = 0$, evaluated with $x = 1$, implies $9s^2h'' + 6sh' = 0$ for every $s \in \mathbb{C}$. Write the convergent entire Taylor series $h(s) = \sum_{n\geq 0} a_n s^n$. The coefficient of s^n in this identity is $3n(3n - 1)a_n$. For each integer $n \geq 1$ its scalar factor is nonzero, so every such a_n vanishes. Conversely all derivatives of a constant vanish and therefore $Wh = 0$. Finally, by the splitting, $\Delta\mathcal{I}h \in \mathcal{I}\mathscr{H}$ is equivalent to $(I - P)\Delta\mathcal{I}h = 0$. $\square$

The splitting also supplies a derivative extension with an explicit common domain and codomain:

$$B = \mathcal{I}DE_\eta : \mathscr{G} \longrightarrow \mathscr{G}, \qquad B\mathcal{I} = \mathcal{I}D.$$

The commutator with the Euclidean Laplacian is consequently well typed. For every $h \in \mathscr{H}$ it is

$$\begin{aligned}[\Delta, B]\mathcal{I}h &= \Delta\mathcal{I}h' - \mathcal{I}D(E_\eta \Delta \mathcal{I}h) \\ &= N\mathcal{I}h''' + L\mathcal{I}h'' - \mathcal{I}(\tfrac{1}{4}h''' + 4h'') \\ &= (N - \tfrac{1}{4})\mathcal{I}h''' + (L-4)\mathcal{I}h'' = W(h'). \end{aligned} \tag{165}$$

The just-proved kernel formula gives the exact consequence $[\Delta, B]\mathcal{I}h = 0$ if and only if $h(s) = a+bs$ for some $a, b \in \mathbb{C}$. This conclusion requires no invariance of $\mathcal{I}\mathscr{H}$ under Δ.

Compression does not reproduce repeated diffusion. To compute the feedback exactly, write $H = D_q^2\sigma$. The full product rule gives

$$\begin{aligned} \Delta(N\mathcal{I}h'') &= (\Delta N)\mathcal{I}h'' + 2(g\cdot\nabla N)\mathcal{I}h^{(3)} + NL\mathcal{I}h^{(3)} + N^2\mathcal{I}h^{(4)}, \\ \Delta(L\mathcal{I}h') &= (\Delta L)\mathcal{I}h' + 2(g\cdot\nabla L)\mathcal{I}h'' + L^2\mathcal{I}h'' + LN\mathcal{I}h^{(3)}, \\ \partial_j N &= 2\sum_i g_i H_{ij}, \\ \Delta N &= 2\sum_{i,j} H_{ij}^2 + 2g\cdot\nabla L, \qquad g\cdot\nabla N = 2g^{\mathsf{T}}Hg. \end{aligned}$$

Adding gives the identity on every entire h,

$$\begin{aligned} \Delta^2\mathcal{I}h = {} & N^2\mathcal{I}h^{(4)} + (4g^{\mathsf{T}}Hg + 2NL)\mathcal{I}h^{(3)} \\ & + (2\sum_{i,j} H_{ij}^2 + 4g\cdot\nabla L + L^2)\mathcal{I}h'' + (\Delta L)\mathcal{I}h'. \end{aligned}$$

Differentiating the original g_i, L and evaluating at the same section $\eta(s)$ gives

$$H(\eta(s)) = \begin{pmatrix} 0 & 3s & 0 \\ 3s & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \sum_{i,j} H_{ij}(\eta(s))^2 = 18s^2 + 16,$$

$$(g^{\mathsf{T}}Hg)(\eta(s)) = 0, \quad (g\cdot\nabla L)(\eta(s)) = 0, \quad (\Delta L)(\eta(s)) = 24s.$$

For the last equality, the term $3w(x^2+y^2)$ in L contributes $12w = 24s$ after the two derivatives and substitution; all other terms give zero there. Therefore

$$\begin{aligned} E_\eta\Delta^2\mathcal{I} &= \tfrac{1}{16}D^4 + 2D^3 + (36s^2+48)D^2 + 24sD, \\ E_\eta\Delta^2\mathcal{I} - A^2 &= (36s^2+32)D^2 + 24sD = E_\eta\Delta W. \end{aligned} \tag{166}$$

The last equality also follows directly from $\Delta\mathcal{I} = \mathcal{I}A + W$. It retains the component that leaves the pullback subspace and returns at the second diffusion. It is an identity of differential operators; no Euclidean heat semigroup on all of $\mathscr{G}$ is presumed.

Finally let $h \in \mathcal{H}_{\le 1}$ and use the established entire evolution $h_t = U_t h$, $\partial_t h_t = -D^2 h_t/4$. For $\nu > 0$ put

$$f(\tau, q) = h_{-4\nu\tau}(\sigma(q)).$$

The time chain rule gives $\partial_\tau f = \nu h''_{-4\nu\tau}(\sigma(q))$. Thus the exact scalar viscous residual is

$$(\partial_\tau - \nu\Delta)f = \nu\big[(1-N(q))h''_{-4\nu\tau}(\sigma(q)) - L(q)h'_{-4\nu\tau}(\sigma(q))\big]. \tag{167}$$

For a specified smooth spatial advection field V, its complete additional term is $(V\cdot\nabla\sigma)h'_{-4\nu\tau}(\sigma(q))$. In particular the retained material identity $V\cdot\nabla\sigma=1$ gives coefficient $1-\nu L$ for the first derivative. Setting $h=\Xi$ recovers $Z_{-4\nu\tau}$ with the factor -4ν and the original Cartesian metric intact. These are scalar equations with explicitly computed forcing. They do not assert the vector momentum equation, its pressure, divergence constraint, or the support, initial-value, and energy conditions of a Navier–Stokes solution.

11 The published fluid input and its complete arithmetic profile map

We now apply the public fluid theorem recorded in Section 10.5 through a proved map. The complete companion calculation [8] was read. The construction and every estimate used here are proved below, including the polynomial coordinates, real flow, pressure, forcing, physical metric, arithmetic Fourier constants, and inverse. The final existence assertion cites the public Theorem 1.1 as an external mathematical input; it is not an independent verification of that theorem's complete proof.

11.1 The original three coordinates and their full inverse differential

In the original real coordinates $\mathbf{x}=(x,y,w)$ put

$$\begin{aligned}F_1&=(1+xy)^3w+y^2(1+xy)(4+3xy),\\F_2&=y+3x(1+xy)^2w+3xy^2(4+3xy),\\F_3&=2x-3x^2y-x^3w,\\\mathbf{S}&=(F_1/2,F_2,F_3)^{\mathsf T},\qquad J_S=D\mathbf{S}.\end{aligned}\tag{168}$$

Thus $S_1=\sigma$ is exactly the earlier spatial coordinate. The bold notation distinguishes this spatial three-vector from the original quotient multiplication operator S.

Here is a complete determinant calculation. On $x\neq0$ set $v=1+xy$, $h=2-3xy-x^2w$, and $A=v^2+v-v^3h$, $B=4v+2-3v^2h$. Direct substitution gives $F=(A/x^2,B/x,xh)$ and

$$\det\frac{\partial(x,v,h)}{\partial(x,y,w)}=-x^3,\qquad \det\frac{\partial F}{\partial(x,v,h)}=x^{-3}\det\begin{pmatrix}-2A&2v+1-3v^2h&-v^3\\-B&4-6vh&-3v^2\\h&0&1\end{pmatrix}=2x^{-3}.$$

For the last equality the determinant's numerator is

$$(4-6vh)(-2v^2-2v+3v^3h)+(2v+1-3v^2h)(4v+2-6v^2h)=2,$$

as expansion verifies. Thus $\det DF=-2$ on $x\neq0$ and, by polynomial identity, everywhere. Consequently $\det J_S=-1$ everywhere. Define the polynomial columns

$$\begin{aligned}W_1&=-\tfrac12\nabla F_2\times\nabla F_3,\quad W_2=-\tfrac12\nabla F_3\times\nabla F_1,\quad W_3=-\tfrac12\nabla F_1\times\nabla F_2,\\(DF)^{-1}&=[\,W_1\ W_2\ W_3\,],\qquad J_S^{-1}=[\,2W_1\ W_2\ W_3\,].\end{aligned}\tag{169}$$

The scalar triple products with the three rows ∇F_i give $DF\,W_i=e_i$, proving both inverse identities. In particular $\ker(d\sigma)_{\mathbf{x}}=\operatorname{span}_{\mathbb{R}}\{W_2(\mathbf{x}),W_3(\mathbf{x})\}$: write any vector in the basis (W_1,W_2,W_3) and use $d\sigma(W_i)=\delta_{1i}/2$. No global inverse of F or $\mathbf{S}$ is presumed.

11.2 The complete real flow and its volume-preserving inverse

Fix $K \subset \mathbb{R}^3$ compact, $0 < T < \infty$ and $\nu > 0$. Let $\mathcal{U}_{K,T,\nu}$ consist of smooth real triples (u, p, f) on $[0, T) \times \mathbb{R}^3$ satisfying

$$\partial_t u_i + \sum_{\ell=1}^{3} u_\ell \partial_\ell u_i = \nu \Delta_{\mathbf{x}} u_i - \partial_i p + f_i, \qquad \sum_i \partial_i u_i = 0, \qquad \operatorname{supp} u(t, \cdot) \subseteq K. \tag{170}$$

There is no restriction on p, f beyond this definition for the correspondence; the public theorem supplies the stronger support and initial conditions in its stated example. For any such triple define

$$\partial_t X_t(q) = u(t, X_t(q)), \qquad X_0(q) = q, \qquad A_X(t, q) = D_q X_t(q), \qquad q \in \mathbb{R}^3.$$

For each $T_0 < T$, every spatial derivative of u is bounded on $[0, T_0] \times K$ and vanishes outside K. In particular its global spatial Lipschitz constant is uniform on this time interval. On an interval of length less than its reciprocal, the integral map $Y \mapsto q + \int u(s, Y(s))\, ds$ is a contraction on continuous curves in the uniform norm. Successive such intervals construct the solution to T_0; boundedness of u prevents finite escape. The same construction backward from each endpoint constructs its inverse. Differentiating the integral equation gives a linear integral equation for each first label derivative, and then successively for every higher label and time derivative. The same local contraction and induction prove smoothness and uniqueness of these derivatives. Thus X_t is a smooth global diffeomorphism for every $t < T$.

Differentiation and the determinant product rule give

$$\partial_t A_X = Du(t, X_t) A_X, \quad A_X(0, q) = I_3, \qquad \partial_t \det A_X = (\operatorname{div} u)(t, X_t) \det A_X = 0.$$

The determinant identity first holds where the matrix is invertible, including a neighbourhood of $t = 0$; its value one prevents any first loss of invertibility and continues the identity. Hence $\det A_X = 1$ with positive orientation. Every $q \notin K$ has its constant trajectory. An endpoint outside K also has a constant backward trajectory, so inverse uniqueness proves

$$X_t(K) = K, \qquad X_t(q) = q \ (q \notin K), \qquad X_t(L) \subseteq K \cup L \quad (L \subset \mathbb{R}^3 \text{ compact}). \tag{171}$$

All these conclusions hold on each preterminal interval without a uniform bound on $|u|$ as t approaches T.

11.3 The full complex fibre and both time derivatives

Set $a_j(t, q) = S_j(X_t(q))$ and $v_j(t, q) = (\nabla S_j \cdot u)(t, X_t(q))$. Then a_j is real and $\partial_t a_j = v_j$. At fixed t the map on the actual real and complex variables is

$$(q, \zeta) \longmapsto (q, a_j(t, q) + \zeta), \qquad (q, s) \longmapsto (q, s - a_j(t, q)).$$

These are exact inverse maps, smooth in q and holomorphic on each complex fibre. Write $\mathcal{B} = C^\infty(\mathbb{R}^3; \mathcal{O}(\mathbb{C}_\zeta))$ with seminorms consisting of suprema of finitely many label and spectral derivatives on compact products. The map

$$\mathcal{P}_t : \mathcal{O}(\mathbb{C})^3 \longrightarrow \mathcal{B}^3, \qquad \mathcal{P}_t(h_j)_j = (h_j(a_j(t, q) + \zeta))_j$$

has the left inverse, for any fixed q_*, $(\mathcal{R}_t G)_j(s) = G_j(q_*, s - a_j(t, q_*))$. Substitution proves $\mathcal{R}_t \mathcal{P}_t = I$. Images of compact products have compact spectral argument; the chain rule involves only finitely many bounded derivatives of a_j there. This proves continuity of $\mathcal{P}_t$, while evaluation and translation prove continuity of $\mathcal{R}_t$. Therefore $\mathcal{P}_t$ is a topological isomorphism onto its image with the inherited topology. At $\zeta = 0$ its first component is exactly $h_1(\sigma(X_t(q))) = X_t^* \sigma^* h_1(q)$.

Keep a smooth, explicitly chosen real Newman-time function $\vartheta : [0, T) \to \mathbb{R}$, independently of physical t. For either original profile $h_\theta = Z_\theta$ or K_θ, define

$$\Psi_j(t, q, \zeta) = h_{\vartheta(t)}(\zeta + a_j(t, q)).$$

The previously proved heat identity gives

$$\partial_t \Psi_j = -\frac{\vartheta'}{4} \partial_\zeta^2 \Psi_j + v_j \partial_\zeta \Psi_j. \tag{172}$$

Since v_j has no spectral dependence, another complete differentiation gives

$$\partial_t^2 \Psi_j = \frac{(\vartheta')^2}{16} \partial_\zeta^4 \Psi_j - \frac{\vartheta' v_j}{2} \partial_\zeta^3 \Psi_j + \left(v_j^2 - \frac{\vartheta''}{4}\right) \partial_\zeta^2 \Psi_j + b_j \partial_\zeta \Psi_j, \tag{173}$$

where, with every original momentum term,

$$b_j = \partial_t v_j = \left[\sum_i (\nu \Delta u_i - \partial_i p + f_i) \partial_i S_j + \sum_{i,\ell} u_i u_\ell \partial_i \partial_\ell S_j\right] (t, X_t(q)). \tag{174}$$

Indeed the material derivative of $\nabla S_j \cdot u$ gives the Hessian term and the full material acceleration of u; substituting (170) gives (174). In the second derivative the mixed cubic term occurs twice, each with coefficient $-\vartheta' v_j/4$, proving the factor $-1/2$.

The full Euclidean viscous operator in label coordinates is

$$G_X = A_X^{-1} A_X^{-\mathsf{T}}, \qquad \mathcal{L}_t = \sum_{\alpha,\beta} \partial_{q_\alpha} (G_X^{\alpha\beta} \partial_{q_\beta}).$$

For every smooth scalar b one has $\mathcal{L}_t X_t^* b = X_t^* \Delta b$. To prove it with all first-order coefficient derivatives, test against $\phi \in C_c^\infty(\mathbb{R}^3)$ and use the gradient chain rule and the change of variables with determinant one:

$$\begin{aligned}\int \phi\, \mathcal{L}_t X_t^* b \, dq &= -\int \nabla_q \phi \cdot A_X^{-1} (\nabla b) \circ X_t \, dq \\ &= -\int \nabla_{\mathbf{x}} (\phi \circ X_t^{-1}) \cdot \nabla b \, d\mathbf{x} = \int \phi\, X_t^* \Delta b \, dq.\end{aligned}$$

Smoothness makes this distributional identity pointwise. The ordinary spatial chain rule now proves

$$\begin{gathered}\mathcal{L}_t \Psi_j = (|\nabla S_j|^2 \circ X_t) \partial_\zeta^2 \Psi_j + (\Delta S_j \circ X_t) \partial_\zeta \Psi_j, \\ (\partial_t - \nu \mathcal{L}_t) \Psi_j = -\left(\frac{\vartheta'}{4} + \nu |\nabla S_j|^2 \circ X_t\right) \partial_\zeta^2 \Psi_j + (v_j - \nu \Delta S_j \circ X_t) \partial_\zeta \Psi_j.\end{gathered} \tag{175}$$

11.4 Original Fourier constants and exact velocity recovery

Define, with the original arithmetic kernel,

$$h_*(r) = \frac{\pi}{2} r^2 (2\pi r^2 - 3) e^{-\pi r^2}, \qquad k(\omega) = e^{\omega/2} \sum_{n \geq 1} h_*(n e^\omega).$$

For the Fourier convention with exponential $e^{-2\pi i r\xi}$, the second and fourth Gaussian derivative formulas are

$$\widehat{r^2 e^{-\pi r^2}} = (1/(2\pi) - \xi^2)e^{-\pi\xi^2}, \quad \widehat{r^4 e^{-\pi r^2}} = (\xi^4 - 3\xi^2/\pi + 3/(4\pi^2))e^{-\pi\xi^2}.$$

They follow by differentiating the Gaussian transform twice and four times and dividing by $(-2\pi i)^2$ and $(-2\pi i)^4$, respectively. The coefficients π^2 and $-3\pi/2$ then give $\widehat{h}_* = h_*$ and $h_*(0) = 0$. Poisson summation gives $\sum_{n\geq 1} h_*(n\lambda) = \lambda^{-1}\sum_{n\geq 1} h_*(n/\lambda)$, and hence $k(\omega) = k(-\omega)$. Poisson summation for this Schwartz function follows directly by periodizing it: its Fourier coefficients are the displayed Fourier transform at integer frequencies; the function and coefficient series converge absolutely with derivatives, so evaluation at zero is valid.

Each fixed number of derivatives of the defining series is locally uniformly convergent. At $\omega \geq 0$ its Gaussian terms give $|k^{(r)}(\omega)| \leq C_r e^{C_r\omega} e^{-\pi e^{2\omega}}$; factor the $n = 1$ exponential and bound the remaining polynomially weighted series by $\sum n^{M_r} e^{-\pi(n^2-1)} < \infty$. Evenness gives the same bound with $|\omega|$ at the other end. Expanding the definition proves $\Phi(u) = 2k(2u)$ with every factor retained. Consequently

$$\begin{aligned} Z_\theta(s) &= 4\int_{\mathbb{R}} k(\omega)e^{\theta\omega^2/4}e^{is\omega}\, d\omega, \\ K_\theta(s) &= 4\int_{\mathbb{R}} m(\omega)k(\omega)e^{\theta\omega^2/4}e^{is\omega}\, d\omega, \end{aligned} \tag{176}$$

where m is exactly the three-shift, four-term multiplier in (108), including $-i\omega/336$ at shift 3. All derivatives of these amplitudes are Schwartz for real θ, locally uniformly in θ; the same is true of their Fourier transforms by integration by parts.

Set $d_h(\theta)^2 = \int_{\mathbb{R}} |h'_\theta(s)|^2\, ds$. The exact constants are

$$\begin{aligned} d_Z(\theta)^2 &= 32\pi\int_{\mathbb{R}} \omega^2 k(\omega)^2 e^{\theta\omega^2/2}\, d\omega, \\ d_K(\theta)^2 &= 32\pi\int_{\mathbb{R}} \omega^2 |m(\omega)|^2 k(\omega)^2 e^{\theta\omega^2/2}\, d\omega. \end{aligned} \tag{177}$$

For completeness, insert $e^{-\epsilon s^2}$ in the squared norm of $4\int b(\omega)e^{is\omega}\, d\omega$. Fubini and the Gaussian integral give

$$16\sqrt{\pi/\epsilon}\iint b(\omega)\overline{b(\eta)}e^{-(\omega-\eta)^2/(4\epsilon)}\, d\omega\, d\eta.$$

The Gaussian convolution has mass 2π and tends to $2\pi b$ in this integral: split at fixed distance from zero, use uniform continuity on the near part and its Gaussian tail on the far part, with Schwartz domination. On the original side dominated convergence applies because the Fourier transform is Schwartz. The result is $32\pi\int |b|^2$. Taking $b = i\omega k e^{\theta\omega^2/4}$ or $i\omega m k e^{\theta\omega^2/4}$ proves (177). Both integrals are strictly positive: $k(0) = \Phi(0)/2 > 0$ and $m(0) = 127/219024 > 0$ give an interval containing nonzero frequencies where the integrands' non-Gaussian factors are positive. The two-ended bounds prove continuity in θ by domination on compact parameter intervals. Thus on each compact $I \subset \mathbb{R}$ there are $0 < d_- \leq d_h(\theta) \leq d_+ < \infty$.

Define the full arithmetic temporal defect on the real spectral line by

$$\mathscr{D}_j = \partial_t\Psi_j + \frac{\vartheta'}{4}\partial_s^2\Psi_j = v_j h'_\vartheta(s + a_j).$$

Translation by the real a_j preserves Lebesgue measure. Therefore

$$\sum_j \|\mathscr{D}_j(t,q,\cdot)\|_2^2 = d_h(\vartheta(t))^2 |v(t,q)|^2,$$

$$v_j(t,q) = \frac{\int_{\mathbb{R}} \overline{h'_\vartheta(s+a_j)}\, \mathscr{D}_j(t,q,s)\, ds}{d_h(\vartheta)^2}, \tag{178}$$

$$u(t, X_t(q)) = 2W_1(X_t(q))v_1 + W_2(X_t(q))v_2 + W_3(X_t(q))v_3.$$

These formulas prove an inverse correspondence $(X_t, u \circ X_t) \leftrightarrow (X_t, \mathscr{D})$ on the three real profile lines at each (t, q). The flow and its inverse sheet are retained in both sides.

For nonempty K, put $M_J = \max_K \|J_S\|_{\mathrm{op}}$ and $N_J = \max_K \|J_S^{-1}\|_{\mathrm{op}}$. They are finite and positive. Define

$$\mathcal{A}_\infty(t) = \sup_q \left(\sum_j \|\mathscr{D}_j(t,q,\cdot)\|_2^2 \right)^{1/2}, \quad \mathcal{A}_2(t) = \left(\sum_j \int_{\mathbb{R}^3} \int_{\mathbb{R}} |\mathscr{D}_j|^2\, ds\, dq \right)^{1/2}.$$

When $\vartheta([0,T)) \subseteq I$, the complete estimates are

$$\frac{d_-}{N_J}\|u(t)\|_\infty \le \mathcal{A}_\infty(t) \le d_+ M_J \|u(t)\|_\infty, \qquad \frac{d_-}{N_J}\|u(t)\|_2 \le \mathcal{A}_2(t) \le d_+ M_J \|u(t)\|_2. \tag{179}$$

Indeed $v = J_S(X_t)(u \circ X_t)$, both vanish off K, and $|J_S u| \le M_J|u|$, $|u| \le N_J|J_S u|$ on K. The supremum is preserved by the diffeomorphism X_t. For the integral, its determinant one gives the exact identity

$$\mathcal{A}_2(t)^2 = d_h(\vartheta(t))^2 \int_{\mathbb{R}^3} |J_S(\mathbf{x})u(t,\mathbf{x})|^2\, d\mathbf{x}.$$

This proves both bounds without discarding any spatial or spectral component. For a fixed original incidence coefficient a_{ef}, the observation $a_{ef}K_\theta$ multiplies $\mathscr{D}^K$ by a_{ef} and its squared norm by $|a_{ef}|^2$; inversion uses its actual nonzero value, and the zero incidence has exactly the zero observation.

11.5 Application to the stated public singularity theorem

Theorem 11.1 (Arithmetic consequence of the cited public input)**.** *Take the velocity, pressure, force and compact set supplied by Theorem 1.1 of [6], with its original $\nu > 0$. Choose the explicit Newman time $\vartheta(t) = 0$ and the actual arithmetic profile $h_0 = \Xi$, on $0 \le t < 1$. Then the constructed entire spectral fields satisfy*

$$\mathscr{D}_j = \partial_t \Psi_j, \qquad \mathscr{D}_j(0,q,s) = 0, \qquad \sup_{t<1} \mathcal{A}_2(t) < \infty, \qquad \limsup_{t \uparrow 1} \mathcal{A}_\infty(t) = \infty. \tag{180}$$

Every $\mathscr{D}_j$ is supported in $q \in K$. For every compact $L \subset \mathbb{R}^3$, compact $E_0 \subset \mathbb{C}$ and integer $r \ge 0$,

$$\sup_{t<1,\ q\in L,\ \zeta \in E_0} |\partial_\zeta^r \Psi_j(t,q,\zeta)| < \infty. \tag{181}$$

The analogous temporal-defect conclusions hold for the original full profile K_0 with its exact $d_K(0)$.

Proof. The cited public theorem provides a triple in (170) with $T = 1$, $u(0) = 0$, uniformly bounded $\|u(t)\|_2$, and unbounded terminal $\|u(t)\|_\infty$. The preceding explicit construction therefore supplies $X_t, \Psi, \mathscr{D}$. With the chosen constant Newman time the heat part of the temporal defect vanishes exactly, and $u(0) = 0$ gives $\mathscr{D}(0) = 0$. Set $d_- = d_+ = d_Z(0) > 0$ in (179). Its two inequalities give the energy bound and the unbounded terminal supremum in (180). Support follows from $v = 0$ off K. For the last bound, (171) places the argument of $\Xi^{(r)}$ in the fixed compact set $S_j(K \cup L) + E_0$. Entire continuity gives its finite maximum. All arguments apply to K_0 using the already proved positive constant $d_K(0)$ and its entire derivatives. This proof uses the external existence theorem exactly at its first step; all transfer formulas and estimates are proved within this fragment. □

11.6 The actual cutoff-summed source witness and its arithmetic image

11.6.1 The precise imported witness and its unchanged viscosity

We now use the particular field constructed in the 166-page revision of [6]. The original cutoff-summed fields, the smooth extension of the force, and their all-order estimates are the imported source input. Its actual closing equations (9.21), (10.4)–(10.14) and (10.20)–(10.23) were read directly. The complete companion specialization [9] was also read; all arithmetic, flow, energy and terminal maps used here have their proofs below. No independent verification of every lemma in the public 166-page existence proof is claimed.

Use a star to retain the source's viscosity-one fields, rather than overwriting the physical viscosity $\nu > 0$. In source equation (9.21), page 115, the actual summed Cartesian vector potential, azimuthal field, and pressure are denoted by $A_*, B_* e_\theta, p_{*,\mathrm{loc}}$. These are the cutoff-summed corrected fields of that equation, including all stages and the finite initialization. With the source's own radial-time coordinate denoted q_{src} to distinguish it from our material label q, its exact sums are

$$
\begin{aligned}
A_* &= A_{*,0} + \sum_{j\ge 1} \chi(a_j q_{\mathrm{src}}) A_{*,j},\\
B_* e_\theta &= B_{*,0} e_\theta + \sum_{j\ge 1} \chi(a_j q_{\mathrm{src}}) B_{*,j},\\
p_{*,\mathrm{loc}} &= p_{*,0} + \sum_{j\ge 1} \chi(a_j q_{\mathrm{src}}) p_{*,j}.
\end{aligned}
$$

The displayed a_j, χ and all finite-state representatives are the actual choices in source (9.21); the subscript star marks viscosity one and introduces no replacement of them. Source equation (10.4), page 118, constructs

$$
u_* = \operatorname{curl}(c_* A_*) + c_* B_* e_\theta, \qquad p_* = c_* p_{*,\mathrm{loc}}, \qquad c_* = \chi_x \chi_t.
$$

Here the source spatial cutoff is smooth in (r^2, z), supported in $r < r_0,\ |z| < z_0$, and equals one on $r \le r_0/2,\ |z| \le z_0/2$. Its time cutoff is zero for $1 - t \ge \tau_0$ and one for $0 \le 1 - t \le \tau_0/2$, where $0 < \tau_0 < 1/2$. Retain the actual source choices of these parameters. The source force before time one is exactly

$$
f_* = \partial_t u_* + (u_* \cdot \nabla) u_* - \Delta u_* + \nabla p_*.
$$

Thus every cutoff derivative belongs to the force. Its smooth extension through time one is source (10.11), page 120, with all its prescribed Taylor coefficients and cutoff sequence. The source proves that its support is contained in $K_* \times [0, 2]$, where $K_* = \operatorname{supp} \chi_x$, and it vanishes near time zero.

The pressure is the displayed compactly supported source pressure; no divergence-free condition is imposed on f_*.

For the physical parameter $\nu > 0$, take precisely source (10.22):

$$y = x/\sqrt{\nu}, \qquad x = \sqrt{\nu}\, y,$$
$$u_\nu(x,t) = \sqrt{\nu}\, u_*(x/\sqrt{\nu}, t), \qquad p_\nu(x,t) = \nu p_*(x/\sqrt{\nu}, t),$$
$$f_\nu(x,t) = \sqrt{\nu}\, f_*(x/\sqrt{\nu}, t), \qquad K_\nu = \sqrt{\nu}\, K_*.$$

The inverse multiplies the velocity and force by $\nu^{-1/2}$, the pressure by ν^{-1}, and substitutes $x = \sqrt{\nu}\, y$; physical time is unchanged. The chain rule gives

$$\partial_t u_\nu + (u_\nu \cdot \nabla_x) u_\nu - \nu \Delta_x u_\nu + \nabla_x p_\nu = \sqrt{\nu}\, [\partial_t u_* + (u_* \cdot \nabla_y) u_* - \Delta_y u_* + \nabla_y p_*](y,t) = f_\nu.$$

Also $\operatorname{div}_x u_\nu = \operatorname{div}_y u_* = 0$, $u_\nu(0) = 0$, and u_ν, p_ν have support in K_ν for every $t < 1$. These are exactly the inputs of Section 11 with $T = 1$, with all coordinate and parameter maps specified.

Let

$$F_\nu(t) = \int_0^t \|f_\nu(s)\|_{L^2(\mathbb{R}^3)}\, ds, \qquad E_\nu = F_\nu(1) = \nu^{5/4} F_*(1) < \infty.$$

The last exponent follows from the factor $\sqrt{\nu}$ in f_ν and $dx = \nu^{3/2} dy$. For completeness the energy estimate used below is proved directly for this input. Compact support and incompressibility give, by integrating the equation against u_ν,

$$\frac{1}{2}\frac{d}{dt}\|u_\nu(t)\|_2^2 + \nu \|\nabla u_\nu(t)\|_2^2 = \langle f_\nu(t), u_\nu(t)\rangle.$$

The convection integral is the integral of $\operatorname{div}(u_\nu |u_\nu|^2/2)$, and the pressure integral is minus the integral of $p_\nu \operatorname{div} u_\nu$; both vanish. Divide by $(\|u_\nu\|_2^2 + \delta^2)^{1/2}$, use Cauchy–Schwarz, integrate from zero, and let $\delta \downarrow 0$. This proves $\|u_\nu(t)\|_2 \le F_\nu(t)$. Substitution back into the integrated identity proves, with its full dissipation coefficient,

$$\|u_\nu(t)\|_2^2 + 2\nu \int_0^t \|\nabla u_\nu(s)\|_2^2 ds \le 2 \int_0^t F_\nu'(s) F_\nu(s) ds = F_\nu(t)^2.$$

11.6.2 The source's actual growing component in two arithmetic channels

The source fixes $0 < h < 1/100$, $X_{\text{in}} > 0$, and $e_0 > 0$ through its construction. Its closing proof, (10.20)–(10.21), page 124, gives the actual corrected swirl along

$$x_{*,\tau} = (\sqrt{2X_{\text{in}}\tau}, 0, 0), \qquad t = 1 - \tau, \qquad (u_*)_2(x_{*,\tau}, 1-\tau) = \tau^{-1/2-h}(e_0 + R_*(\tau)).$$

The original remainder is retained as the function $R_*(\tau) = \tau^{1/2+h}(u_*)_2(x_{*,\tau}, 1-\tau) - e_0$. The source estimate supplies constants $C_* > 0, \tau_* > 0$ with $|R_*(\tau)| \le C_* \tau^{2h}$ for $0 < \tau < \tau_*$. At these points the cylindrical angle is zero, so $e_\theta = (0,1,0)$ and the displayed Cartesian component is precisely the source swirl. The annular and higher-order corrections have already been accounted for in this remainder. The parameter h is the chosen source parameter; no claim that an arbitrary prescribed h works is needed or made.

Define

$$r_{\nu,\tau} = \sqrt{2\nu X_{\text{in}}\tau}, \qquad x_{\nu,\tau} = (r_{\nu,\tau}, 0, 0).$$

The exact viscosity map proves

$$(u_\nu)_2(x_{\nu,\tau}, 1-\tau) = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)). \tag{182}$$

Use the actual material flow $X_{\nu,t}$ of u_ν proved to exist on each interval before one in Section 11. Put

$$X_{\nu,t}(q) = \sqrt{\nu}\, X_{*,t}(q/\sqrt{\nu}).$$

This is an equality of the actual flows: its right side equals q at zero, and its time derivative is $\sqrt{\nu}\, u_*(t, X_{*,t}(q/\sqrt{\nu})) = u_\nu(t, \sqrt{\nu}\, X_{*,t}(q/\sqrt{\nu}))$. The previously proved uniqueness gives the equality, with inverse $X_{\nu,t}^{-1}(x) = \sqrt{\nu}\, X_{*,t}^{-1}(x/\sqrt{\nu})$. Define the labels

$$q_{\nu,\tau} = X_{\nu,1-\tau}^{-1}(x_{\nu,\tau}).$$

This is the exact inverse between the Eulerian sampling points and their material labels. It is a family of labels depending on τ; the source path has not been asserted to be a single fluid trajectory.

Write $S = \mathbf{S} = (F_1/2, F_2, F_3)$ and $J = J_S = DS$ in the physical Cartesian coordinates x, not the viscosity-one coordinate y. Direct substitution into the original polynomials gives

$$S(r,0,0) = (0,0,2r), \qquad J(r,0,0) = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 1 & 3r \\ 2 & -3r^2 & -r^3 \end{pmatrix}.$$

Indeed the first row comes from differentiating $F_1/2 = ((1+xy)^3 w + y^2(1+xy)(4+3xy))/2$; the second comes from $F_2 = y + 3x(1+xy)^2 w + 3xy^2(4+3xy)$; the third comes from $F_3 = 2x - 3x^2 y - x^3 w$, all evaluated at $(x,y,w) = (r,0,0)$. Every entry, including the off-diagonal powers of r, is retained. Writing $v = J(r,0,0)u$ therefore gives the exact component inverse

$$u_3 = 2v_1, \qquad u_2 = v_2 - 6rv_1,$$
$$u_1 = \tfrac{1}{2}v_3 + \tfrac{3}{2}r^2 v_2 - 8r^3 v_1.$$

Substituting these expressions into all three rows proves both directions of this matrix inverse.

Fix actual arithmetic time zero. Take $H = Z_0 = \Xi$ or $H = K_0$ with the complete four-term coefficient from the preceding proof, and write

$$d_H^2 = \int_{\mathbb{R}} |H'(s)|^2 ds > 0, \qquad \Psi_{\nu,j}(t,q,\zeta) = H(S_j(X_{\nu,t}(q)) + \zeta), \qquad D_{\nu,j} = \partial_t \Psi_{\nu,j}.$$

The exact positive constants are those of (177) at $\theta = 0$, including 32π and, for K_0, the entire factor $|m|^2$. At the particular pair $(t,q) = (1-\tau, q_{\nu,\tau})$, the first two shifts are both zero. Consequently define the actual integral

$$I_{\nu,j}(\tau) = \frac{1}{d_H^2} \int_{\mathbb{R}} \overline{H'(s)} D_{\nu,j}(1-\tau, q_{\nu,\tau}, s) ds, \qquad j = 1,2.$$

The chain rule gives $D_{\nu,j} = v_j H'(s)$ there. It follows, without an asymptotic replacement of the profiles, that

$$I_{\nu,2}(\tau) - 6r_{\nu,\tau} I_{\nu,1}(\tau) = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)). \tag{183}$$

There is also a whole-profile identity, stronger than its projection:

$$\begin{aligned} &D_{\nu,2}(1-\tau, q_{\nu,\tau}, s) - 6r_{\nu,\tau} D_{\nu,1}(1-\tau, q_{\nu,\tau}, s) \\ &\quad = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)) H'(s). \end{aligned} \tag{184}$$

Its squared $L^2(ds)$ norm is exactly $\nu\tau^{-1-2h}(e_0 + R_*(\tau))^2 d_H^2$. Thus the source's actual component and remainder, not just an abstract possibility of growth, have been carried into the original arithmetic profile. The complex conjugation and integration above retain the complex values of K_0 when that channel is used.

Choose any $0 < \tau_{**} \le \tau_*$ such that $C_*\tau_{**}^{2h} \le e_0/2$ and the source cutoffs equal one on the path for $\tau < \tau_{**}$. Cauchy–Schwarz on the two component coefficients $(-6r_{\nu,\tau}, 1)$ yields

$$\left(\sum_{j=1}^{3} \|D_{\nu,j}(1-\tau, q_{\nu,\tau}, \cdot)\|_2^2\right)^{1/2} \ge \frac{d_H\sqrt{\nu}\, e_0}{2\sqrt{1+72\nu X_{\text{in}}\tau}}\,\tau^{-1/2-h}. \tag{185}$$

This lower bound proves divergence in the spatial-supremum, spectral-L^2 norm for the imported constructed field. It is not a matching upper asymptotic for its global supremum. All forcing, pressure, and material diffusion terms remain those of (174) and (175), with the explicit (u_ν, p_ν, f_ν) just given.

11.6.3 A terminal arithmetic profile with a bounded energy norm

The same constructed input gives a terminal value in a precise function space. This assertion can be proved from its energy identity without controlling its unbounded spatial supremum. Let $M_\nu = \max_{K_\nu} \|J\|_{\text{op}}$ and put

$$\mathcal{H}_\nu = L^2(K_\nu \times \mathbb{R}, dq\, ds; \mathbb{C}^3), \qquad B_{\nu,j}(t,q,s) = \Psi_{\nu,j}(t,q,s) - H(S_j(q)+s).$$

Subtracting the displayed initial value defines an affine coordinate with the explicit inverse $B \mapsto B + (H(S_j(q)+s))_j$. It does not discard the initial profile. Its domain and reference element are specified; outside K_ν the difference is identically zero. The full profile on K_ν is also square integrable because real translation gives norm $|K_\nu|^{1/2}\|H\|_2$ in each channel.

The already proved exact identity gives

$$\|D_\nu(t)\|_{\mathcal{H}_\nu}^2 = d_H^2 \int_{K_\nu} |J(x)u_\nu(t,x)|^2 dx \le d_H^2 M_\nu^2 E_\nu^2.$$

The identity follows by integrating the real spectral translation and then using $\det D_q X_{\nu,t} = 1$; no spatial coordinate is removed. On a closed time interval before one the functions are smooth and the Schwartz bounds justify their Hilbert-space derivatives. Integrating $\partial_t B_\nu = D_\nu$ there and using the triangle inequality for the integral yields, for $0 \le a < b < 1$,

$$\|B_\nu(b) - B_\nu(a)\|_{\mathcal{H}_\nu} \le d_H M_\nu E_\nu (b-a). \tag{186}$$

Completeness constructs a unique terminal element $B_\nu(1)$ and the exact bound $\|B_\nu(1) - B_\nu(t)\| \le d_H M_\nu E_\nu(1-t)$. This constructs a Lipschitz extension on $[0,1]$ in the specified Hilbert space, while (185) proves divergence in the stronger spatial-supremum norm of its time derivative.

We also identify the terminal element through an actual map on labels. Volume preservation gives

$$\int_{K_\nu}\int_0^1 |u_\nu(t, X_{\nu,t}(q))|\,dt\,dq = \int_0^1 \|u_\nu(t)\|_{L^1(K_\nu)} dt \le |K_\nu|^{1/2} E_\nu.$$

The nonnegative integral is finite, so for almost every q the full trajectory has finite total length before one. Its integral equation therefore constructs the limit $X_{\nu,1}(q) \in K_\nu$ as $t \uparrow 1$. Set its value arbitrarily in K_ν on the null exceptional set. Also the integral triangle inequality and volume preservation give

$$\|X_{\nu,b} - X_{\nu,a}\|_{L^2(K_\nu)} \le \int_a^b \|u_\nu(t)\|_2 dt \le E_\nu(b-a),$$

and the same estimate with $b = 1$ follows by the constructed limit and Fatou's lemma. For every continuous g on compact K_ν, dominated convergence gives

$$\int_{K_\nu} g(X_{\nu,1}(q))dq = \lim_{t\uparrow 1} \int_{K_\nu} g(X_{\nu,t}(q))dq = \int_{K_\nu} g(x)dx.$$

This identifies the pushforward of Lebesgue measure exactly. To see that continuous tests suffice, for each relatively closed set C use the continuous decreasing approximations $g_n(x) = \max(0, 1 - n\,\mathrm{dist}(x, C))$ to its indicator, then monotone or dominated convergence; complements give open sets and the uniqueness of finite measures on their generated Borel sigma algebra gives the stated pushforward.

For almost every q the pointwise terminal entire fibre is

$$\Psi_{\nu,j}(1, q, \zeta) = H(S_j(X_{\nu,1}(q)) + \zeta).$$

It agrees with the Hilbert-space limit: real translations are continuous in $L^2(ds)$, as follows from $\|H(\cdot + a) - H(\cdot + b)\|_2 \le d_H|a - b|$ by integrating H'. The shifts converge almost everywhere, remain bounded on K_ν, and their translated-profile squared differences are bounded by $4\|H\|_2^2$. Dominated convergence proves the identification in $\mathcal{H}_\nu$. Replacing H by $H^{(n)}$ in this proof proves the same assertions for every pure spectral derivative, with the retained constant $\|H^{(n+1)}\|_2 M_\nu E_\nu$.

The terminal label map has a further exact operator correspondence. On $L^2(K_\nu)$ define $V_t g = g \circ X_{\nu,t}$, including $t = 1$. For $t < 1$ its inverse is composition with $X_{\nu,t}^{-1}$, so it is unitary. The just-proved measure identity makes V_1 an isometry. For continuous g, dominated convergence gives $V_t g \to V_1 g$ in L^2. Continuous functions are dense for Lebesgue measure restricted to a compact set: approximate measurable-set indicators by continuous distance cutoffs between compact subsets and open supersets using finite-measure regularity, then approximate simple functions. The isometric bounds extend the convergence to every $g \in L^2(K_\nu)$. Thus $V_t \to V_1$ strongly. Its Hilbert adjoint satisfies $V_1^* V_1 = I$ by preservation of inner products, and $V_1 V_1^*$ is the orthogonal projection onto its closed range: it is self-adjoint, idempotent, acts as identity on that range, and annihilates its orthogonal complement. These are the proved terminal inverse and range statements. No pointwise inverse of $X_{\nu,1}$, or surjectivity of V_1, has been supplied by this argument.

11.6.4 The actual zero image, including the terminal fibres

For every time before one and almost every terminal label, the shift $a = S_j(X_{\nu,t}(q))$ is real. The full coordinate map is $\zeta \mapsto a + \zeta$ with inverse $s \mapsto s - a$. If $H(s) = (s - z_0)^m A_0(s)$ with $A_0(z_0) \ne 0$, the actual image is

$$H(a + \zeta) = (\zeta - (z_0 - a))^m A_0(a + \zeta).$$

This retains the whole analytic unit and all its derivatives. In particular its zero has imaginary part $\mathrm{Im}\, z_0$ at all these times, even for the source input whose arithmetic time-derivative combination obeys (184). With $H = \Xi$ the precise completed-zeta argument is $\rho = 1/2 + i(a + \zeta)$, and $\mathrm{Re}\, \rho = 1/2 - \mathrm{Im}\, \zeta$. Thus the actual source witness produces the proved derivative singularity and terminal profile above; this explicit map preserves zero heights. A zero-height change is not obtained by substituting a faster-growing velocity into this same real-translation map.

12 The actual source quotient of a flow-driven arithmetic profile

12.1 Translation, heat, and the original spectral generator

For $a, \theta \in \mathbb{C}$ define, with the input coordinate s and output coordinate ζ retained,

$$(T_a h)(\zeta) = h(\zeta + a), \qquad V_{\theta,a} = T_a U_\theta : \mathcal{H}_{\leq 1} \longrightarrow \mathcal{H}_{\leq 1}.$$

The derivative is the complex derivative in the displayed coordinate. Translation preserves the original order class. More precisely, for $1 < p < 2$ and $c > 0$ the already defined growth seminorms satisfy

$$\|T_a h\|_{p,c} \leq e^{c|a|^p} \|h\|_{p,c\,2^{1-p}},$$

because $|\zeta + a|^p \leq 2^{p-1}(|\zeta|^p + |a|^p)$. Thus T_a and T_{-a} are continuous for γ. Ordinary differentiation gives $DT_a = T_a D$; continuity and the convergent heat series give $T_a U_\theta = U_\theta T_a$. Consequently

$$V_{\theta,a}^{-1} = U_{-\theta} T_{-a}, \qquad V_{\theta,a} D = D V_{\theta,a}.$$

These assertions hold on all of $\mathcal{H}_{\leq 1}$ and do not impose continuity of translation or differentiation for the unnamed source topology.

Retain the actual τ, N_{ar} and Q_{ar}, and define their exact transported presentations

$$\tau_{\theta,a} = (V_{\theta,a})_* \tau, \qquad N_{\theta,a} = V_{\theta,a} N_{\mathrm{ar}}, \qquad Q_{\theta,a} = (\mathcal{H}_{\leq 1}, \tau_{\theta,a})/N_{\theta,a}. \tag{187}$$

Then $N_{\theta,a}$ is closed, and

$$\overline{V}_{\theta,a} : Q_{\mathrm{ar}} \longrightarrow Q_{\theta,a}, \qquad [h] \longmapsto [V_{\theta,a} h]$$

is a complex-linear homeomorphism. Its inverse is induced by the displayed inverse on representatives. The original arithmetic multiplication has the exact conjugate

$$B_{\theta,a} = V_{\theta,a} s V_{\theta,a}^{-1} = \zeta + a - \frac{\theta}{2} D_\zeta. \tag{188}$$

Indeed the previous heat commutator gives $U_\theta s U_{-\theta} = s - \theta D/2$; translation sends multiplication by s to multiplication by $\zeta + a$ and commutes with D. This full combination preserves $N_{\theta,a}$ and is continuous for $\tau_{\theta,a}$ by its conjugacy. The induced quotient operator is similar through $\overline{V}_{\theta,a}$ to the source S, so its spectrum is exactly the original s-spectrum. To verify both directions, conjugate a continuous two-sided inverse of $S - \lambda$ by $\overline{V}_{\theta,a}$, or conjugate an inverse of the transported operator back. At $\theta = 0$, multiplication by ζ itself descends, and its spectrum is $\mathrm{Spec}(S) - a$ because $B_{0,a} = \zeta + a$.

Along the retained real flow let

$$a_j(t,q) = S_j(X_t(q)), \qquad v_j(t,q) = \partial_t a_j(t,q), \qquad \Psi_j(t,q,\zeta) = V_{\vartheta(t),a_j(t,q)} \Xi(\zeta).$$

Here S_j denotes the original polynomial target component, whereas the unindexed S above denotes the arithmetic quotient operator. The source quotient is taken fibre by fibre at each fixed (t, q, j). If a topology on the whole family of quotient fibres is needed, give their disjoint union the topology transported from $\mathbb{C}^2 \times Q_{\mathrm{ar}}$ by $((\theta, a), [h]) \mapsto ((\theta, a), [V_{\theta,a} h])$. This specifies joint continuity by an exact trivialization and does not assert joint parameter continuity for the untransported τ.

Because $\Xi \in N_{\mathrm{ar}}$, every profile represents zero in its actual transported quotient:

$$[\Psi_j]_{Q_{\vartheta,a_j}} = 0. \tag{189}$$

Its temporal defect is nevertheless a specific full representative,

$$\mathcal{D}_j = \partial_t \Psi_j + \frac{\vartheta'}{4} D_\zeta^2 \Psi_j = v_j V_{\vartheta,a_j} \Xi'.$$

This is the complete chain rule with the original heat coefficient. Write

$$\beta = [\Xi']_{Q_{\mathrm{ar}}}, \qquad \mathfrak{a}_\beta = \{c \in \mathbb{C} : c\Xi' \in N_{\mathrm{ar}}\}.$$

Then the exact quotient statement and its scalar kernel are

$$[\mathcal{D}_j]_{Q_{\vartheta,a_j}} = \overline{V}_{\vartheta,a_j}(v_j \beta), \qquad \ker(c \mapsto c\beta) = \mathfrak{a}_\beta. \tag{190}$$

The set $\mathfrak{a}_\beta$ is a closed complex linear subspace: it is the inverse image of the closed N_{ar} under the continuous scalar map $c \mapsto c\Xi'$ into $(\mathcal{H}_{\le 1}, \tau)$. Its equality with $\{0\}$ or $\mathbb{C}$ is precisely the unresolved membership question $\Xi' \notin N_{\mathrm{ar}}$ or $\Xi' \in N_{\mathrm{ar}}$. No evaluation functional on the source quotient is used to decide it.

At a fixed physical point $x = X_t(q)$, retain $v = J(x)u(t,x)$ and the full inverse $J^{-1}(x) = [2W_1(x)\ W_2(x)\ W_3(x)]$. Thus the exact kernel of the complexified velocity-to-source-class map is

$$J(x)^{-1}(\mathfrak{a}_\beta^3) \subseteq \mathbb{C}^3;$$

for real velocities take its intersection with $\mathbb{R}^3$. This follows componentwise from (190) and the invertibility of the unchanged J.

The derivative relation explains why a zero profile class does not determine its representative derivative. At a fixed (θ, a) put

$$\mathfrak{D}_{\theta,a} = \{([h],[Dh]) : h \in \mathcal{H}_{\le 1}\} \subset Q_{\theta,a}^2, \qquad \mathcal{W}_{\theta,a} = (DN_{\theta,a} + N_{\theta,a})/N_{\theta,a}.$$

It has full domain and output fibre $[Dh] + \mathcal{W}_{\theta,a}$ at $[h]$: replace h by every $h+n$ with $n \in N_{\theta,a}$. Its presenting map has kernel $\{h \in N_{\theta,a} : Dh \in N_{\theta,a}\}$. Since D commutes with $V_{\theta,a}$, conjugating both components by $\overline{V}_{\theta,a}$ gives exactly the previous source derivative relation and its ambiguity space. In particular $\overline{V}_{\theta,a}\beta$ is an actual element of its multivalued part at zero. The displayed temporal connection is an identity on the full smooth profiles; a derivative on all untransported source quotient representatives is not presumed.

12.2 An exact information-preserving refinement and its residue projection

Retain the explicit analytic ideal $I = \Xi\mathcal{H}_{\le 1}$, already proved closed for γ by the full division estimate, and put

$$H_\cap = N_{\mathrm{ar}} \cap I, \qquad C_\cap = (\mathcal{H}_{\le 1}, \tau \vee \gamma)/H_\cap, \qquad Q_I = (\mathcal{H}_{\le 1}, \gamma)/I. \tag{191}$$

The join topology is precisely the topology induced by $h \mapsto (h,h)$ into $(\mathcal{H}_{\le 1}, \tau) \times (\mathcal{H}_{\le 1}, \gamma)$. Both subspaces in the intersection are closed in this join, so $C_\cap$ is Hausdorff. The identity on representatives induces continuous surjections

$$p_N : C_\cap \longrightarrow Q_{\mathrm{ar}}, \qquad p_I : C_\cap \longrightarrow Q_I, \qquad \ker p_N = N_{\mathrm{ar}}/H_\cap, \quad \ker p_I = I/H_\cap. \tag{192}$$

The kernel assertions are direct membership tests; continuity follows from the quotient universal property. These kernel identifications are also homeomorphisms when N_{ar} and I have their subspace topologies from $\tau \vee \gamma$. To prove this, let $q_\cap$ be the open quotient map and let A denote either subspace. Because $H_\cap \subset A$, $q_\cap(A \cap O) = q_\cap(A) \cap q_\cap(O)$ for every open ambient O. Thus its restriction to A is an open quotient map onto the corresponding kernel with its subspace topology. The joint map (p_N, p_I) is continuous and injective since its kernel is zero. No inference of a topological embedding from those two properties is needed.

Let λ be any actual zero of Ξ, with its actual multiplicity $m_\lambda \geq 1$. Choose a disk containing no other zero and retain the full unit

$$\Xi(s) = (s - \lambda)^{m_\lambda} A_\lambda(s), \qquad A_\lambda(\lambda) \neq 0.$$

For $h \in \mathcal{H}_{\leq 1}$ define

$$R_\lambda(h) = \operatorname{Res}_{s=\lambda} \frac{h(s)}{\Xi(s)} = \frac{1}{2\pi i} \int_{|s-\lambda|=r} \frac{h(s)}{\Xi(s)}\, ds, \tag{193}$$

where the circle has positive orientation and lies in this disk. The integral is independent of the permitted r by Laurent expansion. It is continuous for γ: its absolute value is at most $r \sup_{|s-\lambda|=r} |h(s)| / \min_{|s-\lambda|=r} |\Xi(s)|$, and a compact supremum is bounded by a fixed growth seminorm times the maximum of its exponential weight on the circle. It annihilates I, since h/Ξ is entire there, and therefore annihilates $H_\cap$.

Logarithmic differentiation of the full local factor gives

$$\frac{c\Xi'(s)}{\Xi(s)} = \frac{m_\lambda c}{s - \lambda} + c\frac{A'_\lambda(s)}{A_\lambda(s)}, \qquad R_\lambda(c\Xi') = m_\lambda c. \tag{194}$$

In particular $\Xi' \notin I$: the displayed pole is nonzero. Thus $\beta_I = [\Xi']_{Q_I} \neq 0$, and $\widetilde{\beta} = [\Xi']_{C_\cap} \neq 0$ as well, independent of the status of β in the actual source quotient. The residue functional factors continuously through $C_\cap$; write its induced functional as $\widetilde{R}_\lambda$. The formula

$$\Pi_\lambda(c) = \frac{\widetilde{R}_\lambda(c)}{m_\lambda}\, \widetilde{\beta}, \qquad c \in C_\cap \tag{195}$$

defines a continuous projection onto the line $\mathbb{C}\widetilde{\beta}$. Indeed its square is itself by $\widetilde{R}_\lambda(\widetilde{\beta}) = m_\lambda$, and scalar multiplication on the quotient is continuous. Its inverse coordinate on that line is $\widetilde{R}_\lambda / m_\lambda$, while $c \mapsto c\widetilde{\beta}$ is its continuous inverse. Consequently this line has exactly the ordinary complex topology, and $C_\cap = \mathbb{C}\widetilde{\beta} \oplus \ker \widetilde{R}_\lambda$ is a continuous direct sum. This proves an information-preserving refinement of the actual source map, including a continuous recovery coordinate.

The full principal-part map $h \mapsto (\mathrm{PP}_\lambda(h/\Xi))_{\lambda \in Z(\Xi)}$ has kernel exactly I. Vanishing of every negative Laurent coefficient makes h/Ξ entire at every zero; it is already holomorphic elsewhere, and the proved entire division bound puts it in $\mathcal{H}_{\leq 1}$. The reverse inclusion is immediate. Each finite principal-part coordinate is continuous by the same circle integral with the appropriate power of $s - \lambda$. On the velocity line its entire image at every actual zero is the single term $m_\lambda c/(s - \lambda)$ from (194), with the unchanged analytic unit in the regular part.

For general (θ, a) let $I_{\theta,a} = V_{\theta,a} I$ and transport (191) by $V_{\theta,a}$. Its ambient topology is $\tau_{\theta,a} \vee \gamma$, since $V_{\theta,a}$ is a γ-homeomorphism, and its relation is $N_{\theta,a} \cap I_{\theta,a} = V_{\theta,a} H_\cap$. Every map, kernel and projection above is transported exactly. In particular the transported residue coordinate is

$$R_{\lambda;\theta,a}(h) = \operatorname{Res}_{s=\lambda} \frac{(U_{-\theta} T_{-a} h)(s)}{\Xi(s)}.$$

For the actual temporal defect it gives $R_{\lambda;\vartheta,a_j}(\mathcal{D}_j) = m_\lambda v_j$. The pair of source and analytic classes therefore has the explicit injective common lift $v_j[V_{\vartheta,a_j}\Xi']$ in this transported refinement. For nonzero heat time this formula retains the inverse heat operator; it does not replace $V_{\theta,a} I$ by an unproved ordinary ideal generated by $T_a Z_\theta$.

12.3 The logarithmic connection and the public blowup residue field

Take the explicit Newman-time choice $\vartheta(t) = 0$. For the original real flow set

$$\Psi_j(t, q, \zeta) = \Xi(\zeta + a_j(t, q)), \qquad \mathcal{D}_j = \partial_t \Psi_j = v_j(t, q)\Xi'(\zeta + a_j(t, q)).$$

At every actual zero λ the translated divisor is $\zeta_{\lambda,j}(t, q) = \lambda - a_j(t, q)$, and the full factorization is

$$\Psi_j = (\zeta - \zeta_{\lambda,j})^{m_\lambda} A_\lambda(\zeta + a_j).$$

Thus all multiplicities and the actual analytic unit are retained. Since a_j is real, its imaginary coordinate is exactly $\operatorname{Im}\lambda$. Differentiating its location gives $\partial_t \zeta_{\lambda,j} = -v_j$.

On the complement of the divisor in $[0, T) \times \mathbb{R}^3_q \times \mathbb{C}_\zeta$, define the complex-valued differential form, smooth in the real variables and holomorphic in ζ,

$$\omega_j = \frac{d\Psi_j}{\Psi_j} = \frac{\Xi'(\zeta + a_j)}{\Xi(\zeta + a_j)}(d\zeta + da_j), \qquad da_j = v_j\, dt + \sum_b \partial_{q_b} a_j\, dq_b. \tag{196}$$

All label and time coefficients are displayed. Put $s = \zeta + a_j$. Direct exterior differentiation gives $d\omega_j = (\Xi'/\Xi)'(s)\, ds \wedge ds + (\Xi'/\Xi)(s) d^2 s = 0$. Equivalently the connection $d - \omega_j$ on the trivial complex line has zero curvature and annihilates the section Ψ_j. This calculation is on the complement of its zeros and retains the singular divisor explicitly.

At fixed (t, q) a sufficiently small positively oriented circle around $\zeta_{\lambda,j}$ has

$$\frac{1}{2\pi i} \int \omega_j = m_\lambda,$$

where only its $d\zeta$ restriction is integrated. The pole term in (194) gives this integral; the holomorphic unit term has zero integral by Cauchy's theorem. The time component and its residue are exactly

$$\begin{aligned} \omega_j(\partial_t) &= \frac{\mathcal{D}_j}{\Psi_j} = \frac{m_\lambda v_j}{\zeta - \zeta_{\lambda,j}} + v_j \frac{A'_\lambda(\zeta + a_j)}{A_\lambda(\zeta + a_j)}, \\ r_j(t, q) &:= \operatorname{Res}_{\zeta = \zeta_{\lambda,j}} \frac{\mathcal{D}_j}{\Psi_j} = m_\lambda v_j = -m_\lambda \partial_t \zeta_{\lambda,j}. \end{aligned} \tag{197}$$

The coordinate change $s = \zeta + a_j$ has differential $ds = d\zeta$ on each spectral fibre, so this is precisely $R_{\lambda;0,a_j}(\mathcal{D}_j)$, including its orientation and multiplicity factor.

Write $r = (r_1, r_2, r_3)^\mathsf{T}$, and retain the full original polynomial target and its matrix inverse:

$$S = (F_1/2, F_2, F_3)^\mathsf{T}, \quad J = DS = \operatorname{diag}(1/2, 1, 1)DF, \quad J^{-1} = [2W_1\ W_2\ W_3].$$

The literal recovery and pointwise norm identities are

$$\begin{aligned} u(t, X_t(q)) &= \frac{2W_1(X_t(q))r_1 + W_2(X_t(q))r_2 + W_3(X_t(q))r_3}{m_\lambda}, \\ |r(t, q)|^2 &= m_\lambda^2 |J(X_t(q))u(t, X_t(q))|^2. \end{aligned} \tag{198}$$

They follow by $r = m_\lambda v$, $v = Ju$ and the stated matrix inverse, with no selection of a global inverse sheet of S. The flow X_t is retained as part of the data.

For the nonempty compact velocity support K put

$$M_K = \max_K \|J\|_{\mathrm{op}}, \qquad N_K = \max_K \|J^{-1}\|_{\mathrm{op}}.$$

They are finite and positive by compactness and $\det J = -1$. Since $X_t(K) = K$, X_t is onto, and u, v, r vanish outside the corresponding compact labels, (198) gives

$$\frac{m_\lambda}{N_K}\|u(t)\|_\infty \le \sup_q |r(t,q)| \le m_\lambda M_K \|u(t)\|_\infty. \tag{199}$$

The volume-preserving change of variables $x = X_t(q)$ gives

$$\int_{\mathbb{R}^3} |r(t,q)|^2\, dq = m_\lambda^2 \int_{\mathbb{R}^3} |J(x)u(t,x)|^2\, dx, \tag{200}$$

whose square root is between $(m_\lambda/N_K)\|u(t)\|_2$ and $m_\lambda M_K\|u(t)\|_2$. These are exact spatial estimates for the arithmetic residues. With the previously proved spectral normalization $d_Z(0)^2 = \int_{\mathbb{R}} |\Xi'(s)|^2 ds > 0$, the unchanged real translations also give

$$|r(t,q)| = \frac{m_\lambda}{d_Z(0)}\left(\sum_j \|\mathcal{D}_j(t,q,\cdot)\|^2_{L^2(\mathbb{R})}\right)^{1/2}.$$

Proposition 12.1 (Application of the cited public Navier–Stokes theorem)**.** *For every $\nu > 0$, take the velocity, pressure and force supplied by the public Theorem 1.1 of [6]. The construction above, with $\vartheta = 0$, gives arithmetic logarithmic residue fields satisfying*

$$\sup_{0\le t<1} \int_{\mathbb{R}^3} |r(t,q)|^2\, dq < \infty, \qquad \limsup_{t\uparrow 1} \sup_q |r(t,q)| = \infty.$$

Every residue recovers the full physical velocity by (198), and each translated arithmetic divisor has the same imaginary coordinates as the original zero divisor of Ξ.

Proof. The cited theorem supplies a smooth incompressible velocity on $[0,1)\times\mathbb{R}^3$, a fixed compact support, zero initial velocity, a uniform kinetic-energy bound, and unbounded terminal velocity supremum, with its stated smooth compactly supported force and its pressure. These are the inputs of the proved preterminal flow construction, so its volume and inverse statements apply on every $[0, T_0]$ with $T_0 < 1$. Formula (200) and the upper bound using M_K prove the uniform residue energy bound. The lower bound in (199) proves the unbounded terminal supremum. Recovery and imaginary-coordinate preservation were proved above for the actual same fields. The existence step in this corollary is exactly the cited Theorem 1.1; the correspondence and every displayed estimate are proved here. $\square$

12.4 The published witness's exact growing residue

Take the particular corrected field of the public construction, with the source's fixed $h \in (0, 1/100)$, $X_{\mathrm{in}} > 0$, $e_0 > 0$, and remainder $R_*(\tau)$. Its closing formulas [6, equations (10.20)–(10.22), p. 124], as imported in the preceding witness construction, are

$$|R_*(\tau)| \le C_*\tau^{2h} \quad (0 < \tau < \tau_*), \qquad (u_\nu)_2(x_{\nu,\tau}, 1-\tau) = \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau)),$$

$$r_{\nu,\tau} = \sqrt{2\nu X_{\mathrm{in}}\tau}, \quad x_{\nu,\tau} = (r_{\nu,\tau}, 0, 0), \quad q_{\nu,\tau} = X^{-1}_{\nu,1-\tau}(x_{\nu,\tau}).$$

The remainder is the actual source remainder, including its corrected fields. The parameter h is the one chosen in that construction. The quantity $r_{\nu,\tau}$ here is the displayed physical radius; the map to the original polynomial target is retained below. The labels $q_{\nu,\tau}$ vary with τ according to the exact material inverse just displayed.

Substitution into the full original polynomials gives

$$S(r,0,0)=(0,0,2r), \qquad J(r,0,0)=\begin{pmatrix}0 & 0 & 1/2\\ 0 & 1 & 3r\\ 2 & -3r^2 & -r^3\end{pmatrix}.$$

Here the same original formulas are

$$F_1=Q^3w+y^2Q(4+3xy), \quad F_2=y+3xQ^2w+3xy^2(4+3xy), \quad F_3=2x-3x^2y-x^3w, \quad Q=1+xy.$$

Differentiating $S=(F_1/2,F_2,F_3)$ and setting $(x,y,w)=(r,0,0)$ gives each matrix entry. Its componentwise inverse gives $u_3=2v_1$ and $u_2=v_2-6rv_1$, retaining the off-diagonal $3r$. The first two arithmetic shifts at $(1-\tau,q_{\nu,\tau})$ are zero. Consequently, as an identity of full entire functions of ζ,

$$\begin{aligned}\mathcal{E}_{\nu,\tau}(\zeta) &:= \mathcal{D}_{\nu,2}(1-\tau,q_{\nu,\tau},\zeta)-6r_{\nu,\tau}\mathcal{D}_{\nu,1}(1-\tau,q_{\nu,\tau},\zeta)\\ &= \sqrt{\nu}\,\tau^{-1/2-h}(e_0+R_*(\tau))\Xi'(\zeta).\end{aligned} \tag{201}$$

Indeed each defect is $v_j\Xi'$ because its shift is zero, so their combination is $(u_\nu)_2\Xi'$, and the stated source identity applies.

For every actual zero λ of multiplicity m_λ, its complete local logarithmic image is

$$\frac{\mathcal{E}_{\nu,\tau}(\zeta)}{\Xi(\zeta)}=\sqrt{\nu}\,\tau^{-1/2-h}(e_0+R_*(\tau))\left(\frac{m_\lambda}{\zeta-\lambda}+\frac{A'_\lambda(\zeta)}{A_\lambda(\zeta)}\right). \tag{202}$$

Therefore the actual unshifted arithmetic residue at the source sample is exactly

$$\operatorname{Res}_{\zeta=\lambda}\frac{\mathcal{E}_{\nu,\tau}(\zeta)}{\Xi(\zeta)}=m_\lambda\sqrt{\nu}\,\tau^{-1/2-h}(e_0+R_*(\tau)). \tag{203}$$

All multiplicities, the whole analytic unit, the positive viscosity, the original remainder and the exact material labels remain present. Choose $\tau_{**}>0$ within the source interval and its cutoff region with $C_*\tau_{**}^{2h}\le e_0/2$. For $0<\tau<\tau_{**}$ the absolute value of (203) is at least $m_\lambda\sqrt{\nu}\,e_0\tau^{-1/2-h}/2$, which diverges. This is an explicit pole-coefficient singularity of the published constructed field's arithmetic image.

For the three residue channels $r_{\nu,j}$ of (197), the same evaluation and the two-component Cauchy–Schwarz inequality give the quantitative bound

$$|r_\nu(1-\tau,q_{\nu,\tau})|\ge\frac{m_\lambda\sqrt{\nu}\,e_0}{2\sqrt{1+72\nu X_{\mathrm{in}}\tau}}\,\tau^{-1/2-h}. \tag{204}$$

To verify the denominator, the coefficient vector $(-6r_{\nu,\tau},1)$ has squared norm $1+36r_{\nu,\tau}^2=1+72\nu X_{\mathrm{in}}\tau$. Its product with the first two residue coordinates is exactly the residue in (203); the third coordinate can only increase the Euclidean norm on the left. If $E_\nu=\nu^{5/4}F_*(1)$ is the full source force-integral energy bound and $M_\nu=\max_{K_\nu}\|J\|_{\mathrm{op}}$, the earlier volume identity simultaneously gives

$$\int_{\mathbb{R}^3}|r_\nu(t,q)|^2dq\le m_\lambda^2M_\nu^2E_\nu^2=m_\lambda^2M_\nu^2\nu^{5/2}F_*(1)^2 \quad (t<1).$$

At these particular first two channels, their zero shifts mean the source quotient is the original $Q_{\rm ar}$ itself. The exact classes are

$$[\mathcal{E}_{\nu,\tau}]_{Q_{\rm ar}} = \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau))\beta, \qquad [\mathcal{E}_{\nu,\tau}]_{C_\cap} = \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau))\widetilde{\beta}.$$

The latter line has the explicitly proved continuous inverse $\widetilde{R}_\lambda/m_\lambda$, so its scalar coordinate has exactly the displayed singularity. In particular these common classes leave every bounded subset of $C_\cap$: a continuous linear functional sends bounded sets to bounded scalar sets, whereas their $\widetilde{R}_\lambda$ values diverge. Their projection to the actual source quotient and its exact kernel remain (192); the undecided source membership is not replaced by an identification with the analytic ideal.

There is also an exact relation between the changing sample labels and the time component of the flat logarithmic connection. Let $A_\tau = D_q X_{\nu,1-\tau}(q_{\nu,\tau})$. Differentiating $X_{\nu,1-\tau}(q_{\nu,\tau}) = x_{\nu,\tau}$ gives

$$A_\tau \frac{dq_{\nu,\tau}}{d\tau} = \frac{dx_{\nu,\tau}}{d\tau} + u_\nu(1-\tau, x_{\nu,\tau}), \qquad \frac{dx_{\nu,\tau}}{d\tau} = \frac{r_{\nu,\tau}}{2\tau}(1,0,0).$$

The sign follows from $d(1-\tau)/d\tau = -1$. Since $\nabla_q a_j$ is the jth row of $J(x_{\nu,\tau})A_\tau$, and the first two rows of $J(r,0,0)$ have zero first component,

$$\nabla_q a_j \cdot \frac{dq_{\nu,\tau}}{d\tau} = v_j, \qquad j = 1,2.$$

Thus $a_j(1-\tau, q_{\nu,\tau}) = 0$ has total derivative $-v_j + \nabla_q a_j \cdot dq_{\nu,\tau}/d\tau = 0$, exactly as required. On the spectral complement of the zeros, the pullback of (196) to this curve and fixed ζ is zero in the τ direction, while its original ∂_t component has the residue (197). These statements are related by the displayed full tangent-vector identity; the sample is not silently treated as one fixed material label.

For the full four-term profile $K_\theta = U_\theta K_0$, the corresponding actual source classes are instead

$$\kappa = [K_0]_{Q_{\rm ar}}, \qquad \beta_K = [K_0']_{Q_{\rm ar}}, \qquad [T_a K_\theta] = \overline{V}_{\theta,a}\kappa, \qquad [vT_a K_\theta'] = \overline{V}_{\theta,a}(v\beta_K).$$

These follow by the same commuting translation, heat and derivative maps and retain κ without assigning it the value zero. The residue corollary above uses the actual Ξ profile. The displayed four-term quotient formulas concern the whole K profile and its derivative; they do not identify individual zeros of K with zeros of Ξ.

12.5 The translated modular double

Keep $C_E h(s) = \overline{h(\overline{s})}$, $\tau^c = (C_E)_*\tau$ and $N^c = C_E N_{\rm ar}$ from the explicit modular construction. Direct conjugation gives $C_E T_a = T_{\overline{a}} C_E$, and the established heat identity gives $C_E U_\theta = U_{\overline{\theta}} C_E$. On the two entire sectors define

$$\widehat{V}_{\theta,a} = \operatorname{diag}(T_a U_\theta, T_a U_{-\theta}), \qquad \mathcal{J}(h,k) = (C_E k, C_E h).$$

Then the exact all-parameter identity is

$$\mathcal{J}\widehat{V}_{\theta,a} = \widehat{V}_{-\overline{\theta},\overline{a}}\mathcal{J}. \tag{205}$$

Indeed its first component is $C_E T_a U_{-\theta} k = T_{\overline{a}} U_{-\overline{\theta}} C_E k$, and its second component is the same calculation with θ. For the actual real flow, $a \in \mathbb{R}$; for real Newman time it therefore sends (θ, a) to $(-\theta, a)$.

Give the two ambient sectors the topologies $(T_aU_\theta)_*\tau$ and $(T_aU_{-\theta})_*\tau^c$, and quotient by the closed relation

$$\widehat{N}_{\theta,a} = T_aU_\theta N_{\mathrm{ar}} \oplus T_aU_{-\theta}N^c.$$

Applying (205) both to these relation spaces and to the definitions of their transported topologies proves an antilinear homeomorphism from this actual doubled quotient to the one indexed by $(-\overline{\theta}, \overline{a})$. Its square is the identity, and its conjugacy with the quotient heat and translation maps follows on representatives from the same displayed equality. Thus the Cartan reversal retains the full real flow translation while reversing the radial heat parameter, with both original source and conjugate-source sectors specified.

13 The exact strength of the frozen source spectral information

The primary spectrum statement in [1, Proposition 3.6(ii)] specifies the actual zero set as the spectrum of the Hadamard quotient's multiplication operator. It gives no multiplicity assertion. The cited [2, Proposition 2.24] computes the finite Hermite polynomial image; the discussion after its proof explicitly treats function-space choices. The finite multiplicity calculation in [2, Theorem 2.47 and equations (2.447)–(2.449)] belongs to the separately specified weighted Hilbert quotient. The following exact algebra records both the obstruction available from such data and the limits of transferring a spectrum set alone.

13.1 One-sided heat inclusion and the full generalized chain

The already proved commutator identity (20) and its following heat calculation show that $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ makes every $S - \lambda$ algebraically onto. The single inclusion $U_tN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ at any $t \neq 0$ also makes every $S - \lambda$ onto, using every derivative of $U_t\Xi$ and the exact source division map. Here is a stronger consequence that needs no descended derivative.

Lemma 13.1. *Let $T : V \to V$ be a surjective endomorphism of a complex vector space and let $0 \neq v_0 \in \ker T$. Then for every integer $k \geq 1$, $\dim\ker T^k \geq k$. More precisely one can choose $Tv_j = v_{j-1}$ for every $j \geq 1$, and the vectors $v_0, \ldots, v_n$ are linearly independent for every n.*

Proof. Surjectivity gives the successive preimages. It follows inductively that $T^jv_j = v_0$ and $T^{j+1}v_j = 0$. Applying T^n to a relation $\sum_{j=0}^{n} c_jv_j = 0$ gives $c_nv_0 = 0$, hence $c_n = 0$. Repeating with the remaining highest index gives every coefficient zero. The first k vectors lie in $\ker T^k$, proving the bound. ☐

Applied to $T = S - \lambda$, this proves that either frozen inclusion just stated is incompatible with any nonzero finite-dimensional generalized eigenspace of the actual source operator. The source-independent implication is

$$U_tN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}, \quad t \neq 0, \quad 0 \neq v_0 \in \ker(S - \lambda) \quad \Longrightarrow \quad \dim\ker(S - \lambda)^k \geq k \quad (k \geq 1). \tag{206}$$

With a descended $d[f] = [Df]$, its more specific ladder is $v_j = (-1)^jd^jv_0/j!$; indeed $[d, S] = I$ gives $(S - \lambda)d^jv_0 = -jd^{j-1}v_0$. Thus the complete factorial and sign agree with the general preimage construction. This is a proved algebraic incompatibility; it does not assert that the unnamed source quotient has the finite generalized block needed to apply the incompatibility.

13.2 The actual finite block in the specified Sobolev source

Retain the source's real frequency t, compact character χ_0, weight $\delta > 1$, and actual zero multiplicity m. The vectors in the dual block of the specified source are

$$\eta_{t,a}(x) = \chi_0(x)|x|^{it}(\log|x|)^a, \qquad a \in \mathbb{Z}_{\geq 0}, \quad a < m, \quad a < (\delta - 1)/2.$$

The last strict bound is exactly integrability of $u^{2a}(1+u^2)^{-\delta/2}$ on $\mathbb{R}$: at infinity its power is $2a - \delta$, integrable precisely when it is less than -1; near zero it is locally integrable for all these a. Let r be the number of permitted integers. The source representation at $\lambda > 0$ acts by

$$\vartheta_m^{\mathrm{tr}}(\lambda)\eta_{t,a} = \lambda^{it} \sum_{b=0}^{a} \binom{a}{b} (\log\lambda)^b \eta_{t,a-b}.$$

Differentiating at $\lambda = e^{\epsilon}$, $\epsilon = 0$, proves

$$(D_{\chi_0}^{\mathrm{tr}} - it)\eta_{t,a} = a\eta_{t,a-1}, \qquad (D_{\chi_0}^{\mathrm{tr}} - it)^k \eta_{t,a} = \frac{a!}{(a-k)!}\eta_{t,a-k} \quad (0 \leq k \leq a). \tag{207}$$

For $k > a$ the expression is zero. These vectors are independent: divide a proposed relation by the nonzero character and use that $\log|x|$ ranges over $\mathbb{R}$, making the remaining polynomial identically zero. Thus the block has dimension r, one-dimensional ordinary eigenspace, and nilpotence index r. Its multiplicity is generalized multiplicity. The common-test morphism to the actual Hadamard source was proved earlier with its exact two kernels; this finite block is not silently identified with a block in that different quotient.

13.3 An explicit spectral model and the inverse at zero

Let $\Lambda = Z(\Xi)$ be the actual zero set. On the complex vector space

$$V_\Lambda = \bigoplus_{\lambda\in\Lambda} \bigoplus_{n\geq 0} \mathbb{C}e_{\lambda,n}$$

every vector has finite support. Give it the finest locally convex topology, defined by all seminorms on this vector space. Coordinate functionals separate points, so it is Hausdorff. Every linear map from it to any locally convex space is continuous: each pullback of a continuous seminorm is among these defining seminorms. With $e_{\lambda,-1} = 0$ set

$$S_\Lambda e_{\lambda,n} = \lambda e_{\lambda,n} + e_{\lambda,n-1}, \qquad d_\Lambda e_{\lambda,n} = -(n+1)e_{\lambda,n+1}.$$

Both are continuous, everywhere-defined, and direct evaluation on the basis proves $[d_\Lambda, S_\Lambda] = I$. For $\mu \notin \Lambda$ the full continuous inverse is

$$(S_\Lambda - \mu)^{-1} e_{\lambda,n} = \sum_{j=0}^{n} \frac{(-1)^j}{(\lambda-\mu)^{j+1}} e_{\lambda,n-j}. \tag{208}$$

Multiplication in either order cancels adjacent terms of this finite sum and leaves the basis vector. The formula preserves finite support, so it defines an inverse on the whole space, continuous by its topology. For $\mu \in \Lambda$, the μ-summand is a surjective backward shift and every other summand uses the same finite inverse formula. Thus $S_\Lambda - \mu$ is onto, its kernel is exactly $\mathbb{C}e_{\mu,0}$, and

$$\ker(S_\Lambda - \lambda)^k = \mathrm{span}\{e_{\lambda,0}, \ldots, e_{\lambda,k-1}\}, \quad \mathrm{Spec}_{\mathrm{alg}} S_\Lambda = \mathrm{Spec}_{\mathrm{end}} S_\Lambda = \mathrm{Spec}_{\mathrm{point}} S_\Lambda = \Lambda. \tag{209}$$

Here $\mathrm{Spec}_{\mathrm{end}}$ means failure of a continuous two-sided inverse in the algebra of continuous endomorphisms, not the Hilbert spectral subdivision called continuous spectrum. Since $\Xi(0) \neq 0$, this S_Λ has a continuous inverse at zero.

There is an exact principal-part realization of the model: send $e_{\lambda,n}$ to the class of w^{-n-1} in $\mathbb{C}[w, w^{-1}]/\mathbb{C}[w]$ in its λ-summand. The finite negative Laurent coefficients give its inverse. Multiplication by $\lambda + w$ is S_Λ and differentiation in w is d_Λ, as evaluation on each monomial proves. This complete model shows that the correct zero set as spectrum, geometric multiplicity one, and a continuous inverse at zero can coexist with a continuous Weyl pair. It does not identify this model with Q_{ar}.

For the actual quotient, injectivity of S gives exactly $N_{\mathrm{ar}} \cap s\mathcal{H}_{\leq 1} = sN_{\mathrm{ar}}$: if $sf \in N_{\mathrm{ar}}$, then $S[f] = 0$, so $f \in N_{\mathrm{ar}}$; the converse is polynomial invariance. The original source inverse at zero is explicitly

$$S^{-1}[f] = \left[\frac{f(s) - f(0)\Xi(s)/\Xi(0)}{s}\right]. \tag{210}$$

The singularity is removable, and division by a degree-one polynomial preserves the original order class. Multiplication by s gives $[f]$, and injectivity gives its unique inverse. If f is changed by $n \in N_{\mathrm{ar}}$, the numerator difference belongs to $N_{\mathrm{ar}} \cap s\mathcal{H}_{\leq 1} = sN_{\mathrm{ar}}$, proving representative independence directly. The usual topological reading of the source resolvent also supplies continuity of this quotient inverse, not continuity of its chosen lift. More generally,

$$S - \lambda \text{ injective} \quad \Longleftrightarrow \quad N_{\mathrm{ar}} \cap (s - \lambda)\mathcal{H}_{\leq 1} = (s - \lambda)N_{\mathrm{ar}}$$

by the same calculation. Saturation at zero does not prove saturation at other points. The missing frozen-descent decision can therefore be approached through an actual finite generalized source block or a range obstruction, as well as through identification of the topology. Neither has been supplied by the primary Hadamard spectrum clauses inspected here.

14 Proof scope and reproducibility

The new source-topology assertions proved here are the complete transported closure, homeomorphic quotient, conjugated spectral operator, source-domain differential correspondence, exact coordinate kernels and cokernels, correlated jet extension, finite free cover, compact-test heat ingress with its weighted-moment inverse domains, and exact comparison diagrams. The compact and growth relation subspaces both equal $\Xi\mathcal{H}_{\leq 1}$ by the complete division proof. Their quotient topologies are computed: the compact quotient has its exact jet-image topology and product completion, whereas the growth quotient is complete and the comparison inverse is not continuous. All derivative relation domains and finite ambiguity projections retain the full zero multiplicities and analytic units. The arithmetic principal-part realization has the exact kernel $\Xi\mathcal{H}_{\leq 1}$, every finite image, the same product completion, and explicit original-coordinate spectral blocks. The full Schwartz trace factorization and its order-class intersection are proved without assigning every Schwartz trace to that order class. Its exact Mellin image has a proved inverse and complete finite-strip estimates, including every Taylor residue at the input origin. The exponential trace orbit has its exact open Schwartz boundary. The meromorphic arithmetic heat action retains the full zeta connection, its pole cancellations and the convergent numerator jets determining its Schwartz domain. The full persistent heat and even-derivative cores of $\Xi\mathcal{H}_{\leq 1}$ are zero. Every nonzero input returns to that analytic relation, or to the original Schwartz trace intersection, only at a closed discrete set of times; the complete proof retains every actual zero multiplicity and uses the original Hadamard product. The actual four-term ξ^4 coefficient is proved nonzero from its complete Fourier multiplier, commutes exactly with heat, and has those same discrete arithmetic return restrictions, with its original incidence factor retained. The retained Cartesian bilaplacian has its complete four coefficients and exact scaled residue. That original residue has full complex ambiguity on every actual moving source quotient, including time zero. Its finite trace sections and the precise quotient topology give an explicit homeomorphic residue-zero presentation. Every finite origin jet has a reciprocal-Ξ polynomial section, and at every nonzero time every one-point finite jet is realized by transported original polynomial traces. The complete normal-order correction, nonzero-polynomial obstruction, finite section construction and both quotient maps are proved. The separately transported residue has its full convergent derivative series and the exact heat-conjugate quotient. The Cartan construction now has a proved standard-real-subspace modular double for the original heat multiplier, full spectral domains at both signs of the parameter, and an exact conjugate moving-source quotient diagram. Its involution reverses the radial parameter. The threefold Fourier map identifies the positive Euclidean viscous generator with the original heat coefficient. For the retained nonlinear spatial pullback, the complete splitting, compressed diffusion, constant leakage kernel, affine commutator kernel, and second-diffusion feedback are computed explicitly. The separate public Navier–Stokes theorem is recorded as an imported source claim with distinct revision receipts; its full proof has not been independently verified in this fragment. Its actual cutoff-summed witness is now an explicit imported input to the full arithmetic flow map. The original polynomial target, inverse differential, pressure, forcing, viscosity scaling, material metric, both time generators, and the two exact Fourier norm constants are retained. The source's particular growing swirl produces a complete arithmetic derivative channel and an unbounded residue with its whole source remainder. The joint profile and residue energies stay bounded, and the full profile has a proved terminal limit in its stated Hilbert space. The actual moving source class and its exact scalar kernel are calculated. A common quotient by $N_{\mathrm{ar}} \cap \Xi\mathcal{H}_{\leq 1}$ maps to both source and analytic quotients with proved kernels and a continuous residue recovery projection. The full logarithmic connection, varying source sample labels, and translated Cartan reversal are proved. A separate exact spectral model and generalized-chain argument identify what additional actual source spectral data would decide frozen descent without imposing a replacement topology. The specified source Sobolev cokernel has an exact dense compact-test ingress and a common-test comparison with Q_{ar}, with both kernels and all generator signs proved. Both explicit realizations have proved frozen differential and heat noninvariance and the original-coordinate divisor spectrum. Each heated finite Hermite trace also has its proved unique Möbius inverse, precise Mellin domain and full nontrivial-zero pole orders, with actual failures of boundedness at the origin for suitable original seeds at every nonzero time. The missing identification of the paper's unnamed topology is recorded explicitly and is not replaced by a conditional arithmetic theorem. Nothing in these results establishes an off-critical zero, a negative Weil value, or an arithmetic endpoint by analogy with the inverse-fibre collision. The exact transport through that collision cover and the derivative information it requires have been calculated above.

The companion checker uses rational polynomial arithmetic to verify the heat conjugation on finite polynomial inputs, the retained product terms, the matrix commutator, the iterated branch-domain coefficients, the original-coordinate relation, and sample finite jet multiplication, the multiplicity-sensitive derivative domains, the source heat commutator recurrence, exact Möbius divisor sums, and Mellin principal-part coordinate inverses. The continuation checks the exponential Gaussian input and its generator, inverse Mellin differentiation and arithmetic residues, the Sobolev cutoff and Fourier moments, and the meromorphic heat conjugacy with all local pole terms. It also checks the unchanged reflection composition, time-jet coefficients and positive

Hadamard counting integrand used to compute the persistent core. The final spatial checks retain all Cartesian derivatives, all four bilaplacian coefficients, the residue limit and original four-term coefficient, the reciprocal-jet sections, and every tested normal-order and heat-conjugation correction. The Cartesian replay also checks the full spatial compression, leakage and its returning component, and the original scalar viscous residual. The modular domains and quotient diagram have an independent written proof audit. The new original three-coordinate replay checks its full determinant and cofactor inverse, the actual source sample matrix and inverse, the swirl channel, both time derivatives, all three material accelerations, the translated spectral conjugacy, and every tested logarithmic unit and residue coefficient. The varying material-label tangent signs and the quantitative source lower-bound denominator are retained. These are reproducible checks of the displayed identities, not a substitute for their written proofs. Source hashes and exact reading locators are in the companion provenance receipt; no numerical zero search or Lean run is part of this work.

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